URP: English Figure Atlas

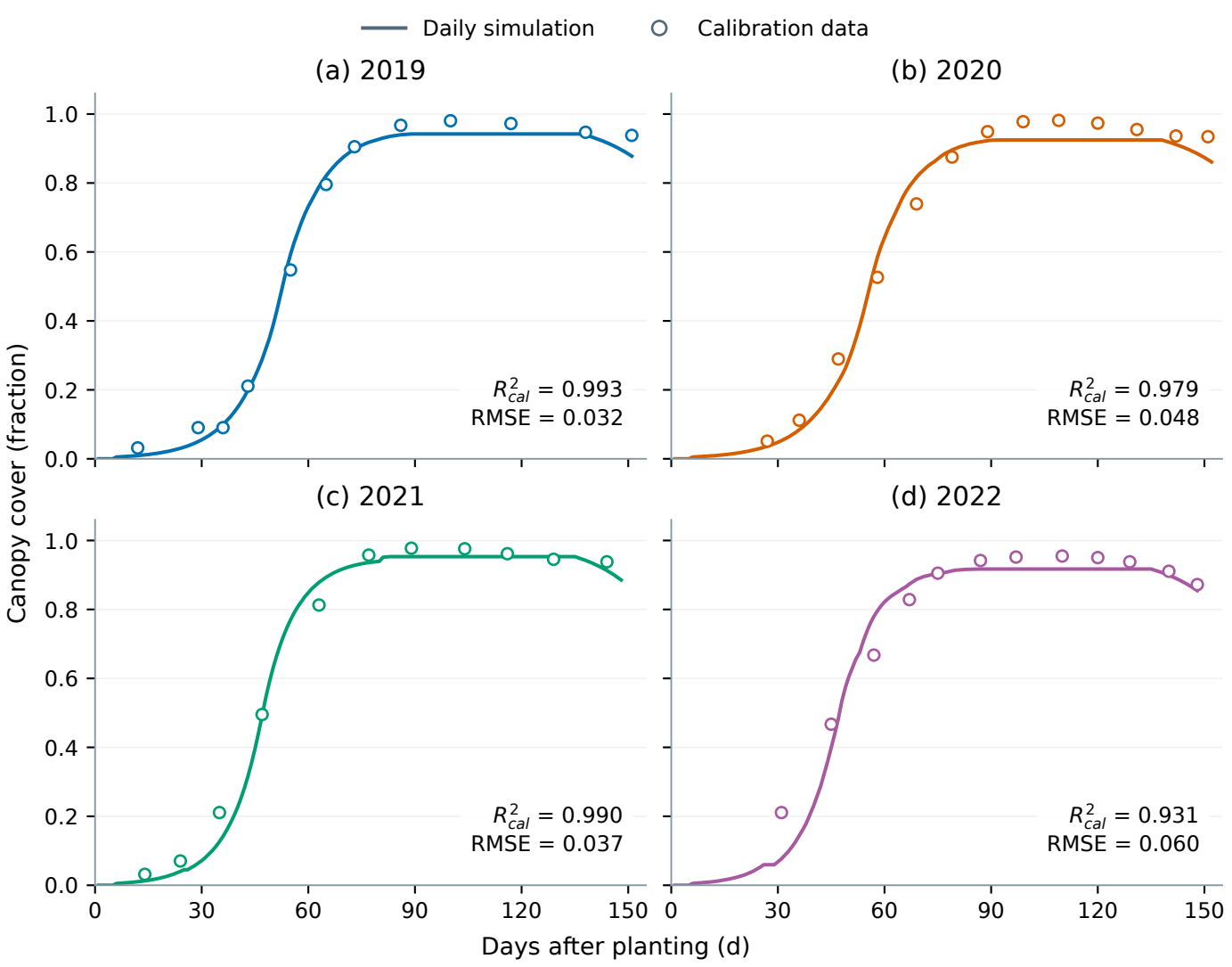
2019-2022 DI maize | 21 October 2026

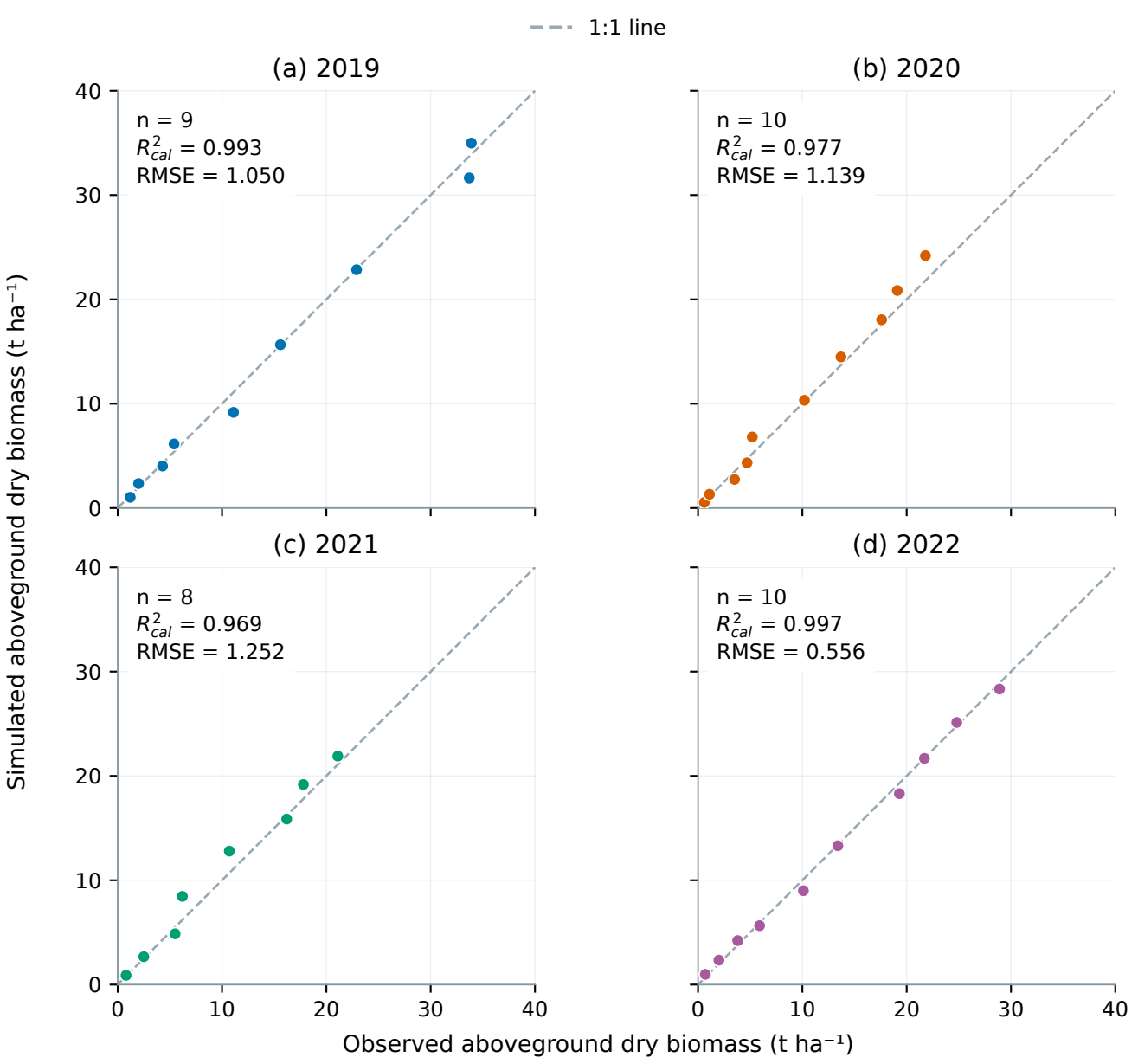
Original IDs are preserved for traceability. Main-manuscript figure numbers differ.

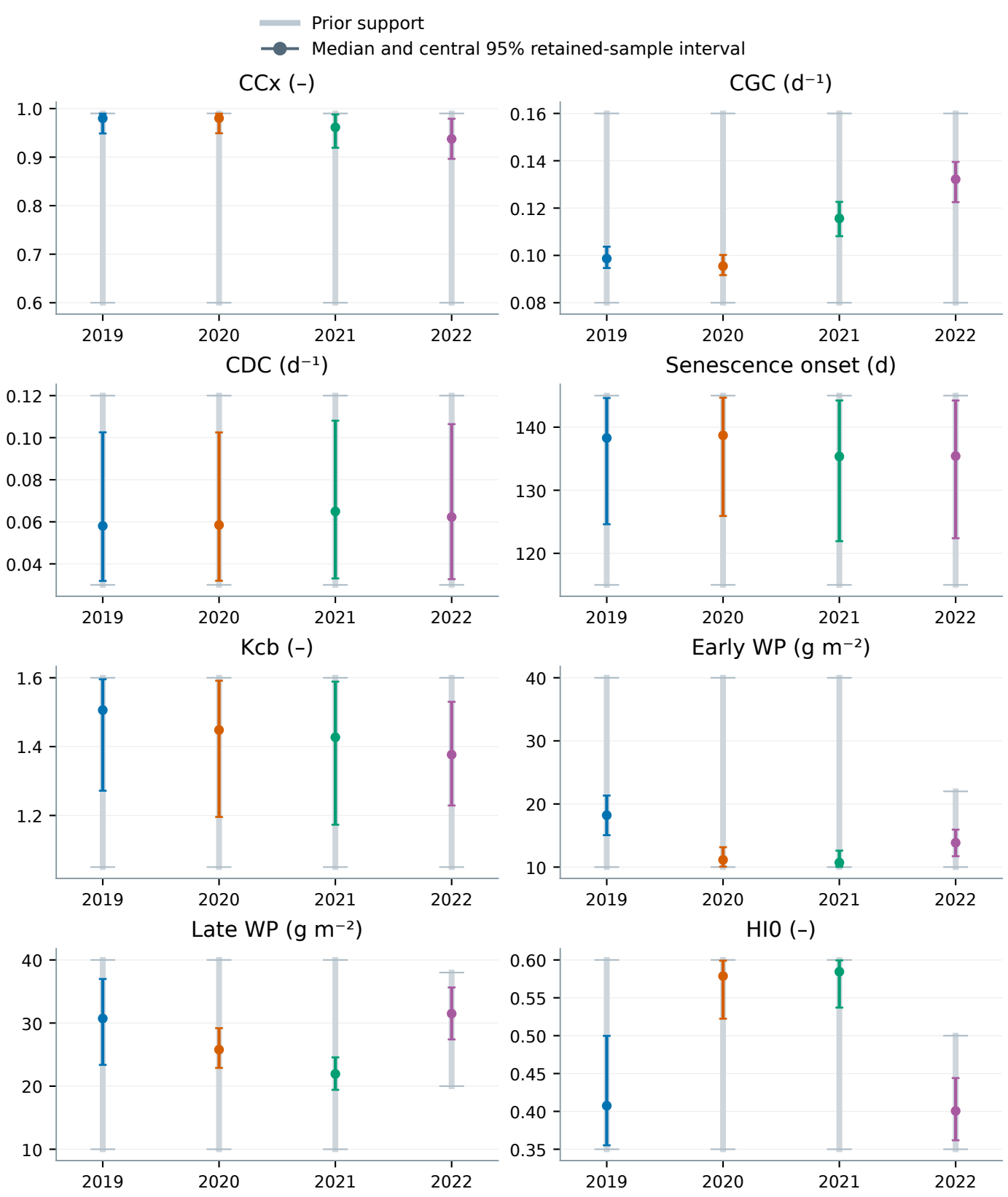
The accompanying CSV/Word guide provides captions, source tables and interpretation boundaries.

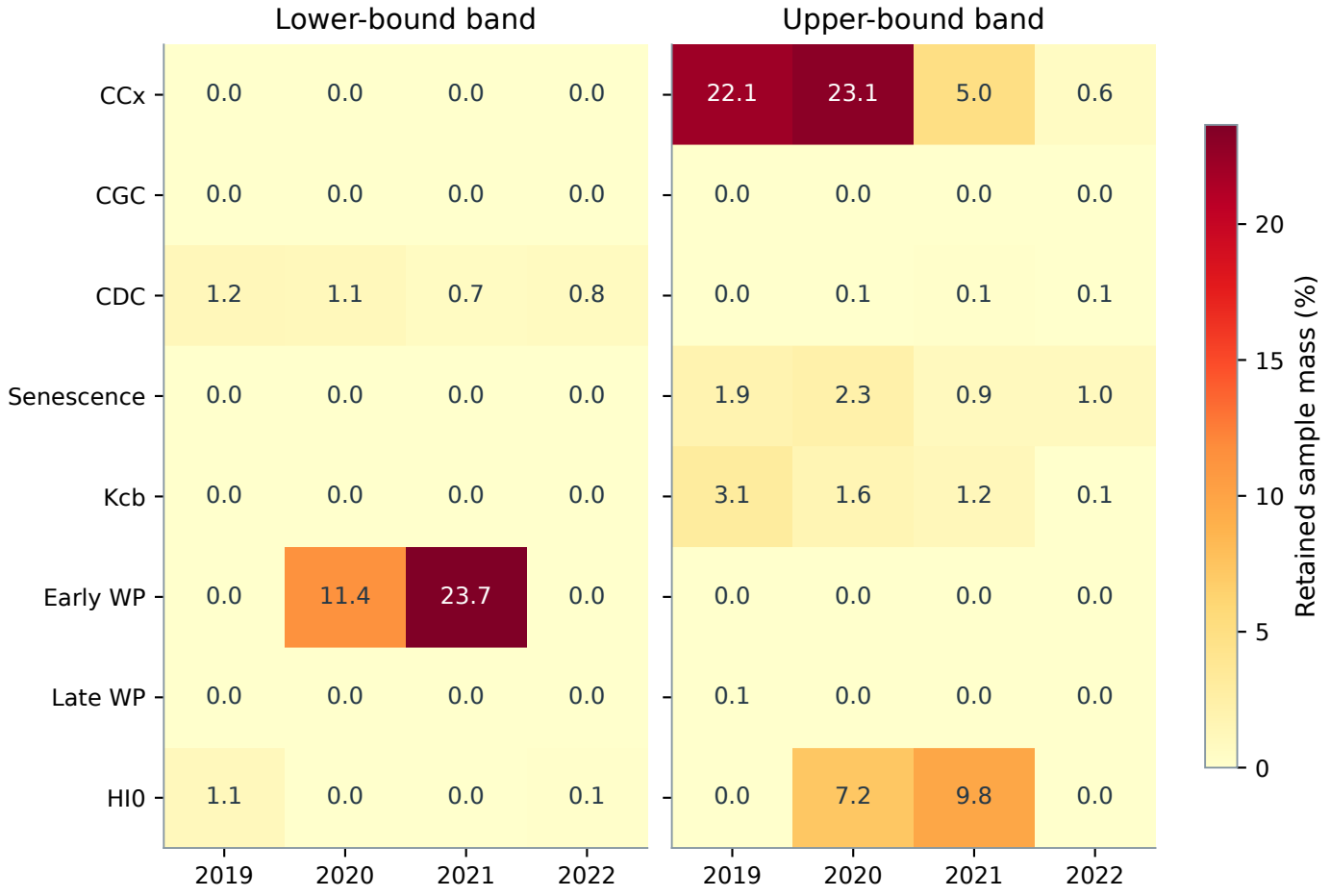
Historical four-year results (Zmax 2.30 m, fMulch 0) are separate from 2021 root-depth/mulch pilots. Numerical verification is not field validation.

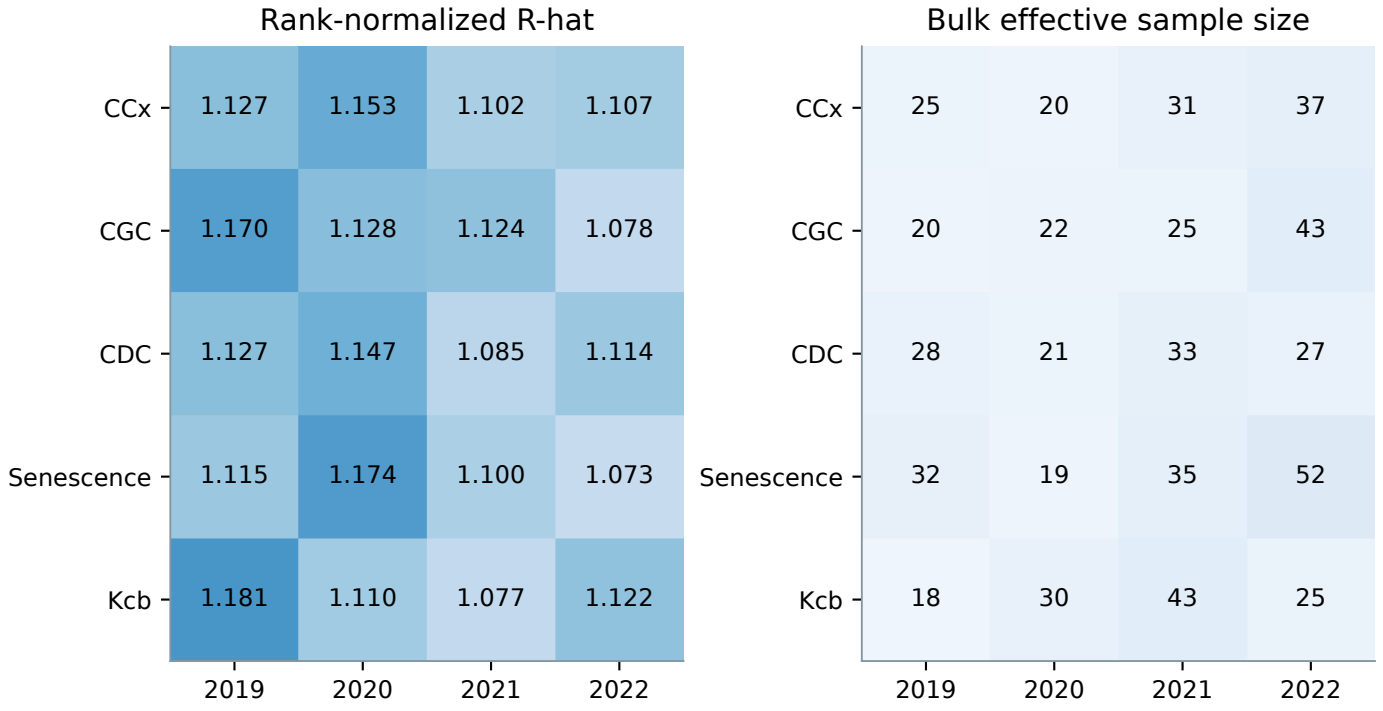
Full-resolution PNG, PDF and SVG files remain in their scientific-stage packages.

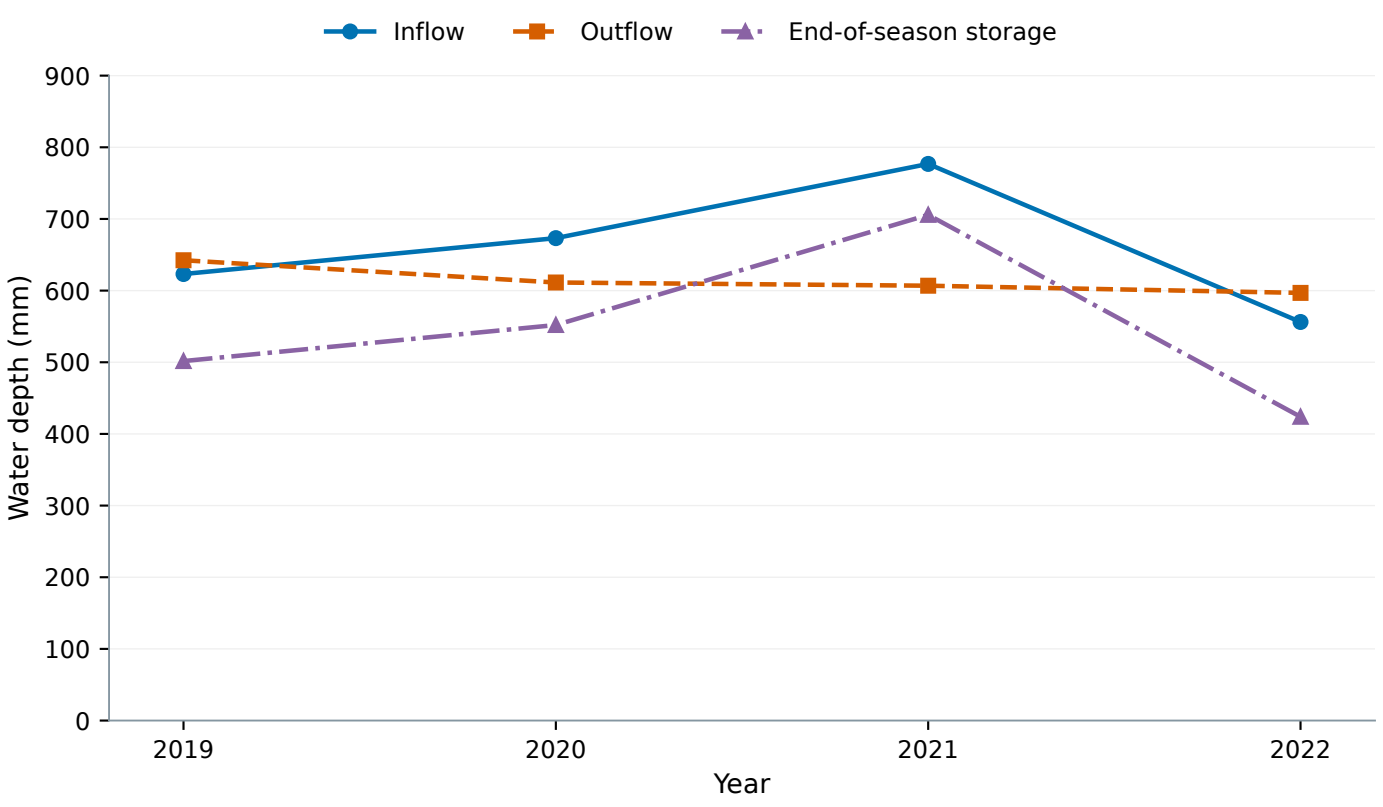

Daily simulation
Calibration data
(a) 2019
(b) 2020
(c) 2021
(d) 2022
R^2_{cal} = 0.993
RMSE = 0.032
R^2_{cal} = 0.979
RMSE = 0.048
R^2_{cal} = 0.990
RMSE = 0.037
R^2_{cal} = 0.931
RMSE = 0.060
Canopy cover (fraction)
Days after planting (d)
0.0
0.2
0.4
0.6
0.8
1.0
0
30
60
90
120
150

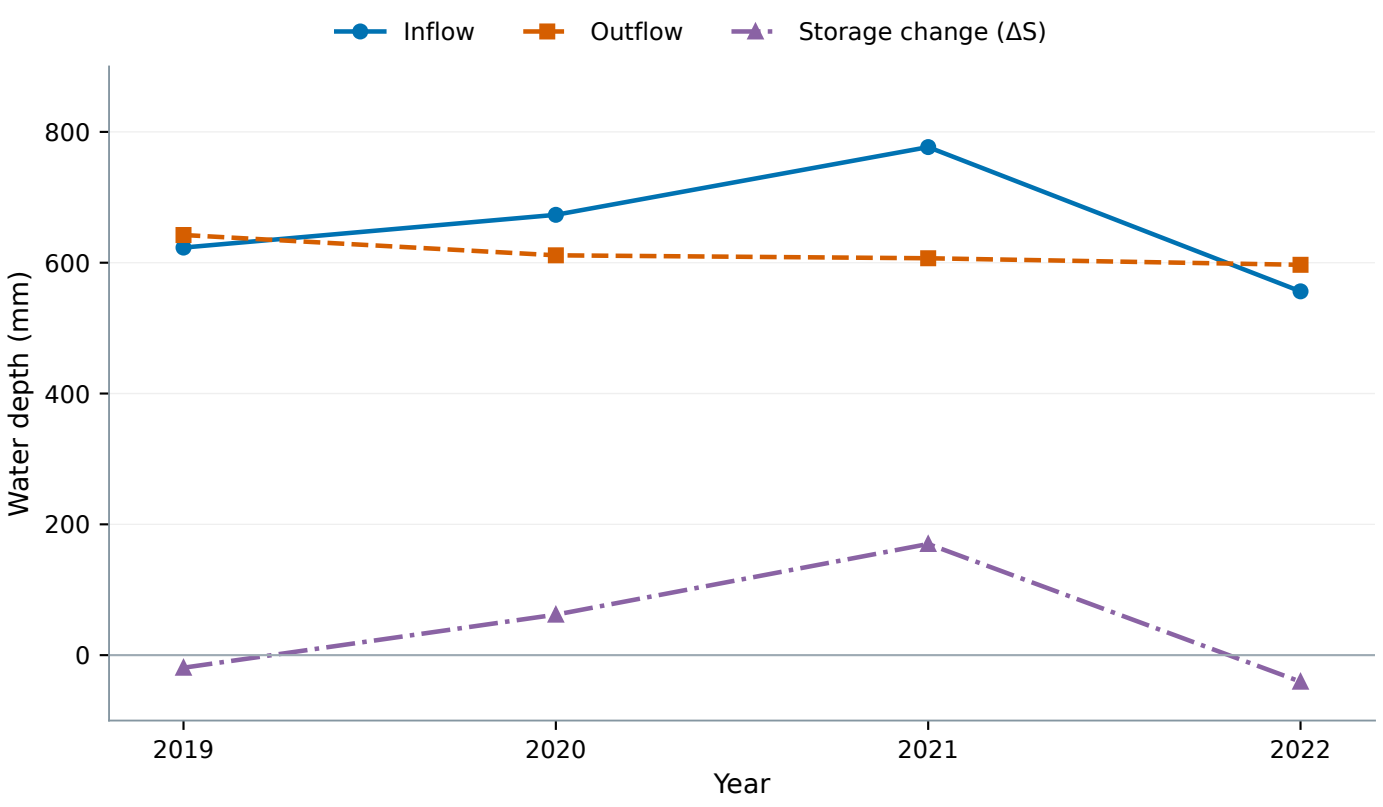

1:1 line
(a) 2019
n = 9
R^2_{cal} = 0.993
RMSE = 1.050
(b) 2020
n = 10
R^2_{cal} = 0.977
RMSE = 1.139
(c) 2021
n = 8
R^2_{cal} = 0.969
RMSE = 1.252
(d) 2022
n = 10
R^2_{cal} = 0.997
RMSE = 0.556
Simulated aboveground dry biomass (t ha⁻¹)
Observed aboveground dry biomass (t ha⁻¹)
0
10
20
30
40

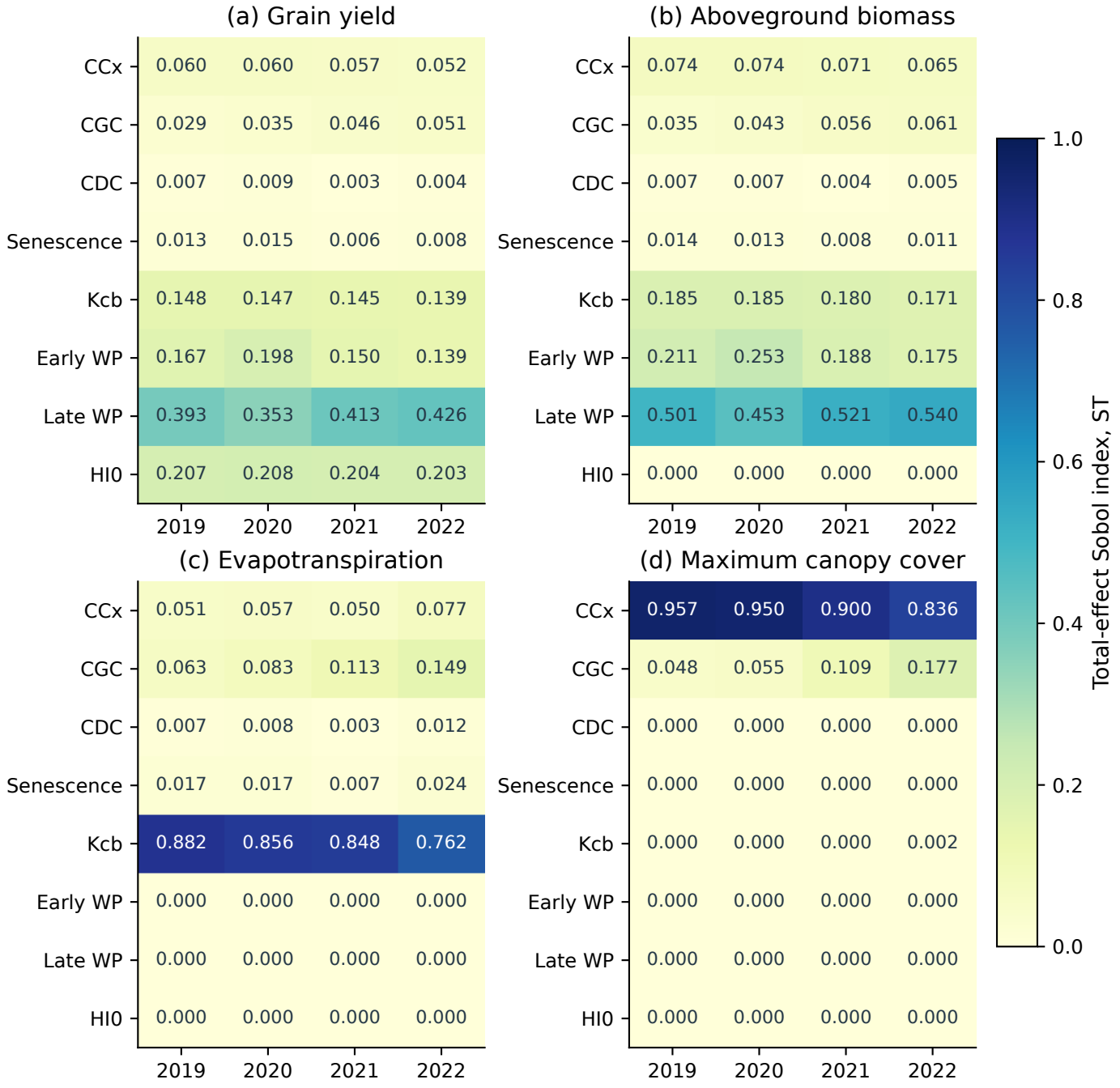

Prior support
Median and central 95% retained-sample interval
CCx (–)
CGC (d⁻¹)
CDC (d⁻¹)
Senescence onset (d)
Kcb (–)
Early WP (g m⁻²)
Late WP (g m⁻²)
HI0 (–)
2019
2020
2021
2022

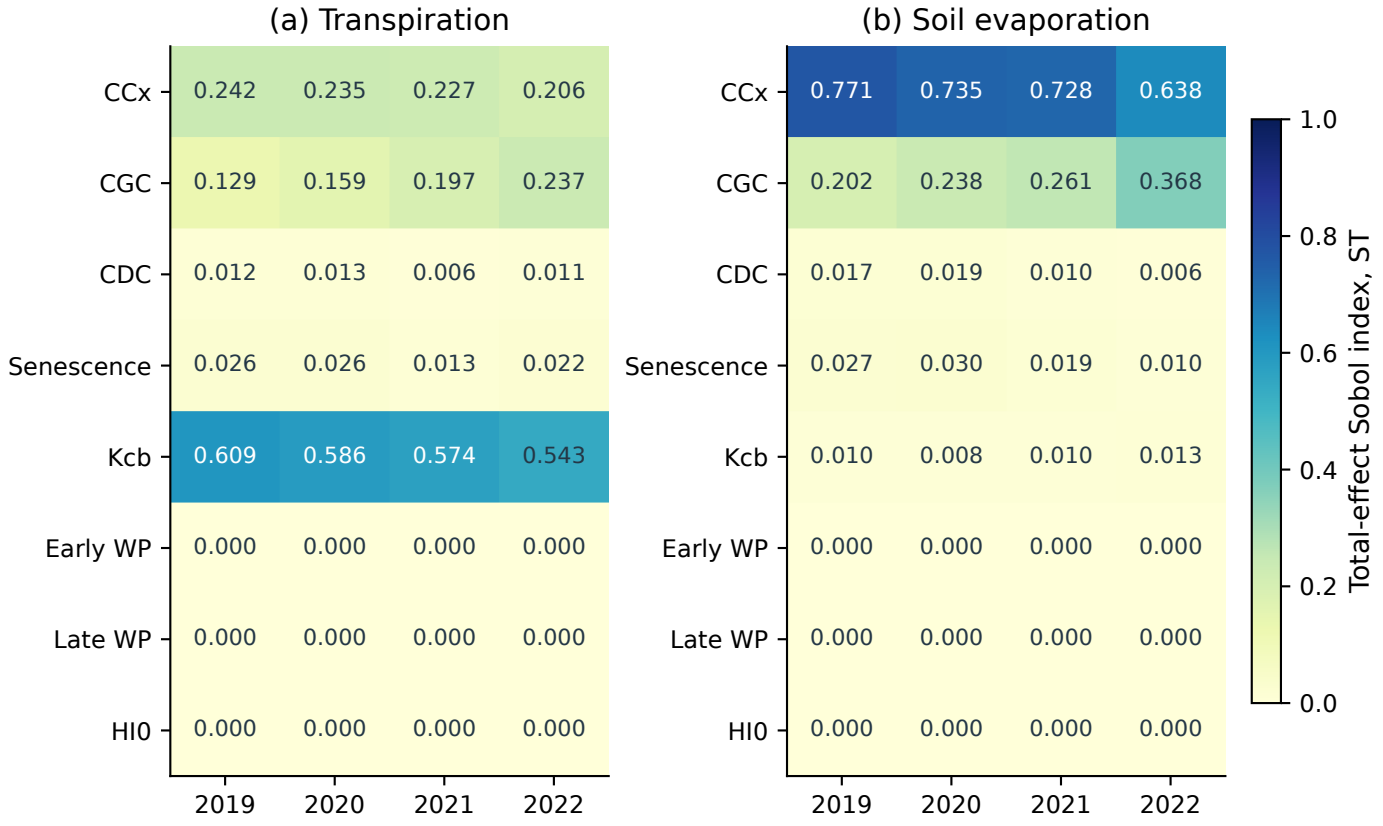

Lower-bound band
Upper-bound band
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.0 0.0 0.0 0.0
0.0 0.0 0.0 0.0
1.2 1.1 0.7 0.8
0.0 0.0 0.0 0.0
0.0 0.0 0.0 0.0
0.0 11.4 23.7 0.0
0.0 0.0 0.0 0.0
1.1 0.0 0.0 0.1
22.1 23.1 5.0 0.6
0.0 0.0 0.0 0.0
0.0 0.1 0.1 0.1
1.9 2.3 0.9 1.0
3.1 1.6 1.2 0.1
0.0 0.0 0.0 0.0
0.1 0.0 0.0 0.0
0.0 7.2 9.8 0.0
2019
2020
2021
2022
Retained sample mass (%)
0
5
10
15
20

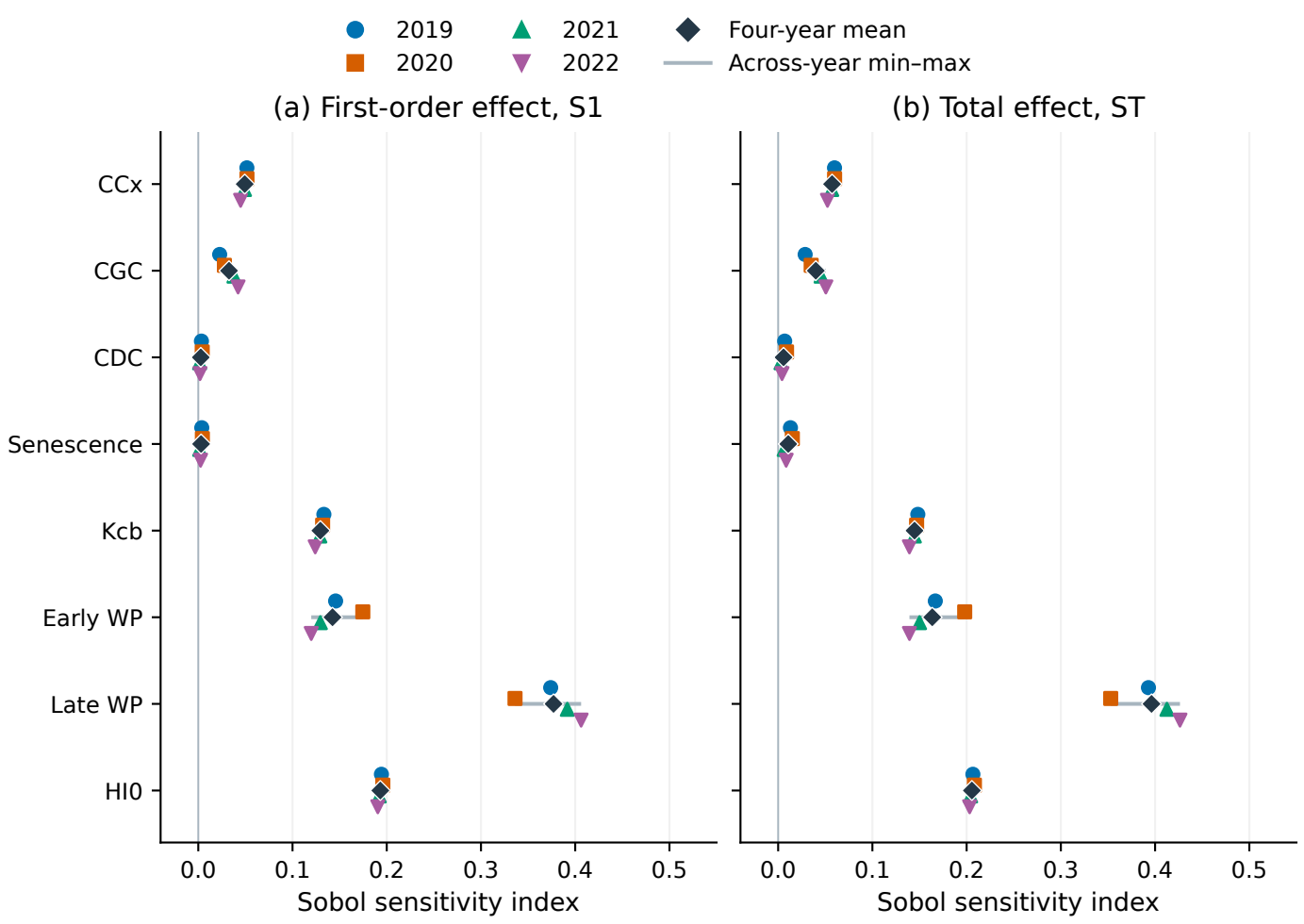

Rank-normalized R-hat
CCx
CGC
CDC
Senescence
Kcb
1.127
1.153
1.102
1.107
1.170
1.128
1.124
1.078
1.127
1.147
1.085
1.114
1.115
1.174
1.100
1.073
1.181
1.110
1.077
1.122
2019
2020
2021
2022
Bulk effective sample size
CCx
CGC
CDC
Senescence
Kcb
25
20
31
37
20
22
25
43
28
21
33
27
32
19
35
52
18
30
43
25
2019
2020
2021
2022

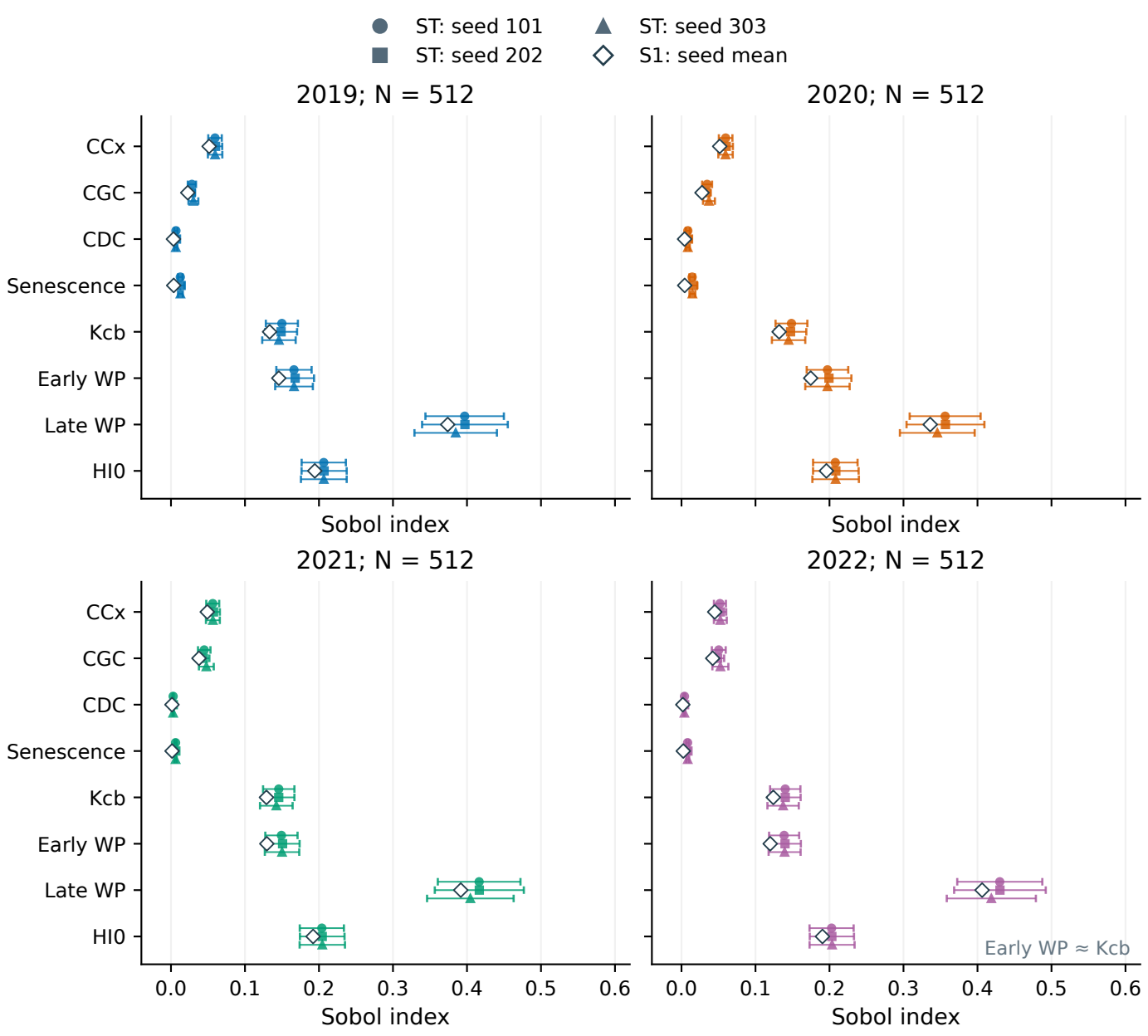

Inflow
Outflow
End-of-season storage
Water depth (mm)
900
800
700
600
500
400
300
200
100
0
2019
2020
2021
2022
Year

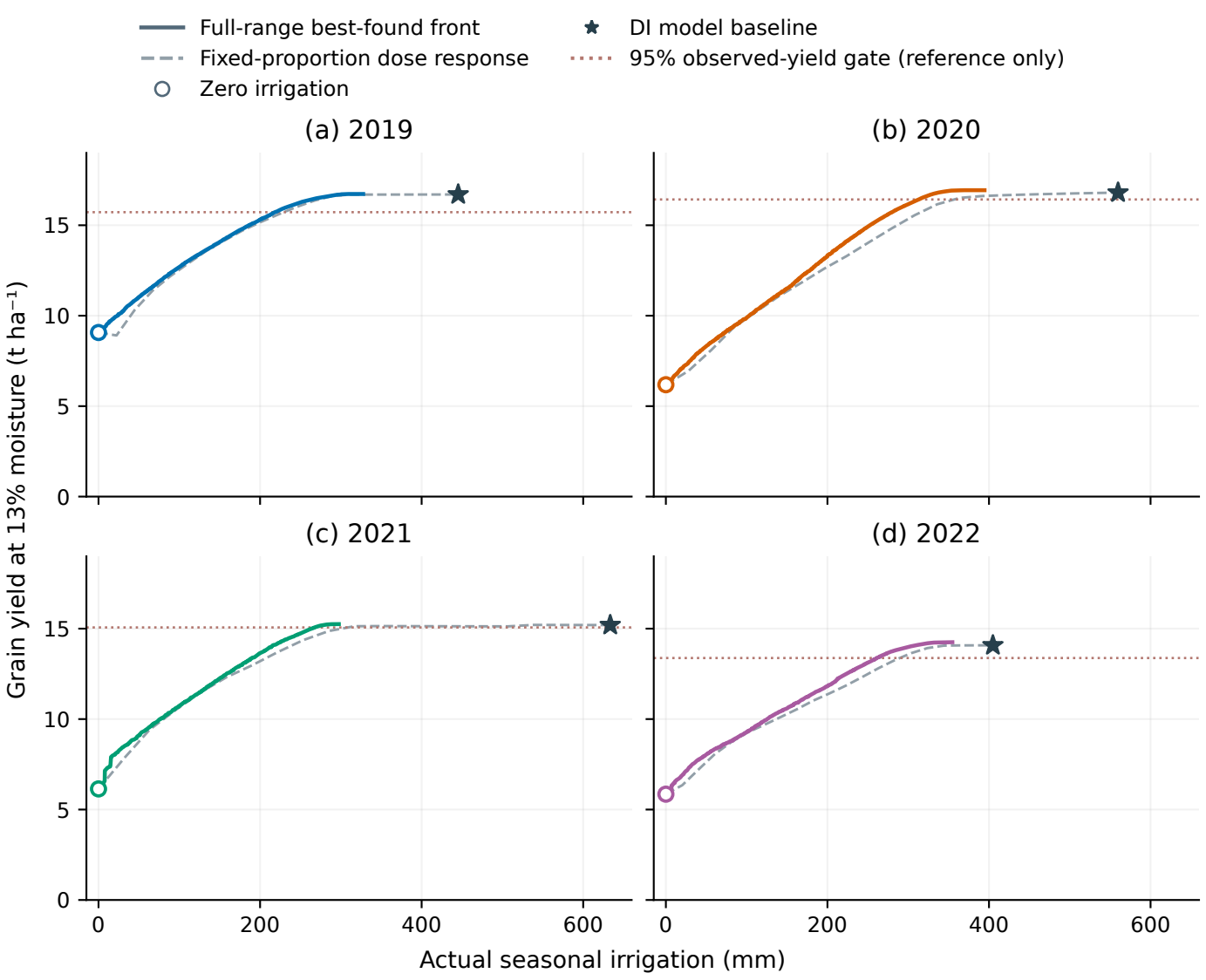

Inflow
Outflow
Storage change (ΔS)
800
600
400
200
0
Water depth (mm)
2019
2020
2021
2022
Year

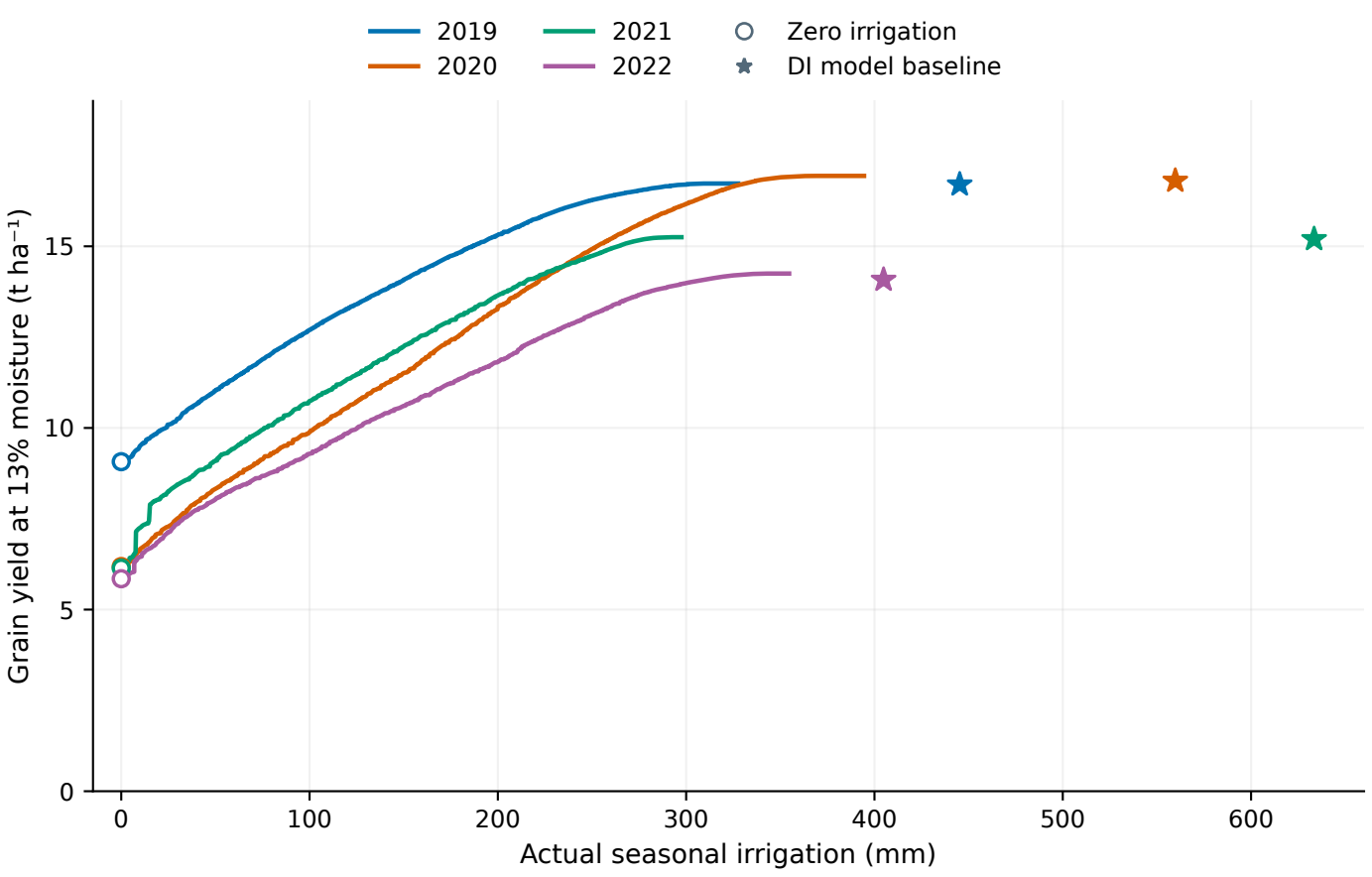

(a) Grain yield
CCx 0.060 0.060 0.057 0.052
CGC 0.029 0.035 0.046 0.051
CDC 0.007 0.009 0.003 0.004
Senescence 0.013 0.015 0.006 0.008
Kcb 0.148 0.147 0.145 0.139
Early WP 0.167 0.198 0.150 0.139
Late WP 0.393 0.353 0.413 0.426
HI0 0.207 0.208 0.204 0.203
2019 2020 2021 2022
(b) Aboveground biomass
CCx 0.074 0.074 0.071 0.065
CGC 0.035 0.043 0.056 0.061
CDC 0.007 0.007 0.004 0.005
Senescence 0.014 0.013 0.008 0.011
Kcb 0.185 0.185 0.180 0.171
Early WP 0.211 0.253 0.188 0.175
Late WP 0.501 0.453 0.521 0.540
HI0 0.000 0.000 0.000 0.000
2019 2020 2021 2022
(c) Evapotranspiration
CCx 0.051 0.057 0.050 0.077
CGC 0.063 0.083 0.113 0.149
CDC 0.007 0.008 0.003 0.012
Senescence 0.017 0.017 0.007 0.024
Kcb 0.882 0.856 0.848 0.762
Early WP 0.000 0.000 0.000 0.000
Late WP 0.000 0.000 0.000 0.000
HI0 0.000 0.000 0.000 0.000
2019 2020 2021 2022
(d) Maximum canopy cover
CCx 0.957 0.950 0.900 0.836
CGC 0.048 0.055 0.109 0.177
CDC 0.000 0.000 0.000 0.000
Senescence 0.000 0.000 0.000 0.000
Kcb 0.000 0.000 0.000 0.002
Early WP 0.000 0.000 0.000 0.000
Late WP 0.000 0.000 0.000 0.000
HI0 0.000 0.000 0.000 0.000
2019 2020 2021 2022
Total-effect Sobol index, ST
1.0
0.8
0.6
0.4
0.2
0.0

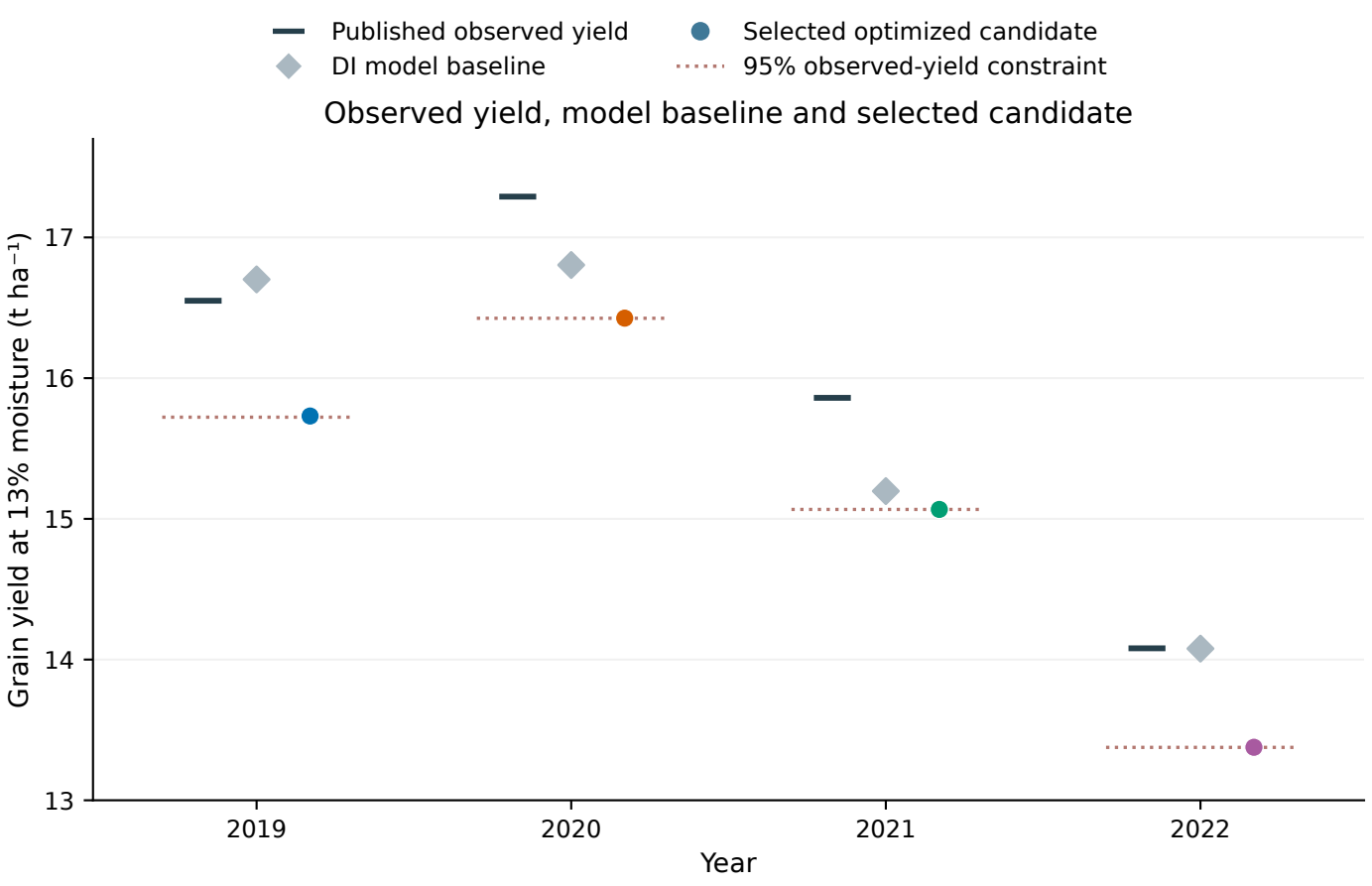

(a) Transpiration
(b) Soil evaporation
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.242 0.235 0.227 0.206
0.129 0.159 0.197 0.237
0.012 0.013 0.006 0.011
0.026 0.026 0.013 0.022
0.609 0.586 0.574 0.543
0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000
0.771 0.735 0.728 0.638
0.202 0.238 0.261 0.368
0.017 0.019 0.010 0.006
0.027 0.030 0.019 0.010
0.010 0.008 0.010 0.013
0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000
0.000 0.000 0.000 0.000
2019
2020
2021
2022
Total-effect Sobol index, ST
1.0
0.8
0.6
0.4
0.2
0.0

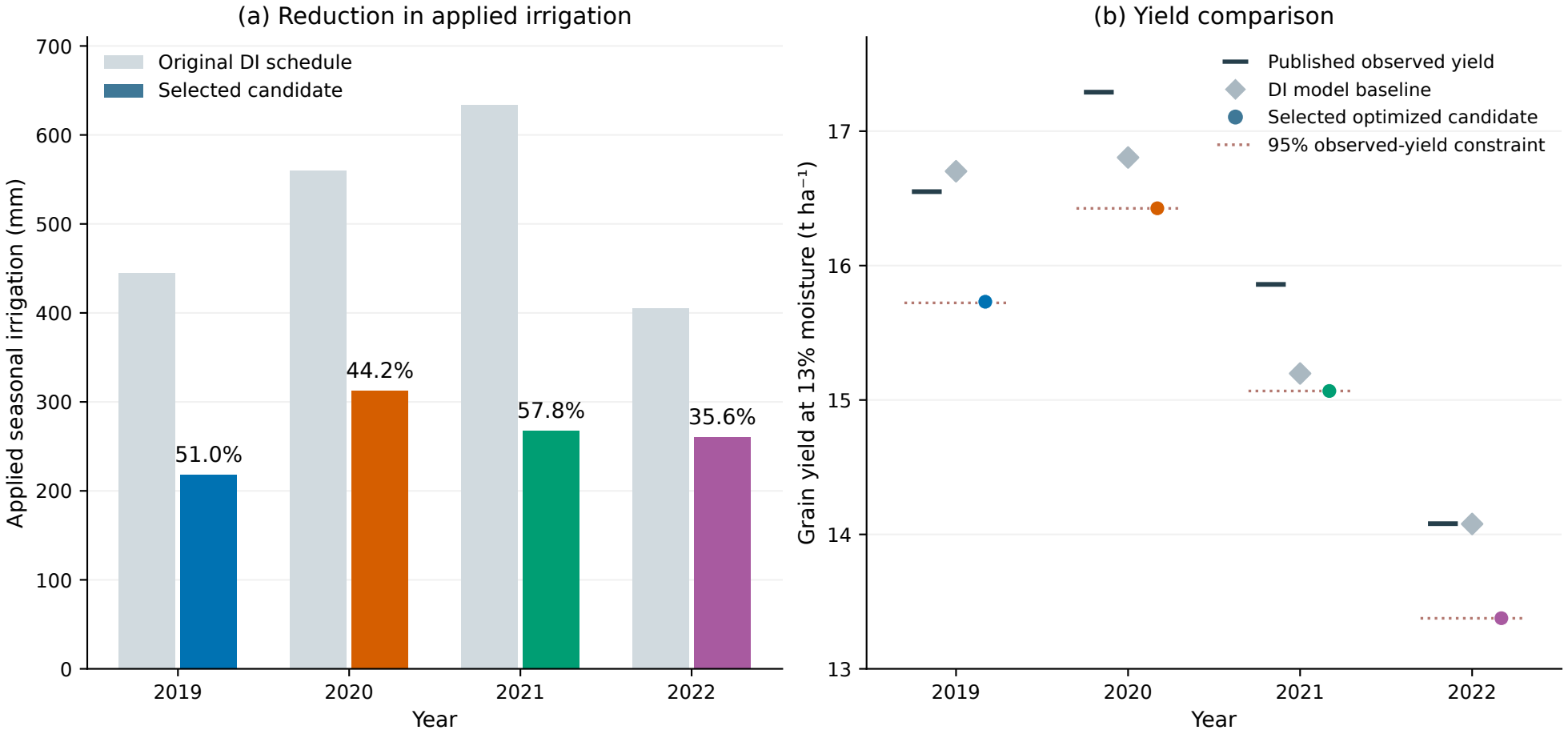

2019
2020
2021
2022
Four-year mean
Across-year min–max
(a) First-order effect, S1
(b) Total effect, ST
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.0
0.1
0.2
0.3
0.4
0.5
Sobol sensitivity index
Sobol sensitivity index

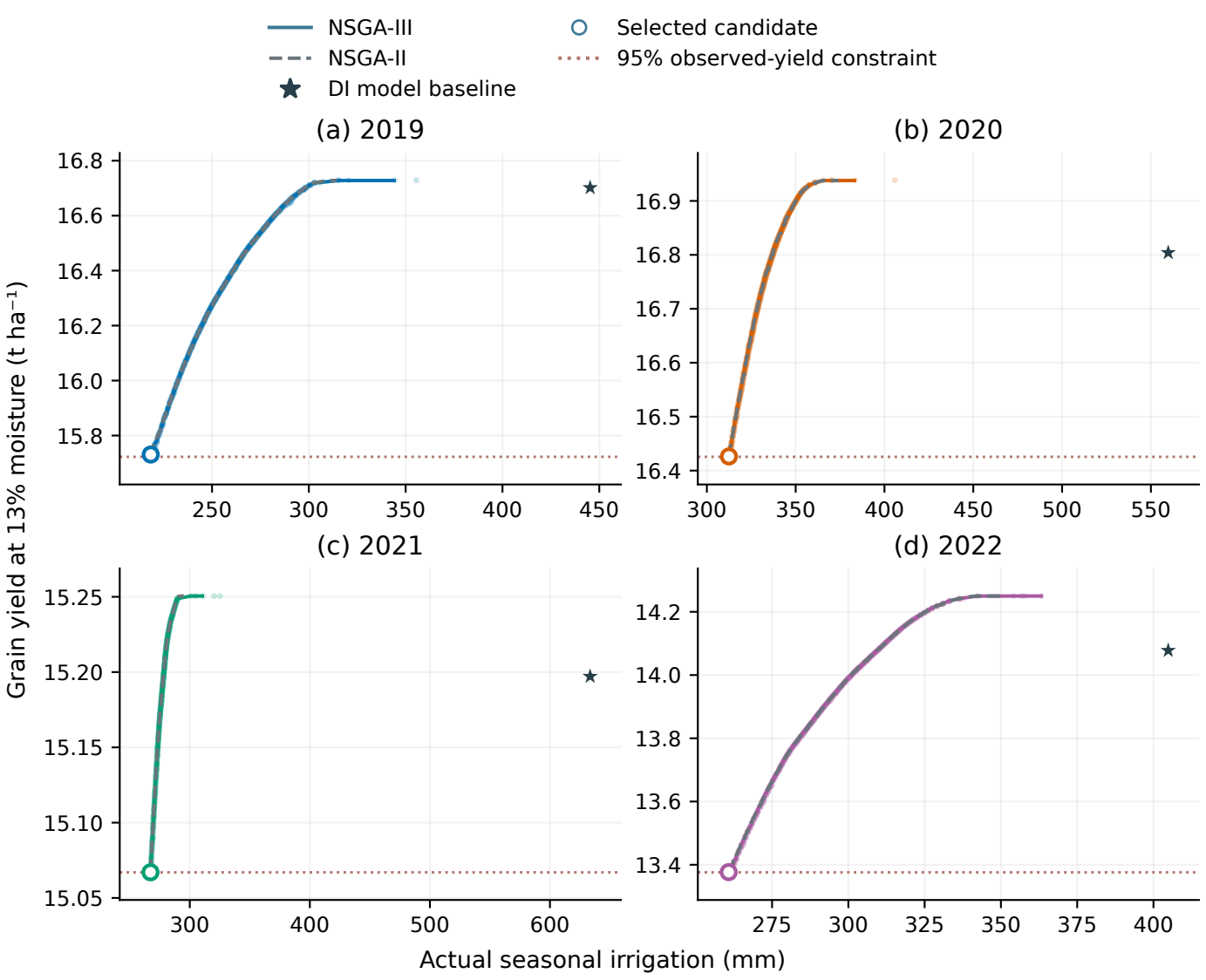

ST: seed 101
ST: seed 202
ST: seed 303
S1: seed mean
2019; N = 512
2020; N = 512
2021; N = 512
2022; N = 512
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
Sobol index
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Early WP ≈ Kcb

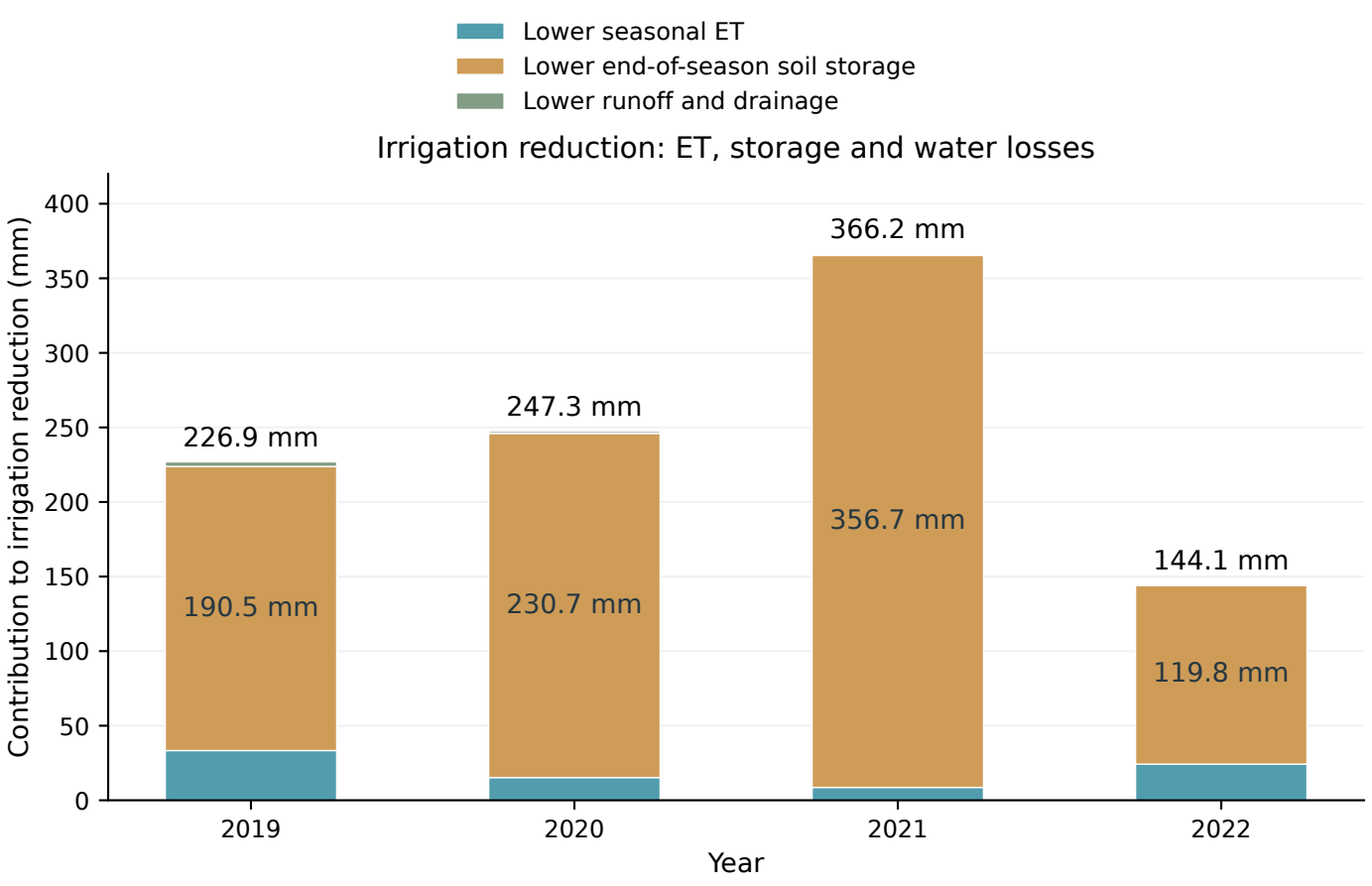

Full-range best-found front
Fixed-proportion dose response
Zero irrigation
DI model baseline
95% observed-yield gate (reference only)
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Grain yield at 13% moisture (t ha⁻¹)
Actual seasonal irrigation (mm)
0
5
10
15
0
200
400
600

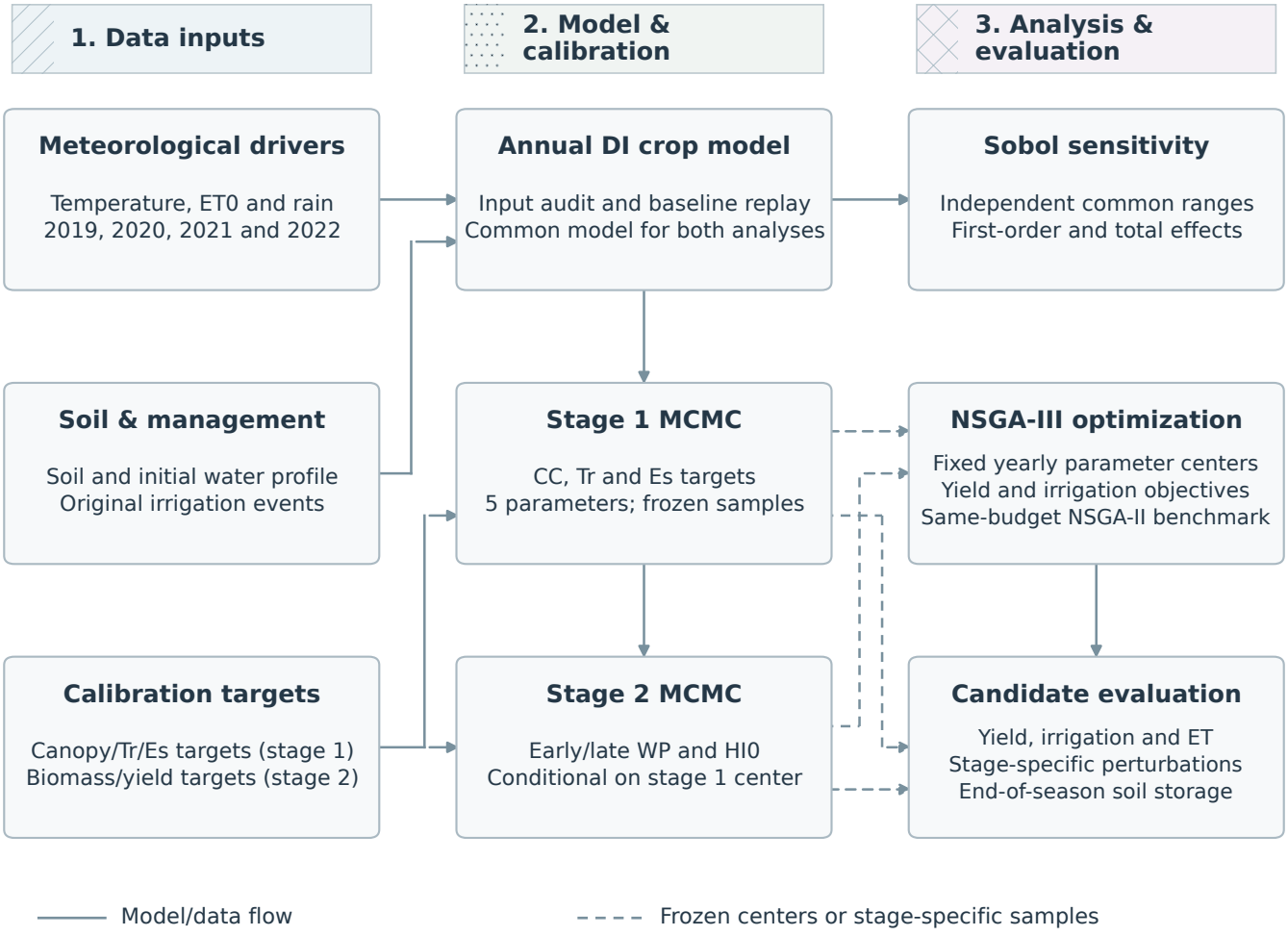

2019
2020
2021
2022
Zero irrigation
DI model baseline
Grain yield at 13% moisture (t ha⁻¹)
0
5
10
15
0
100
200
300
400
500
600
Actual seasonal irrigation (mm)


Published observed yield
DI model baseline
Selected optimized candidate
95% observed-yield constraint
Observed yield, model baseline and selected candidate
Grain yield at 13% moisture (t ha⁻¹)
17
16
15
14
13
2019
2020
2021
2022
Year

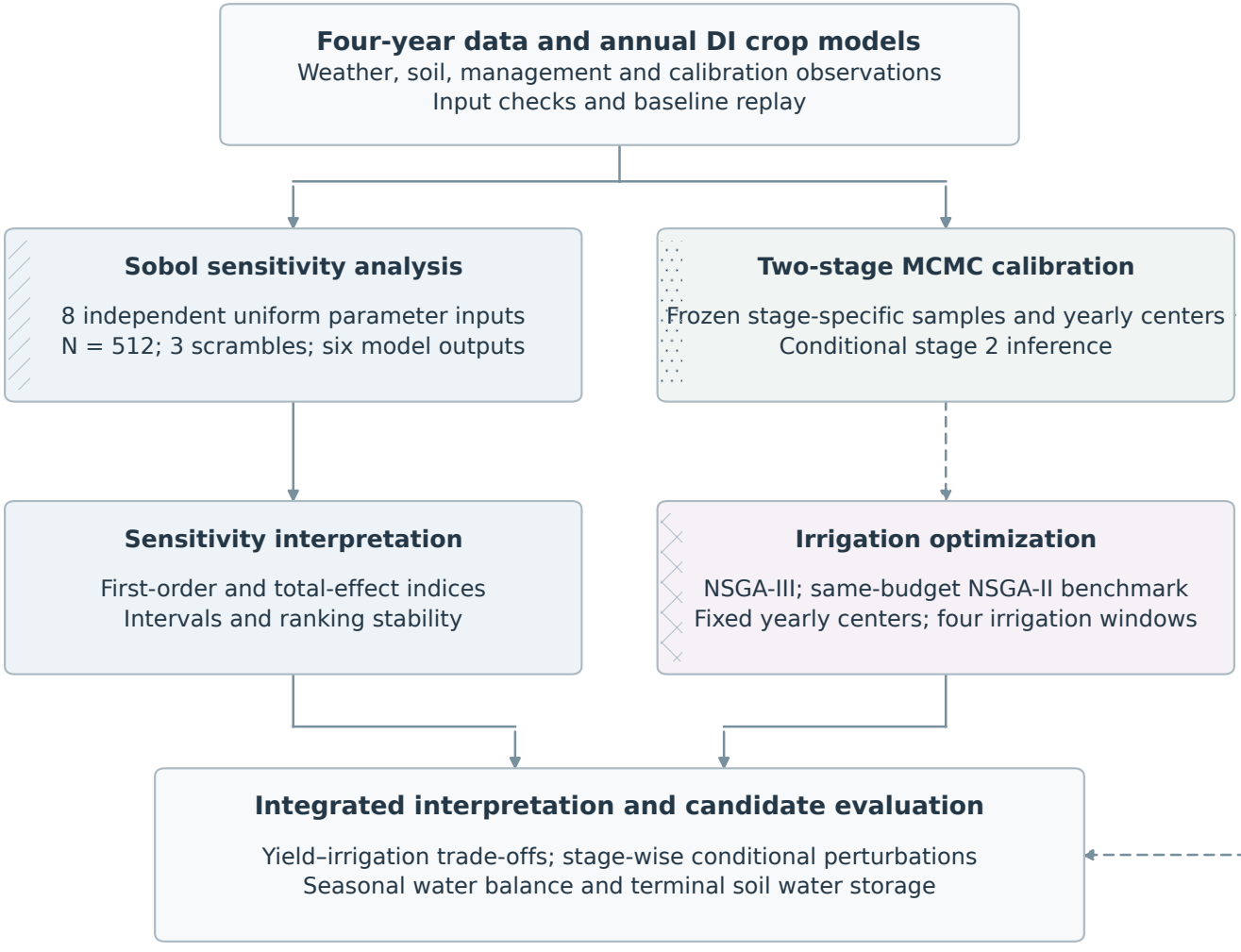

(a) Reduction in applied irrigation
Original DI schedule
Selected candidate
Applied seasonal irrigation (mm)
700
600
500
400
300
200
100
0
51.0%
44.2%
57.8%
35.6%
2019
2020
2021
2022
Year
(b) Yield comparison
Published observed yield
DI model baseline
Selected optimized candidate
95% observed-yield constraint
Grain yield at 13% moisture (t ha⁻¹)
17
16
15
14
13
2019
2020
2021
2022
Year


NSGA-III
NSGA-II
DI model baseline
Selected candidate
95% observed-yield constraint
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Grain yield at 13% moisture (t ha⁻¹)
Actual seasonal irrigation (mm)

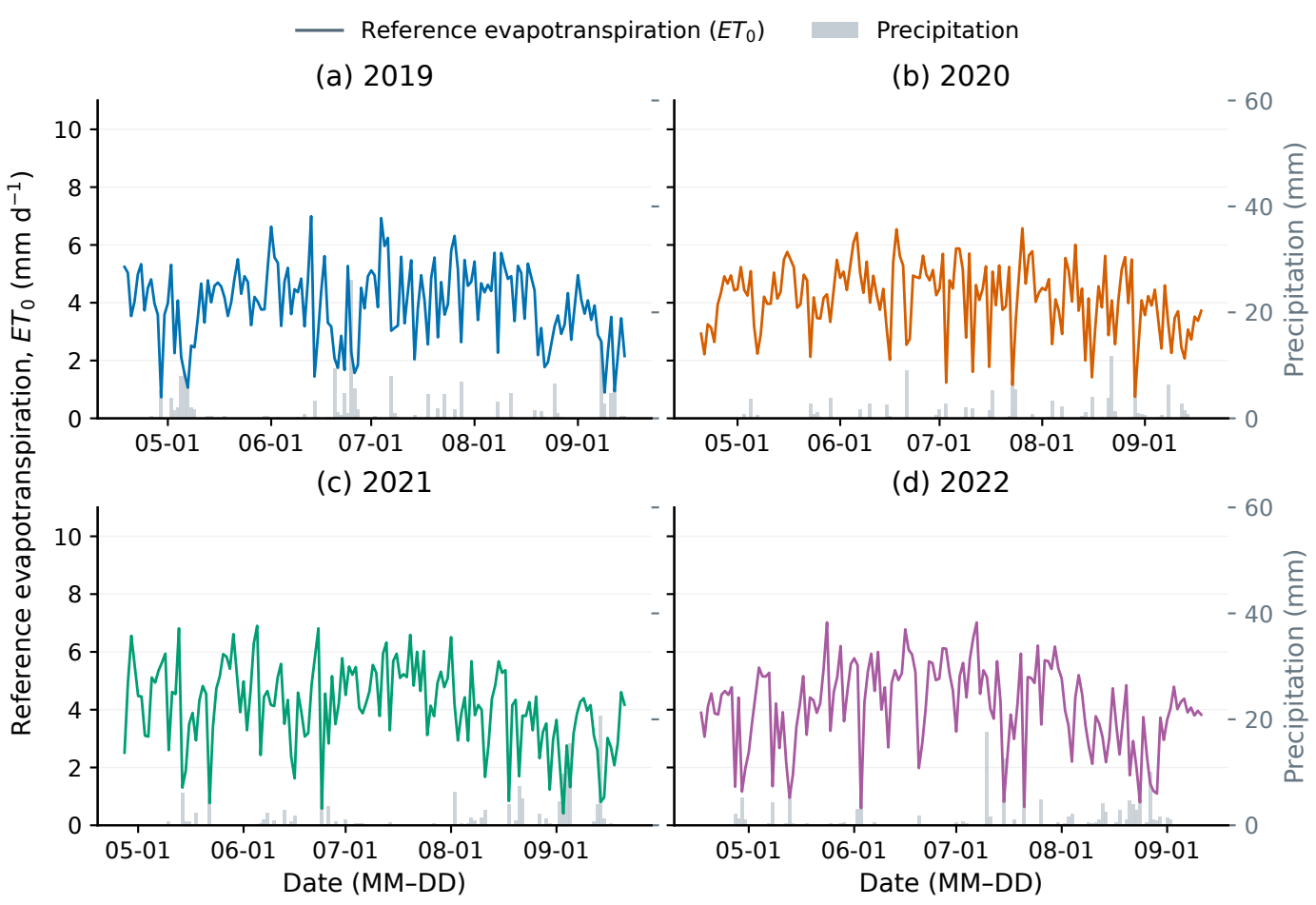

Lower seasonal ET
Lower end-of-season soil storage
Lower runoff and drainage
Irrigation reduction: ET, storage and water losses
Contribution to irrigation reduction (mm)
0
50
100
150
200
250
300
350
400
226.9 mm
190.5 mm
247.3 mm
230.7 mm
366.2 mm
356.7 mm
144.1 mm
119.8 mm
2019
2020
2021
2022
Year

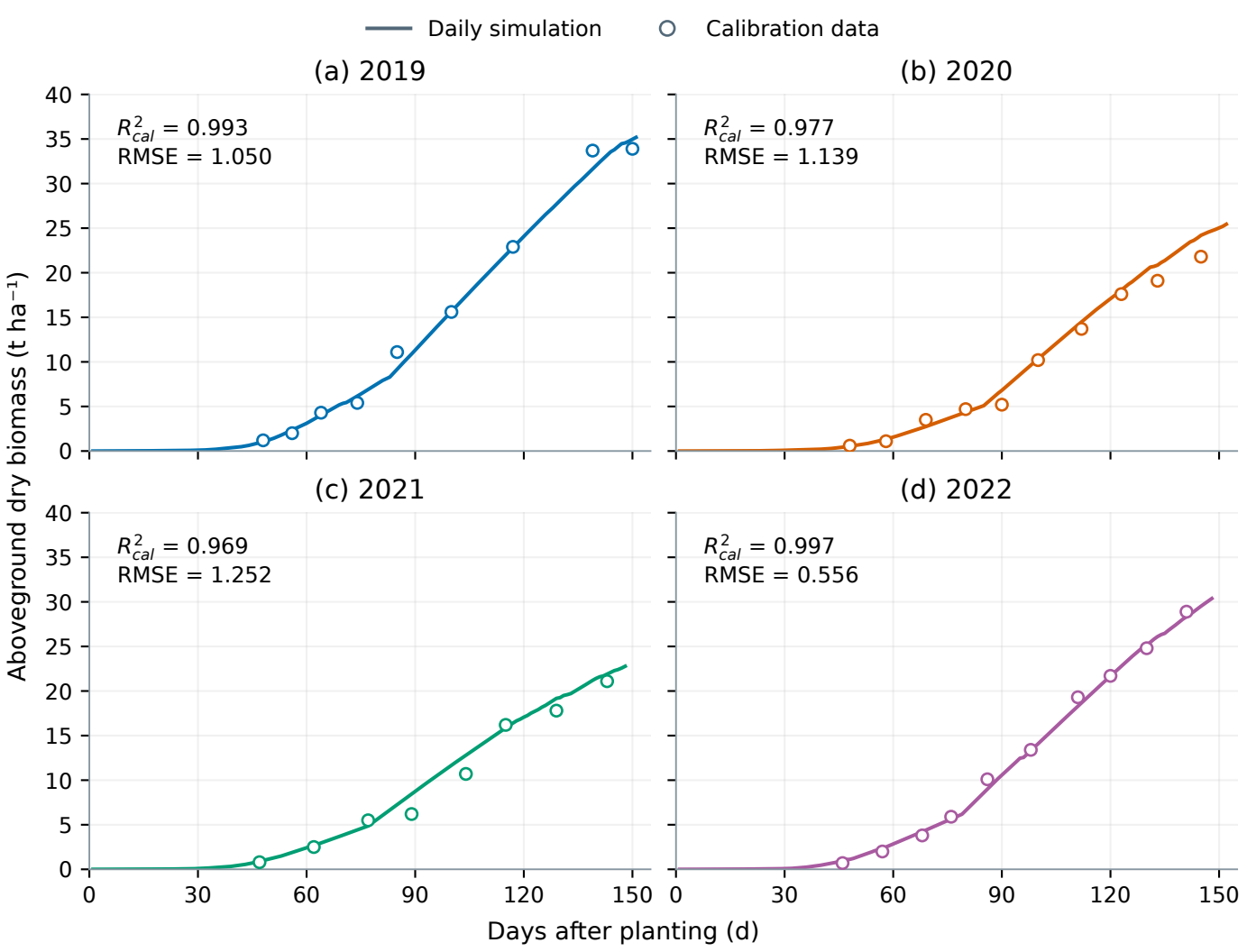


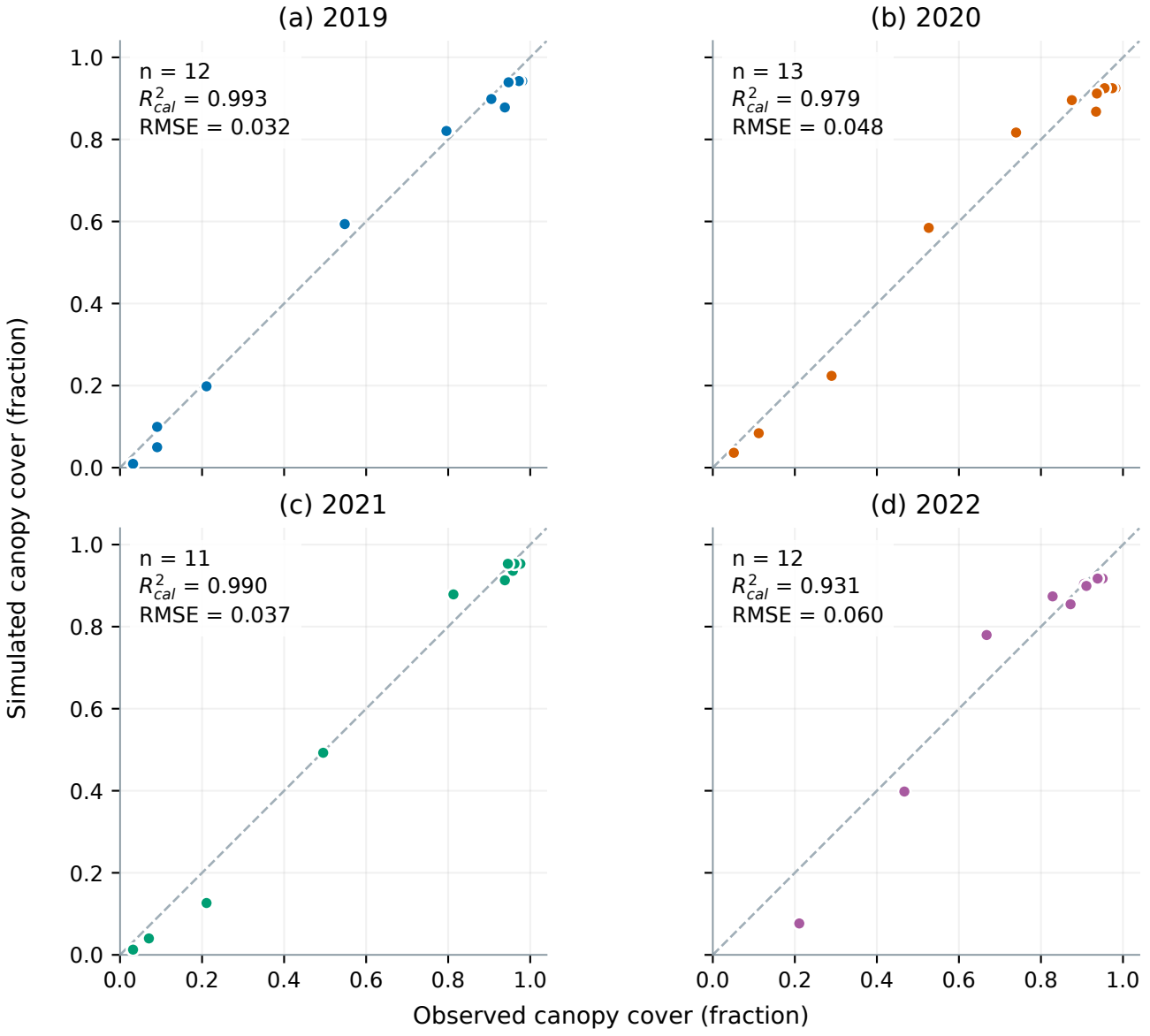
Sensitivity and optimization use the same annual model, with separately defined inputs.

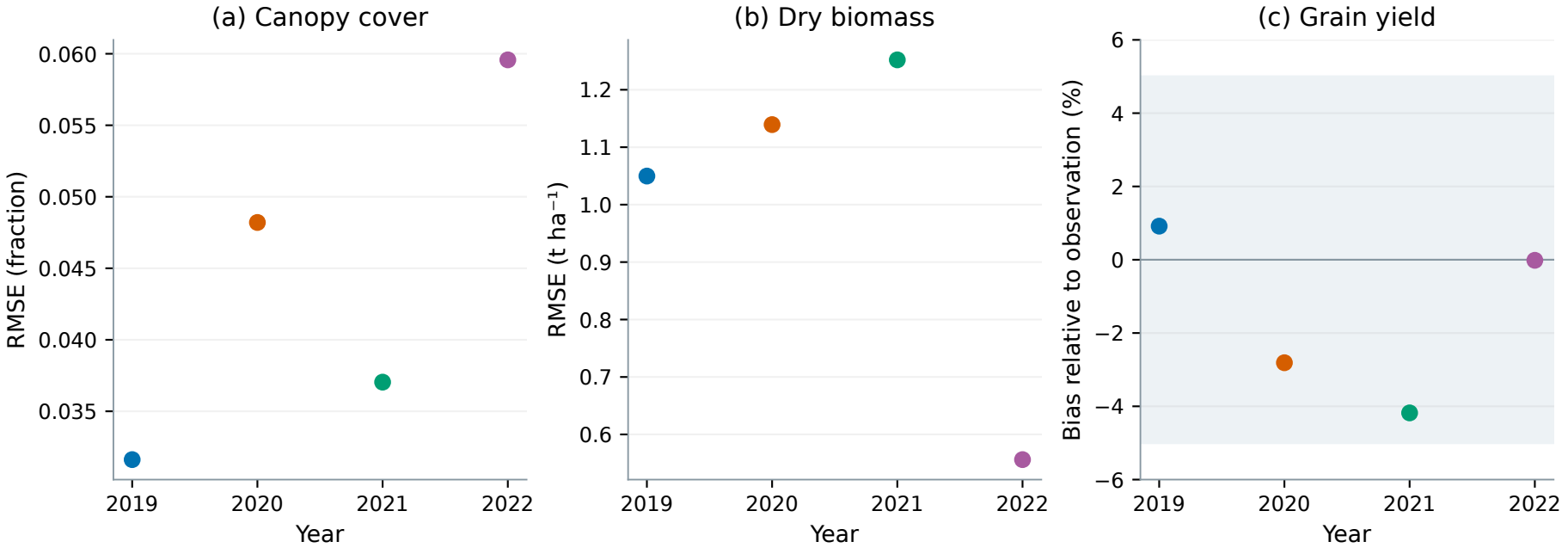


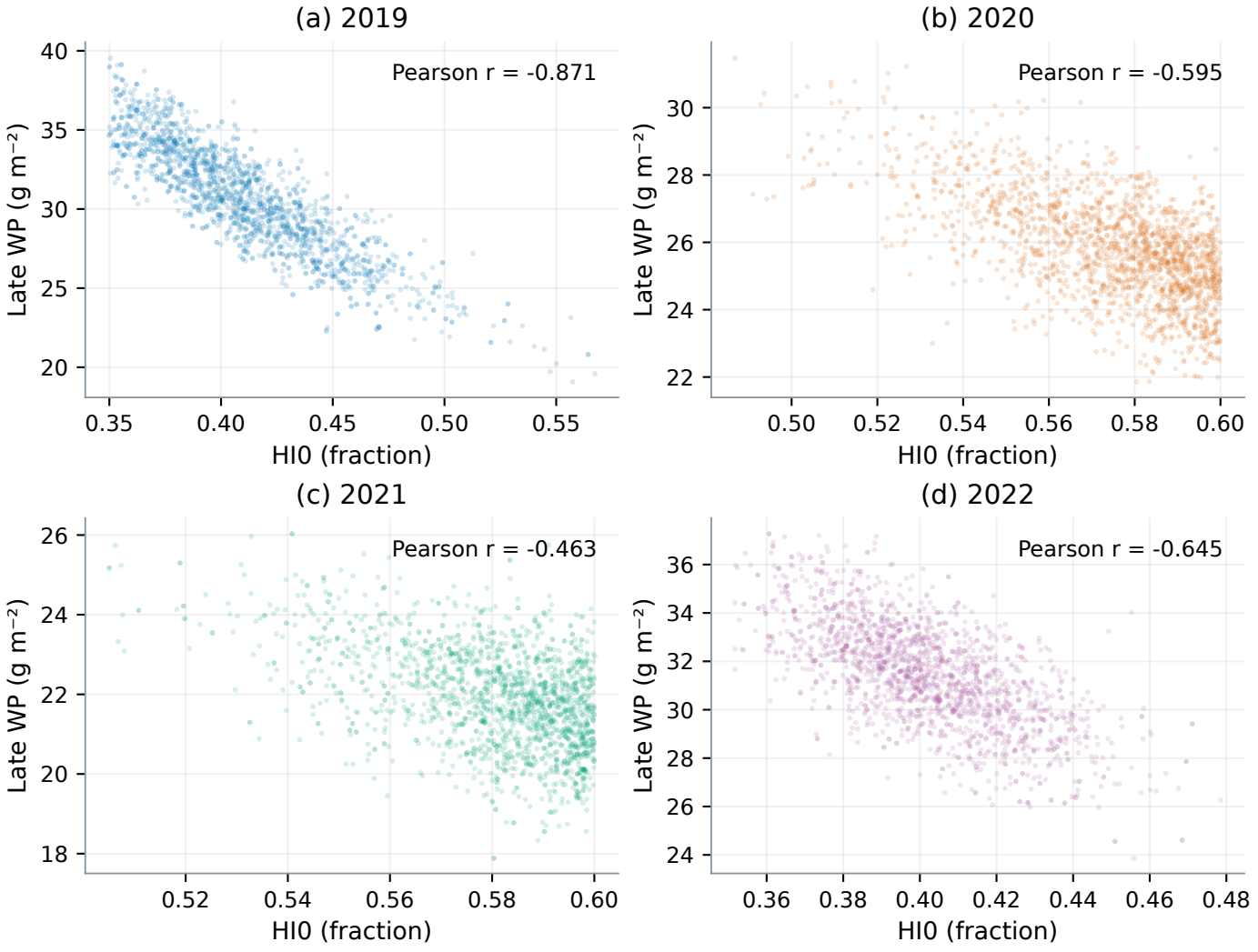
Solid arrows: model/data flow. Dashed arrows: frozen centers or stage-specific samples.

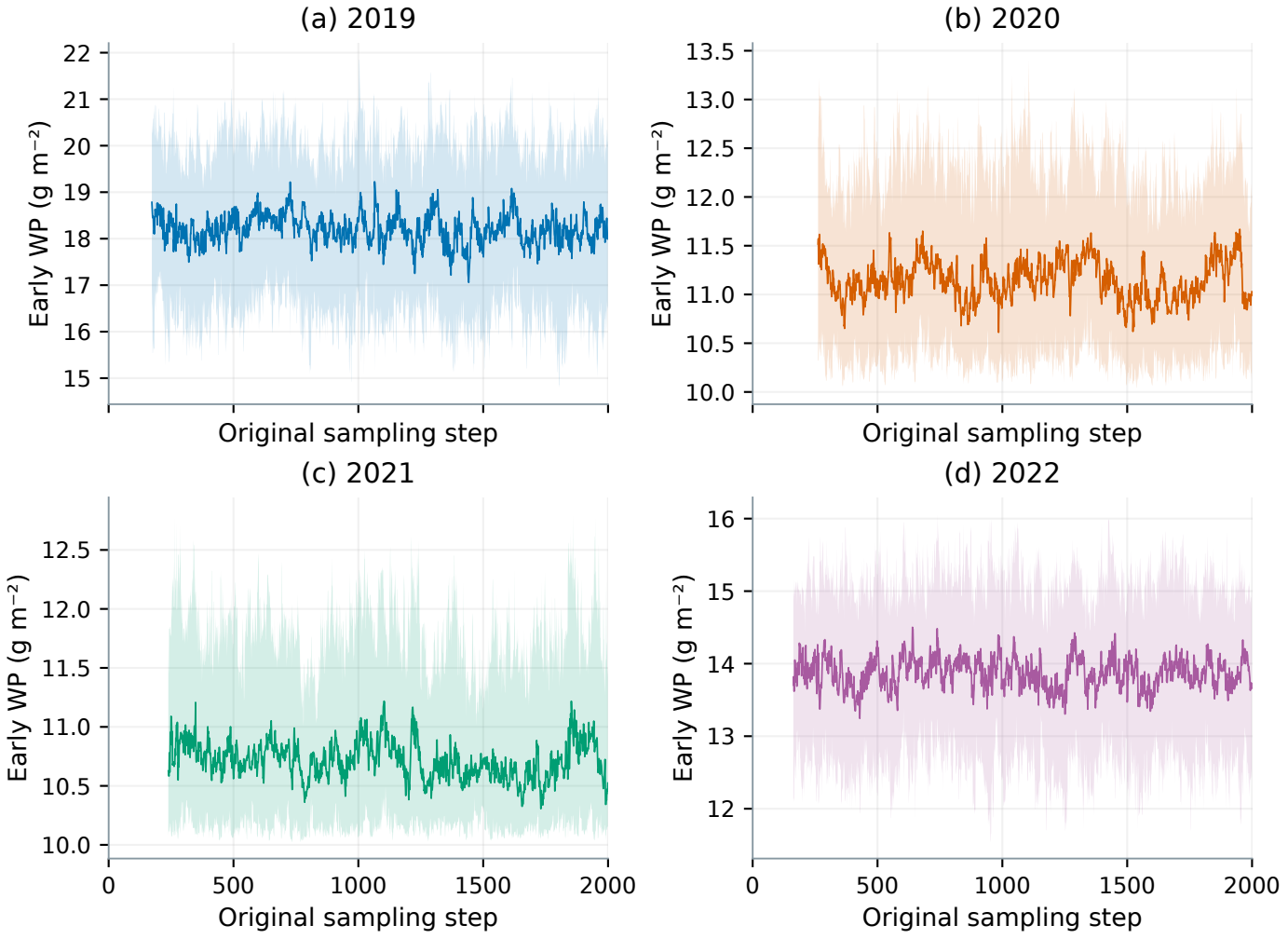

Reference evapotranspiration (ET_0)
Precipitation
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Reference evapotranspiration, ET_0 (mm d^{-1})
Precipitation (mm)
Date (MM–DD)
05-01
06-01
07-01
08-01
09-01
0
2
4
6
8
10
0
20
40
60

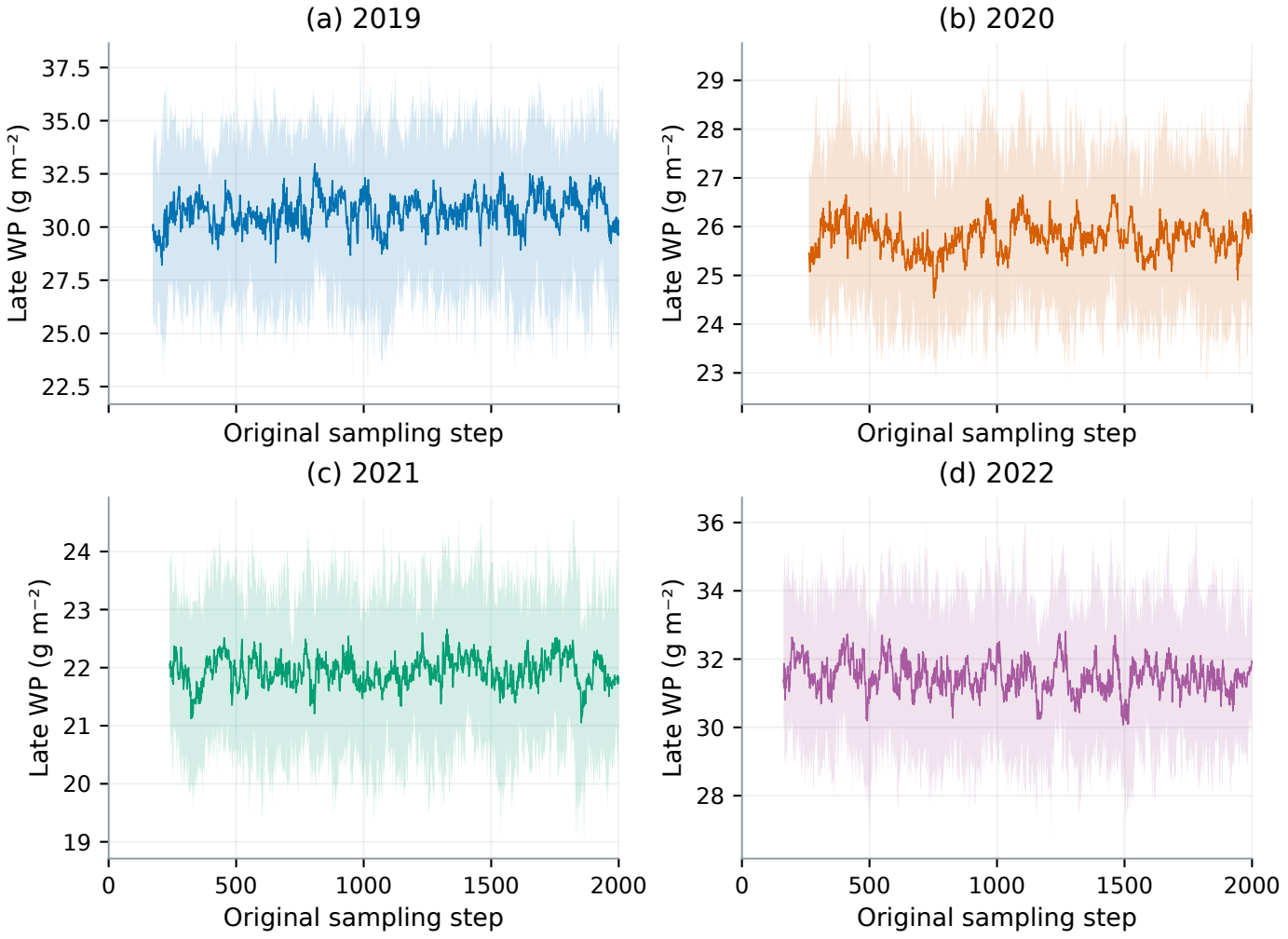

Daily simulation
Calibration data
(a) 2019
R^2_{cal} = 0.993
RMSE = 1.050
(b) 2020
R^2_{cal} = 0.977
RMSE = 1.139
(c) 2021
R^2_{cal} = 0.969
RMSE = 1.252
(d) 2022
R^2_{cal} = 0.997
RMSE = 0.556
Aboveground dry biomass (t ha^{-1})
Days after planting (d)
0
5
10
15
20
25
30
35
40
0
30
60
90
120
150

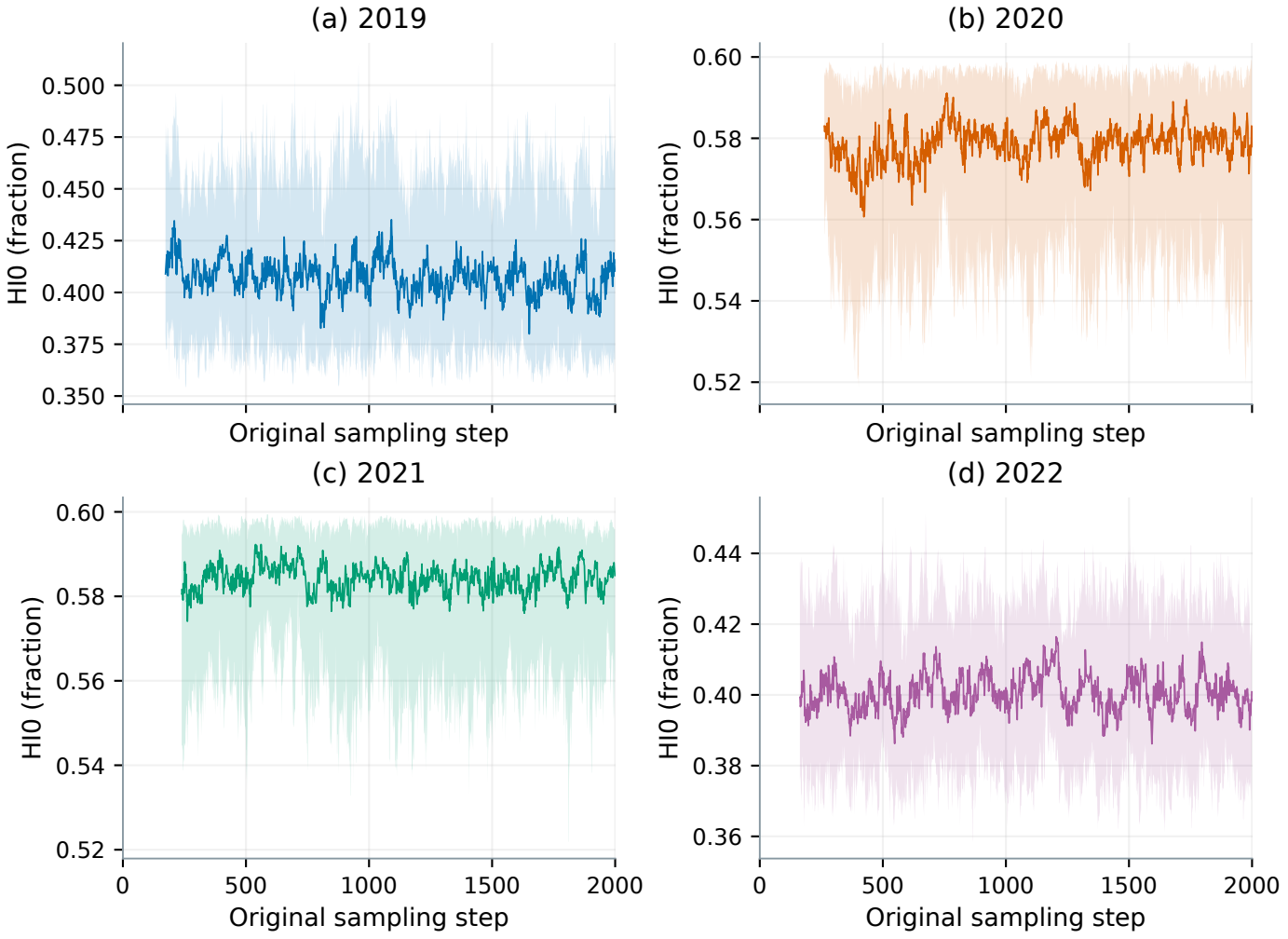

(a) 2019
n = 12
R^2_{cal} = 0.993
RMSE = 0.032
(b) 2020
n = 13
R^2_{cal} = 0.979
RMSE = 0.048
(c) 2021
n = 11
R^2_{cal} = 0.990
RMSE = 0.037
(d) 2022
n = 12
R^2_{cal} = 0.931
RMSE = 0.060
Simulated canopy cover (fraction)
Observed canopy cover (fraction)
0.0
0.2
0.4
0.6
0.8
1.0

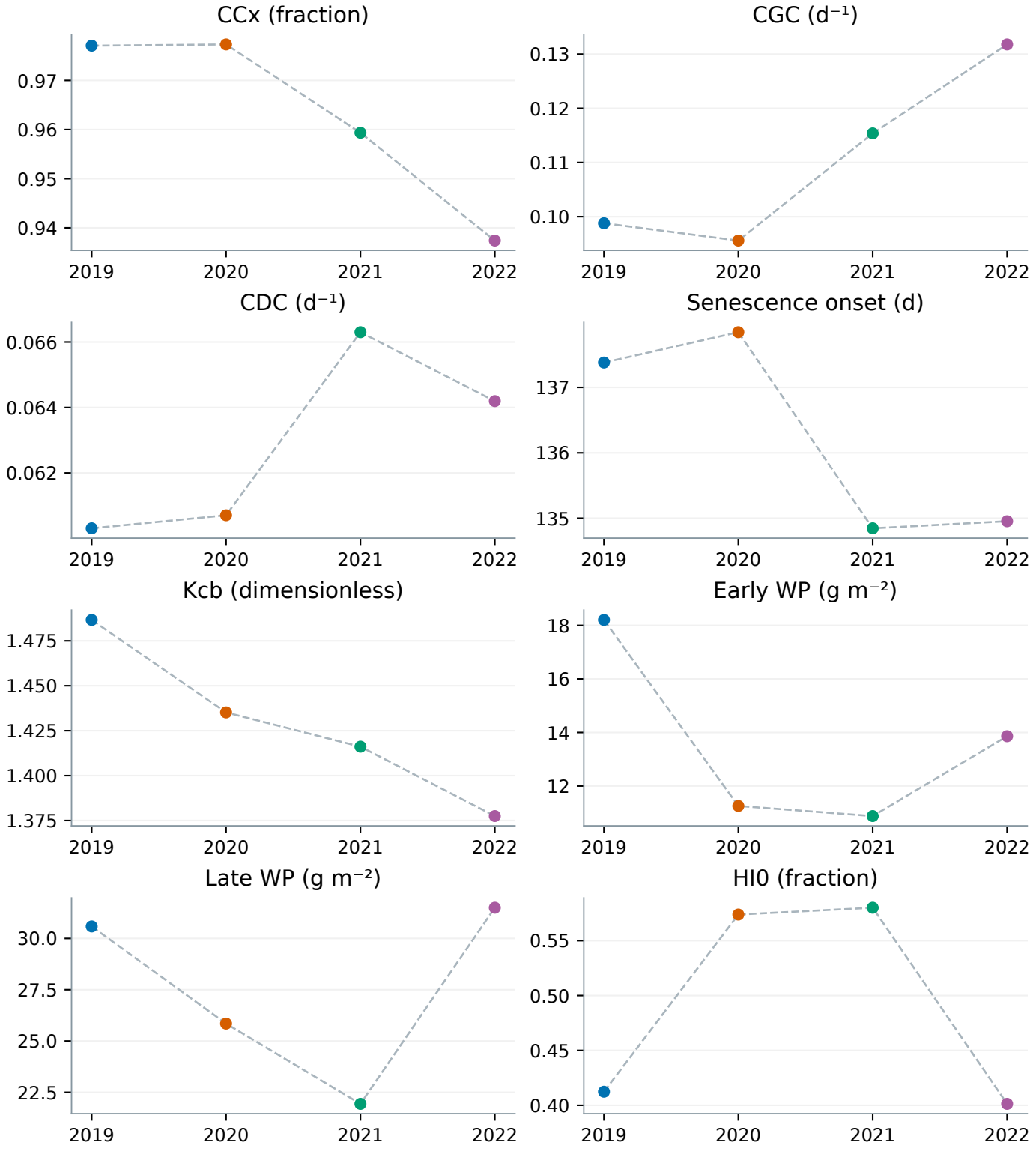

(a) Canopy cover
RMSE (fraction)
0.060
0.055
0.050
0.045
0.040
0.035
2019
2020
2021
2022
Year
(b) Dry biomass
RMSE (t ha⁻¹)
1.2
1.1
1.0
0.9
0.8
0.7
0.6
2019
2020
2021
2022
Year
(c) Grain yield
Bias relative to observation (%)
6
4
2
0
−2
−4
−6
2019
2020
2021
2022
Year

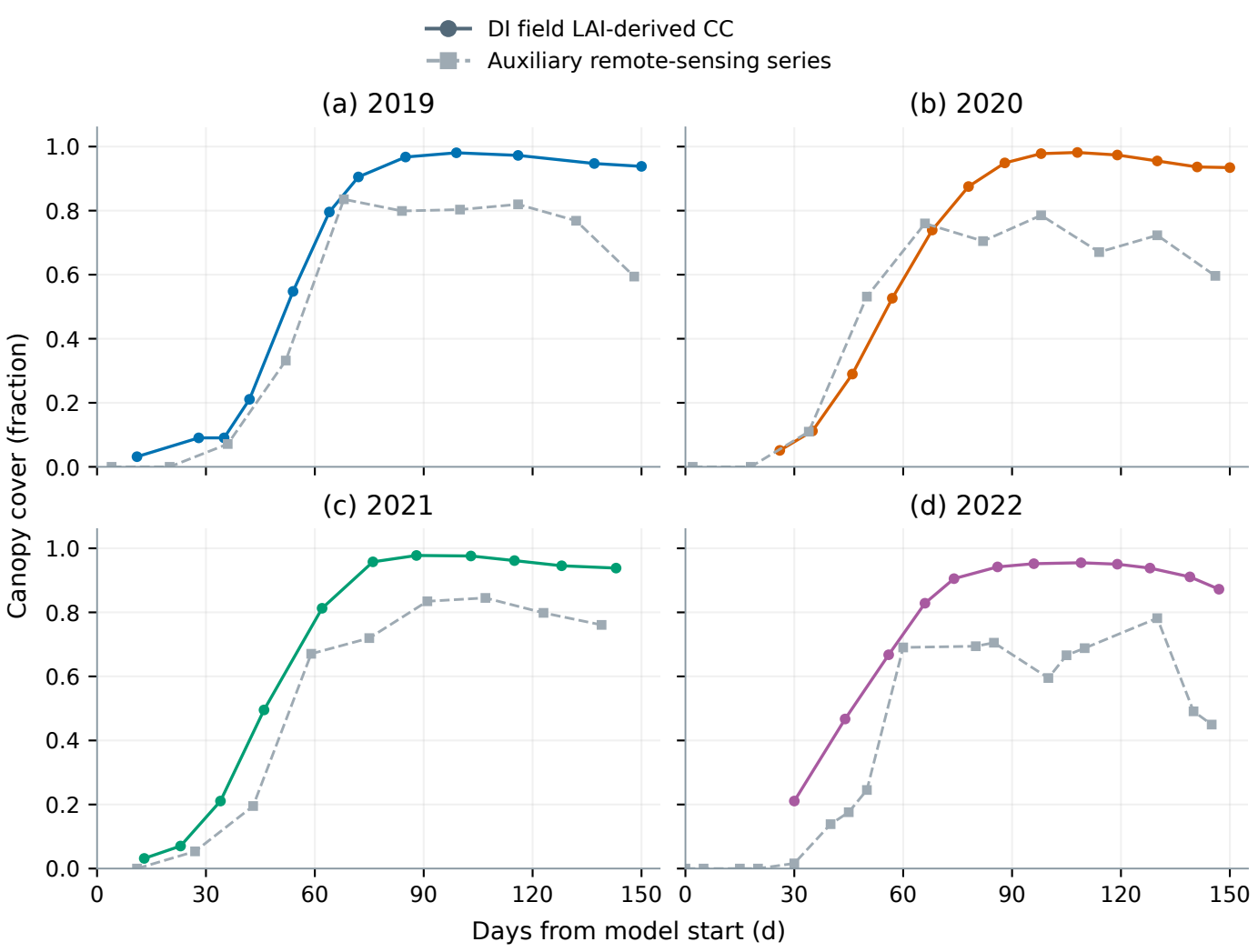

(a) 2019
Pearson r = -0.871
Late WP (g m⁻²)
HI0 (fraction)
(b) 2020
Pearson r = -0.595
Late WP (g m⁻²)
HI0 (fraction)
(c) 2021
Pearson r = -0.463
Late WP (g m⁻²)
HI0 (fraction)
(d) 2022
Pearson r = -0.645
Late WP (g m⁻²)
HI0 (fraction)

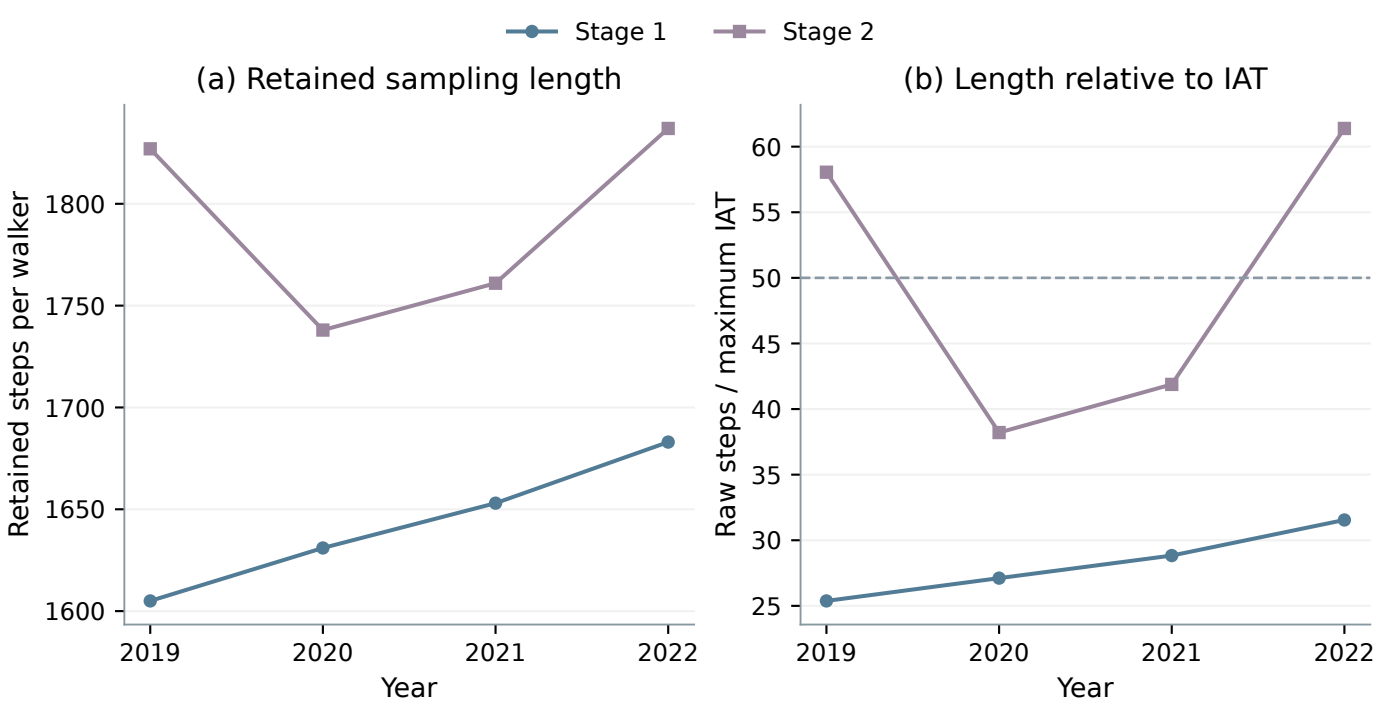

(a) 2019
(b) 2020
(c) 2021
(d) 2022
Early WP (g m⁻²)
Original sampling step
22
21
20
19
18
17
16
15
13.5
13.0
12.5
12.0
11.5
11.0
10.5
10.0
16
15
14
13
12
0
500
1000
1500
2000

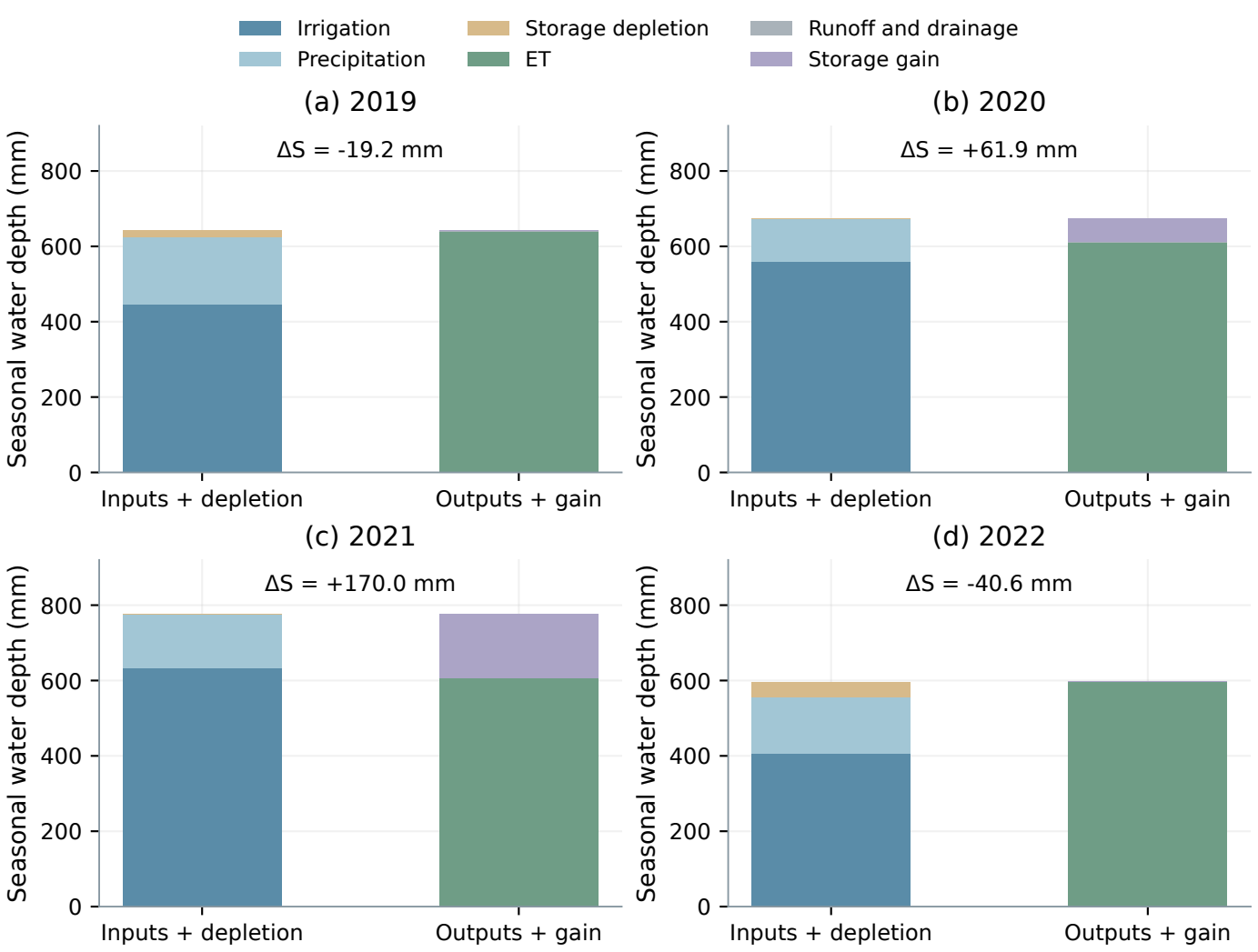

(a) 2019
(b) 2020
(c) 2021
(d) 2022
Late WP (g m⁻²)
Original sampling step
37.5
35.0
32.5
30.0
27.5
25.0
22.5
29
28
27
26
25
24
23
24
23
22
21
20
19
36
34
32
30
28
0
500
1000
1500
2000

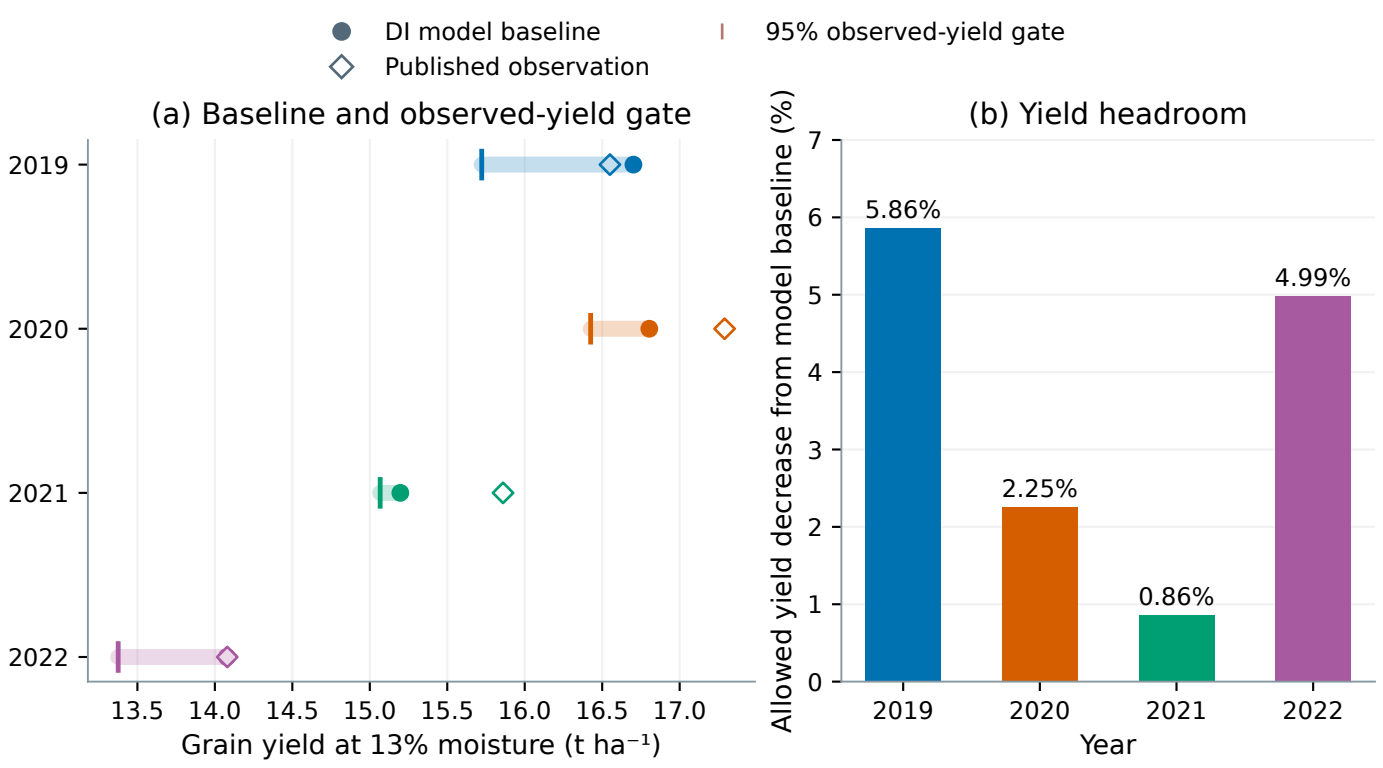

(a) 2019
HI0 (fraction)
0.500
0.475
0.450
0.425
0.400
0.375
0.350
Original sampling step
(b) 2020
HI0 (fraction)
0.60
0.58
0.56
0.54
0.52
Original sampling step
(c) 2021
HI0 (fraction)
0.60
0.58
0.56
0.54
0.52
0
500
1000
1500
2000
Original sampling step
(d) 2022
HI0 (fraction)
0.44
0.42
0.40
0.38
0.36
0
500
1000
1500
2000
Original sampling step

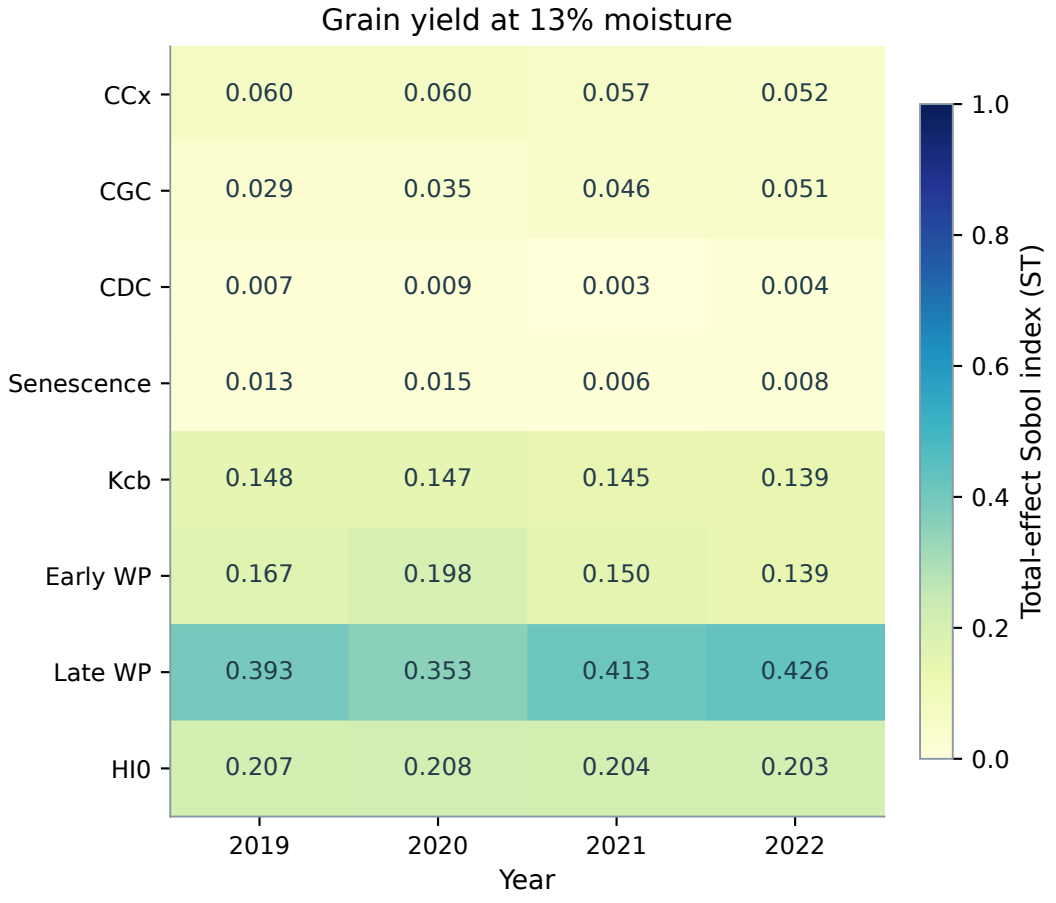

CCx (fraction)
0.97
0.96
0.95
0.94
2019
2020
2021
2022
CGC (d⁻¹)
0.13
0.12
0.11
0.10
2019
2020
2021
2022
CDC (d⁻¹)
0.066
0.064
0.062
2019
2020
2021
2022
Senescence onset (d)
137
136
135
2019
2020
2021
2022
Kcb (dimensionless)
1.475
1.450
1.425
1.400
1.375
2019
2020
2021
2022
Early WP (g m⁻²)
18
16
14
12
2019
2020
2021
2022
Late WP (g m⁻²)
30.0
27.5
25.0
22.5
2019
2020
2021
2022
HI0 (fraction)
0.55
0.50
0.45
0.40
2019
2020
2021
2022

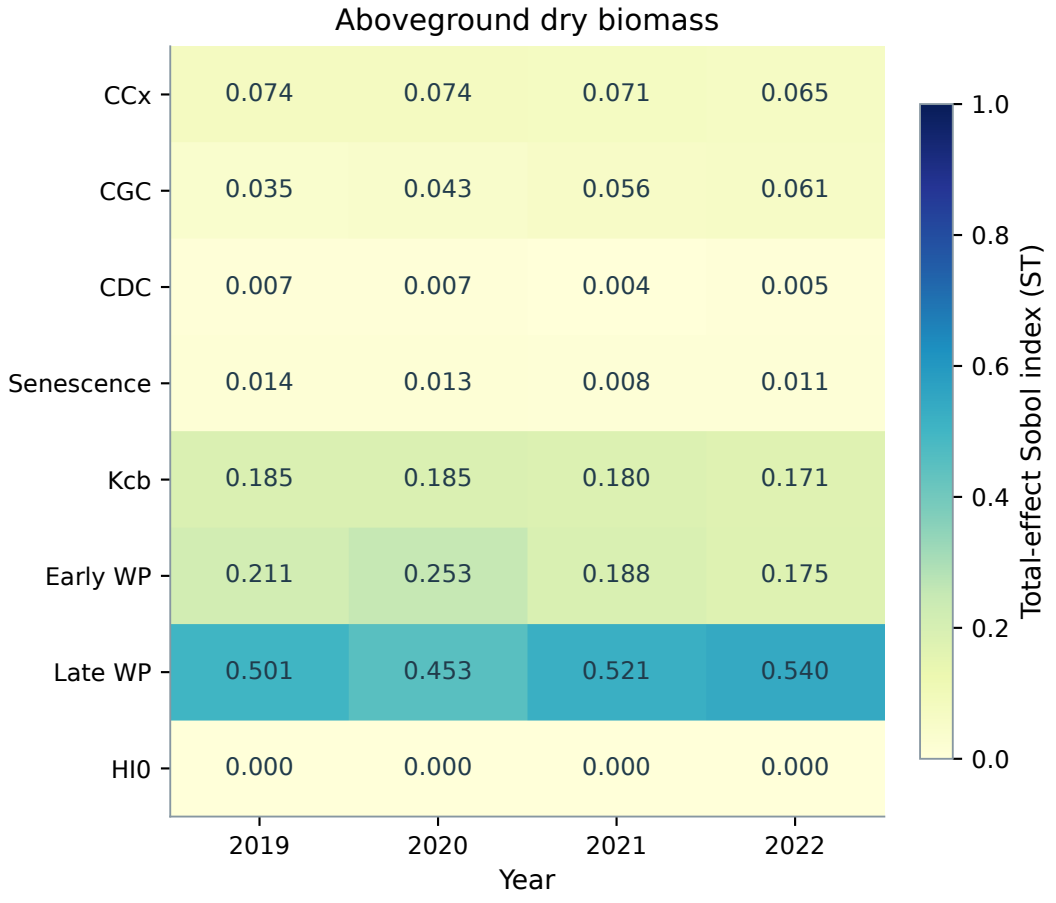

DI field LAI-derived CC
Auxiliary remote-sensing series
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Canopy cover (fraction)
0.0
0.2
0.4
0.6
0.8
1.0
0
30
60
90
120
150
Days from model start (d)

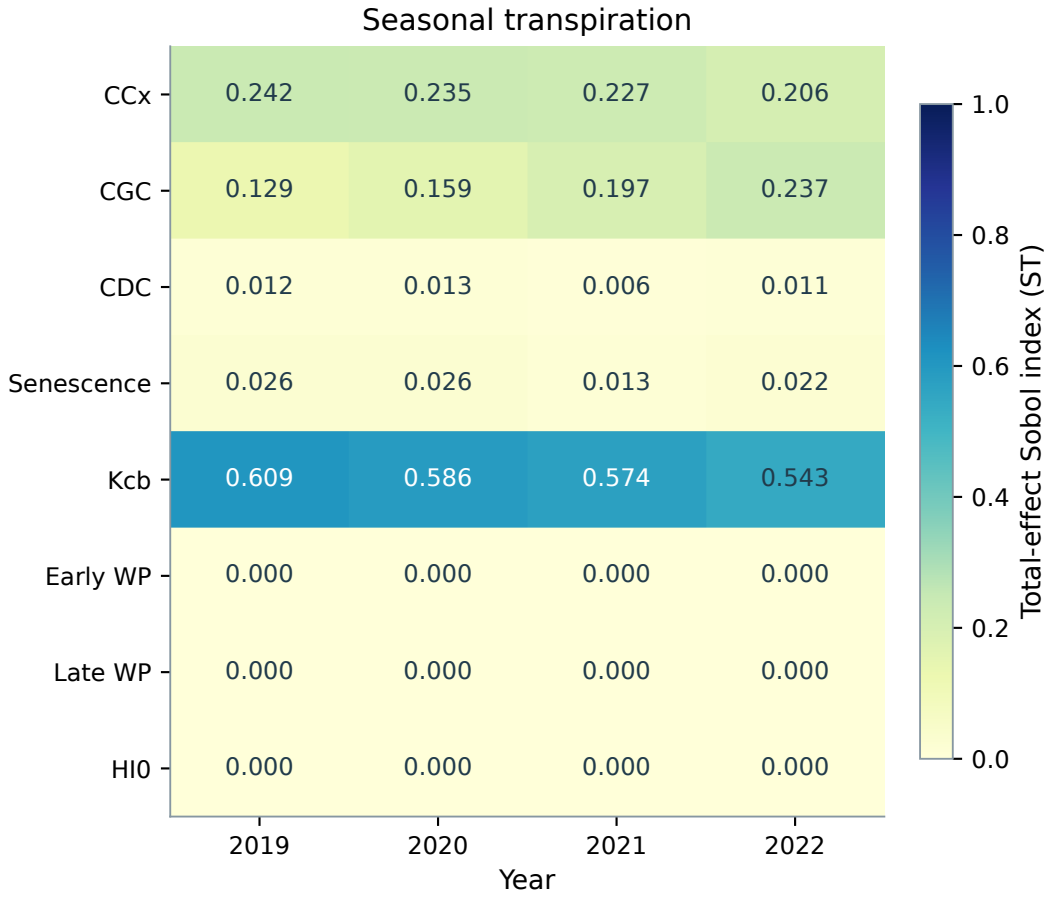

Stage 1
Stage 2
(a) Retained sampling length
Retained steps per walker
1800
1750
1700
1650
1600
2019
2020
2021
2022
Year
(b) Length relative to IAT
Raw steps / maximum IAT
60
55
50
45
40
35
30
25
2019
2020
2021
2022
Year

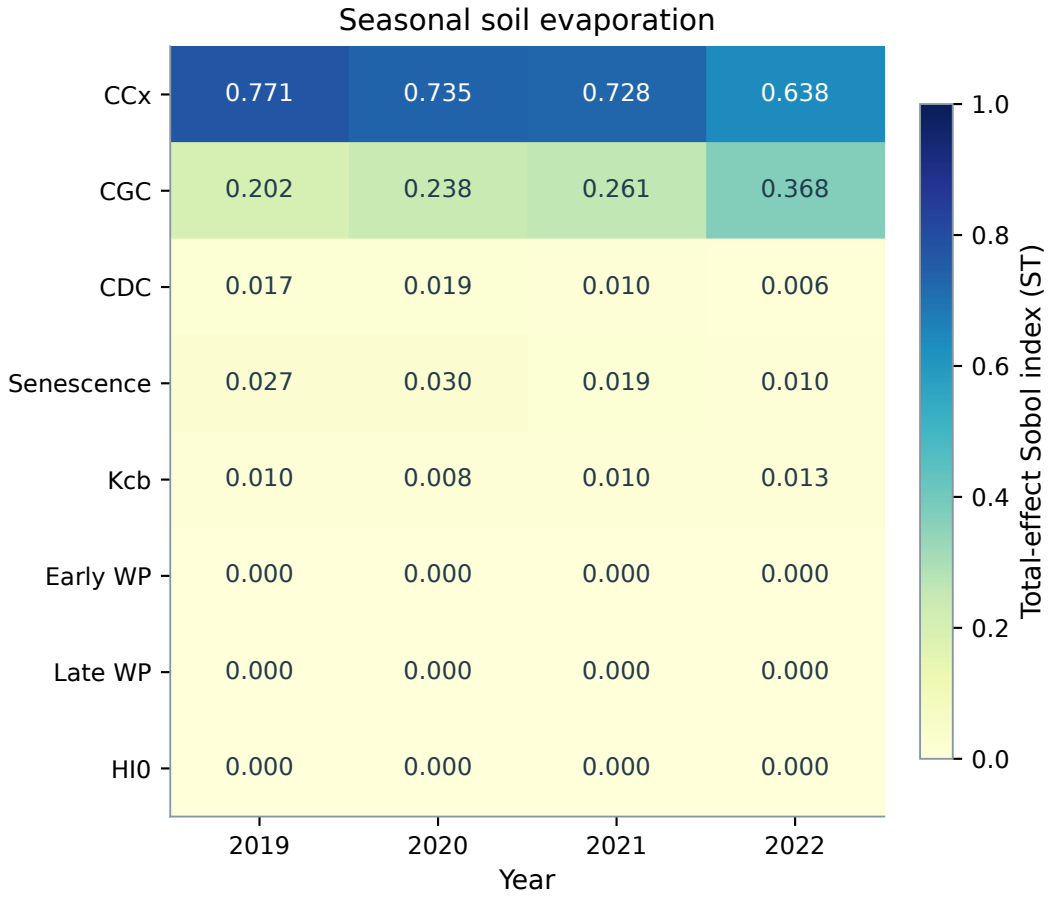

Irrigation
Precipitation
Storage depletion
ET
Runoff and drainage
Storage gain
(a) 2019
ΔS = -19.2 mm
(b) 2020
ΔS = +61.9 mm
(c) 2021
ΔS = +170.0 mm
(d) 2022
ΔS = -40.6 mm
Seasonal water depth (mm)
0
200
400
600
800
Inputs + depletion
Outputs + gain

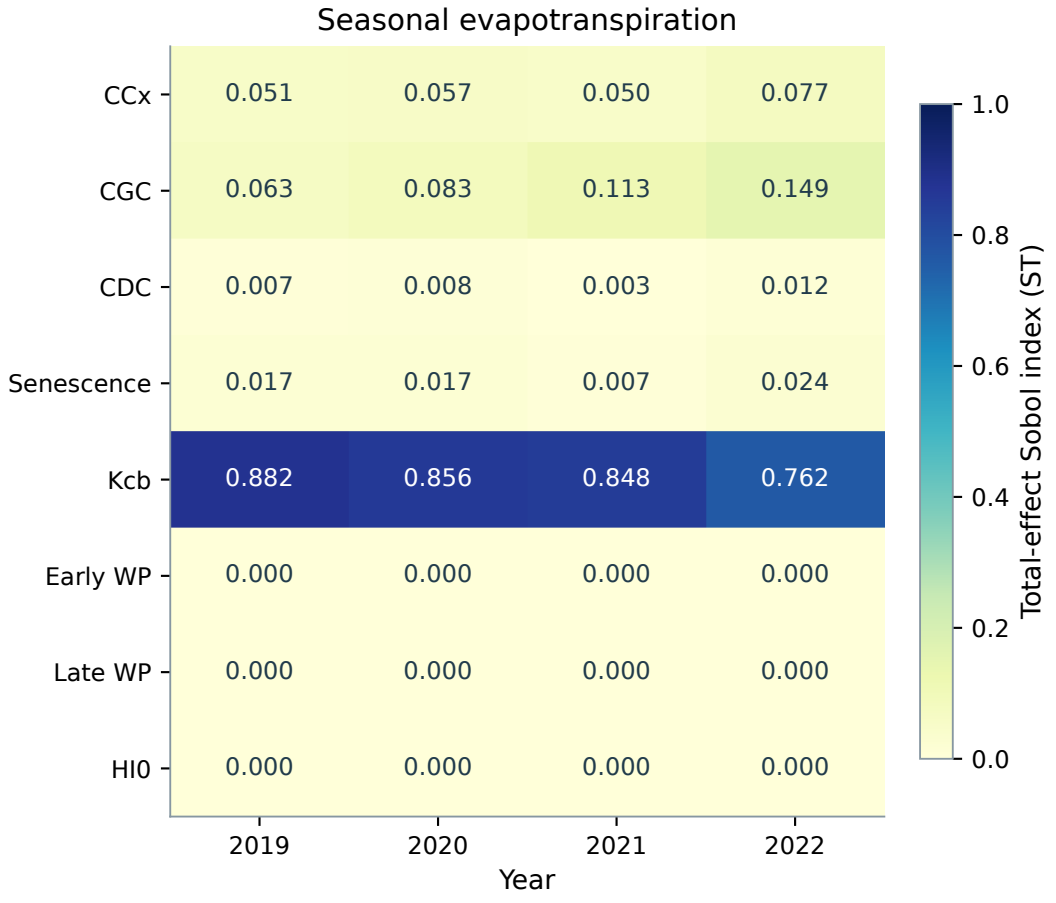

DI model baseline
Published observation
95% observed-yield gate
(a) Baseline and observed-yield gate
2019
2020
2021
2022
13.5
14.0
14.5
15.0
15.5
16.0
16.5
17.0
Grain yield at 13% moisture (t ha⁻¹)
(b) Yield headroom
Allowed yield decrease from model baseline (%)
0
1
2
3
4
5
6
7
5.86%
2.25%
0.86%
4.99%
2019
2020
2021
2022
Year

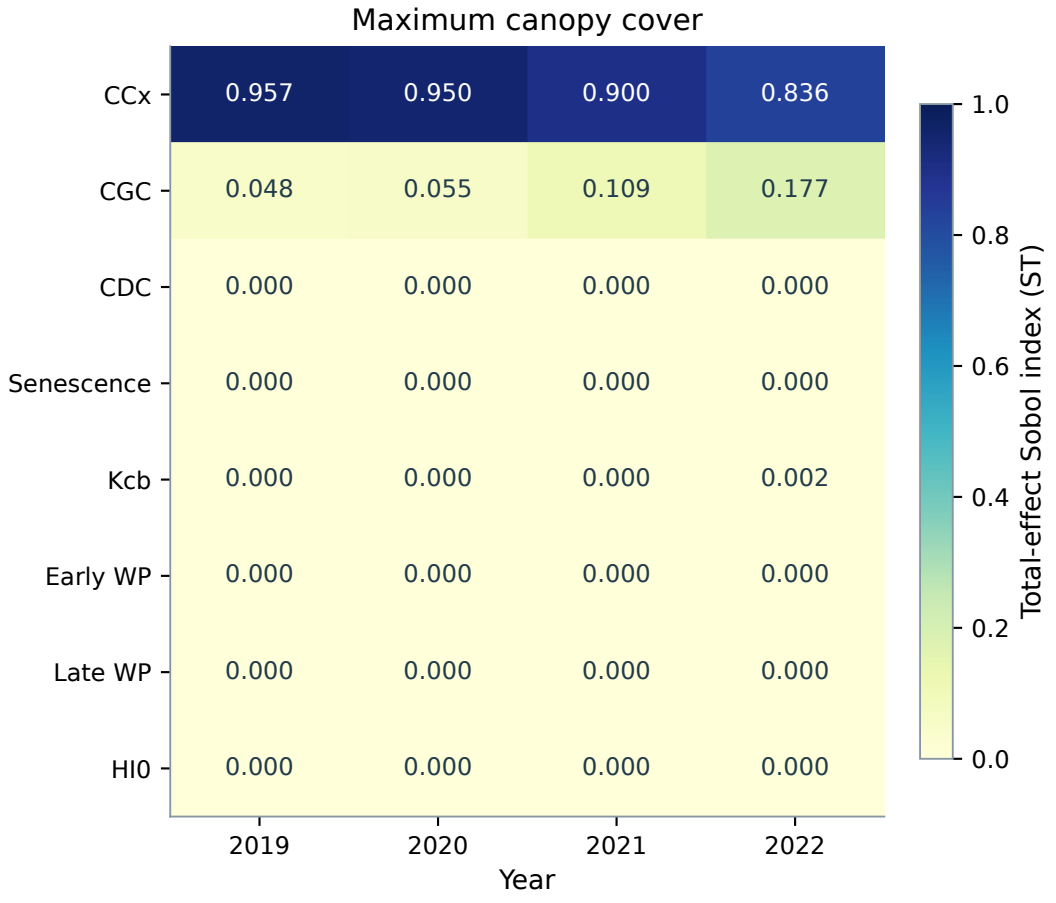

Grain yield at 13% moisture
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.060
0.060
0.057
0.052
0.029
0.035
0.046
0.051
0.007
0.009
0.003
0.004
0.013
0.015
0.006
0.008
0.148
0.147
0.145
0.139
0.167
0.198
0.150
0.139
0.393
0.353
0.413
0.426
0.207
0.208
0.204
0.203
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

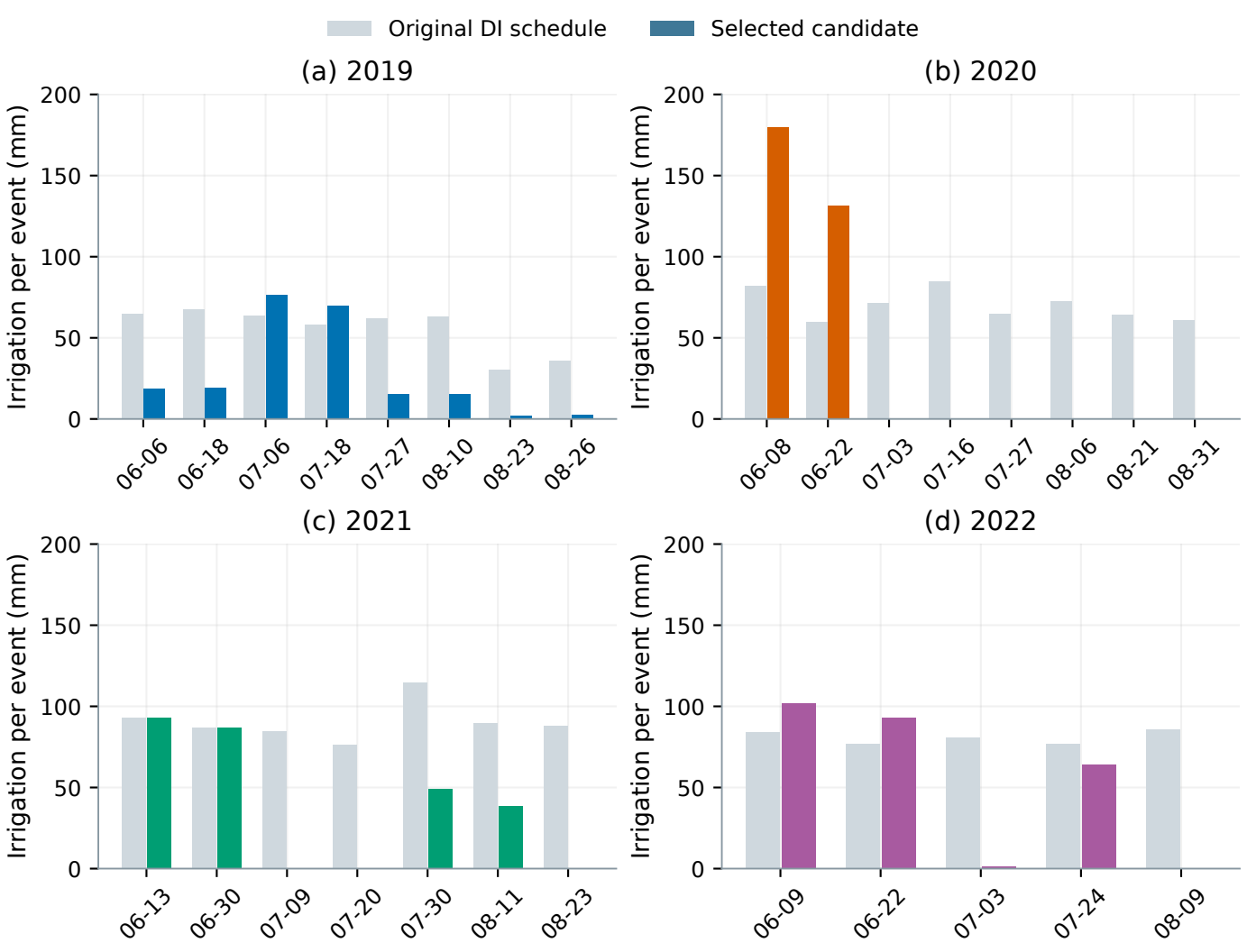

Aboveground dry biomass
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.074 0.074 0.071 0.065
0.035 0.043 0.056 0.061
0.007 0.007 0.004 0.005
0.014 0.013 0.008 0.011
0.185 0.185 0.180 0.171
0.211 0.253 0.188 0.175
0.501 0.453 0.521 0.540
0.000 0.000 0.000 0.000
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

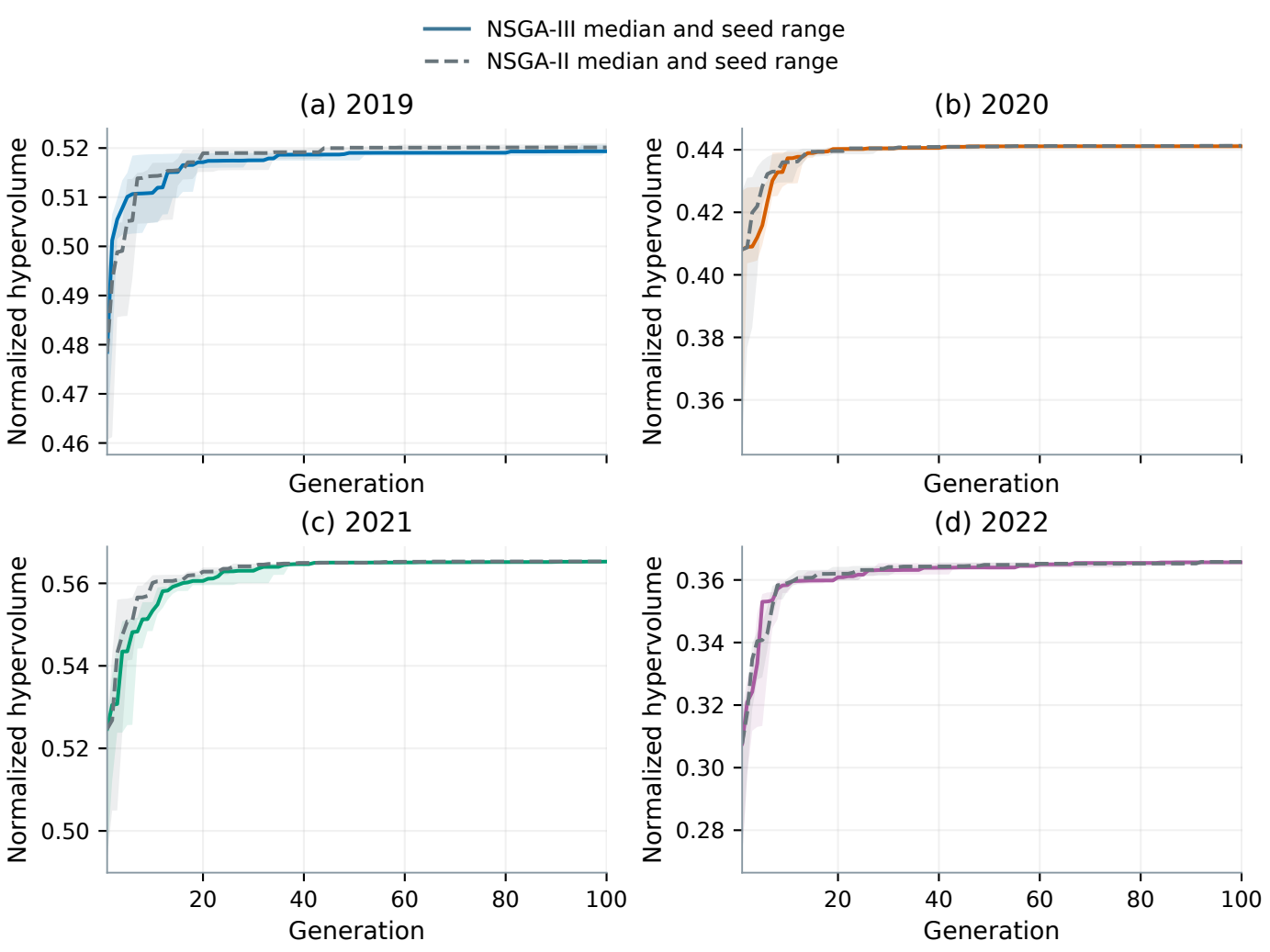

Seasonal transpiration
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.242
0.235
0.227
0.206
0.129
0.159
0.197
0.237
0.012
0.013
0.006
0.011
0.026
0.026
0.013
0.022
0.609
0.586
0.574
0.543
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

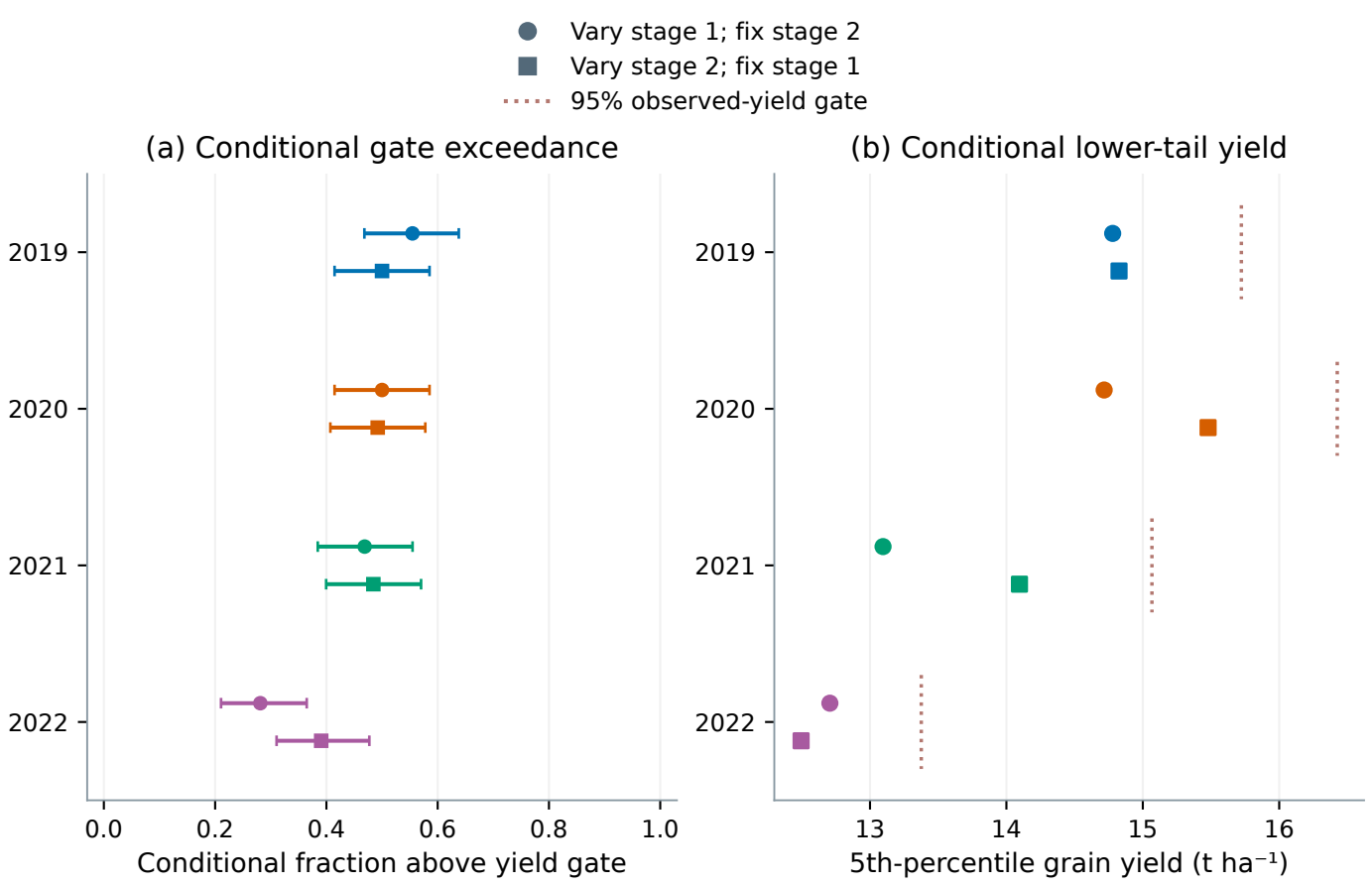

Seasonal soil evaporation
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.771
0.735
0.728
0.638
0.202
0.238
0.261
0.368
0.017
0.019
0.010
0.006
0.027
0.030
0.019
0.010
0.010
0.008
0.010
0.013
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

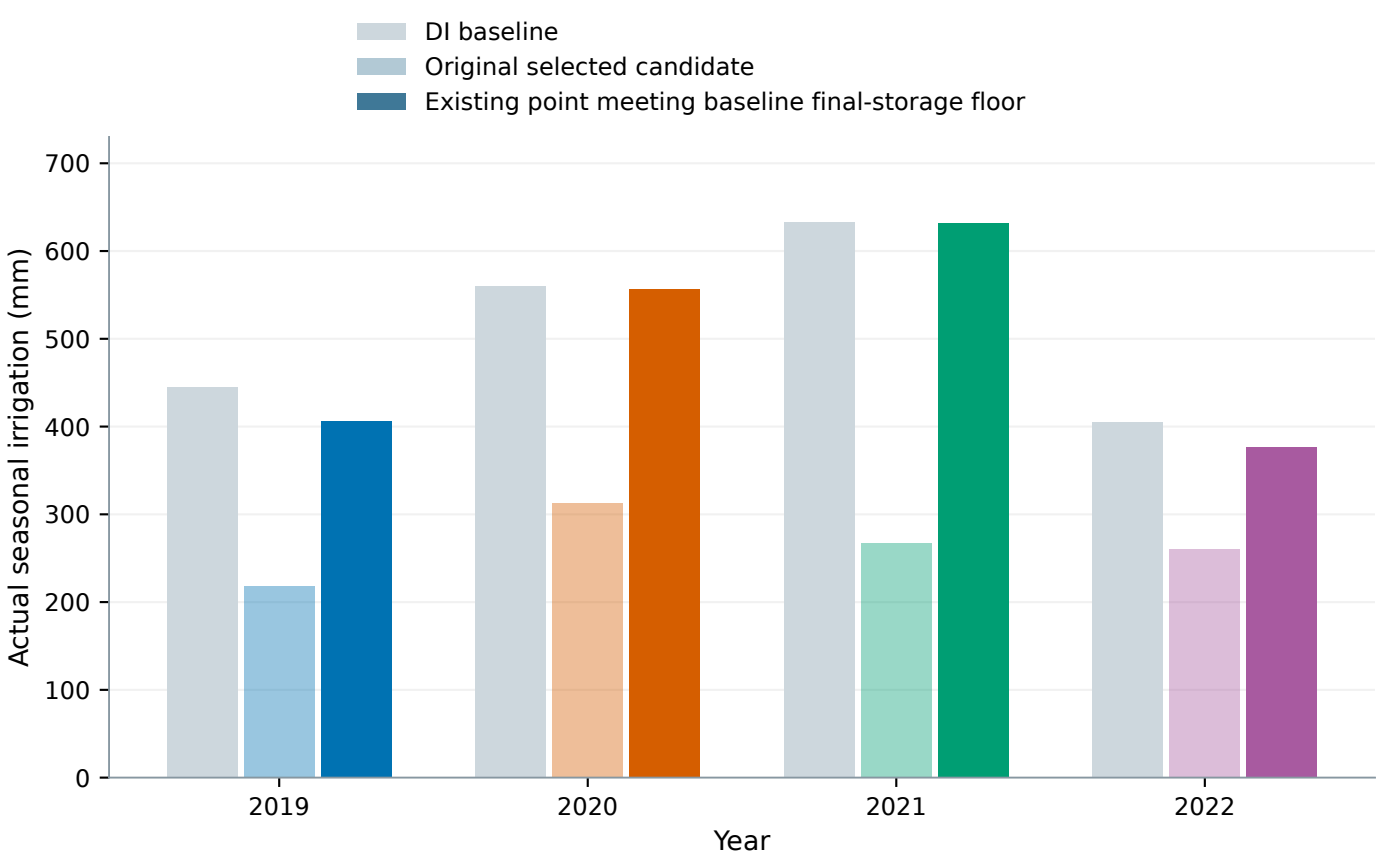

Seasonal evapotranspiration
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.051
0.057
0.050
0.077
0.063
0.083
0.113
0.149
0.007
0.008
0.003
0.012
0.017
0.017
0.007
0.024
0.882
0.856
0.848
0.762
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
0.000
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

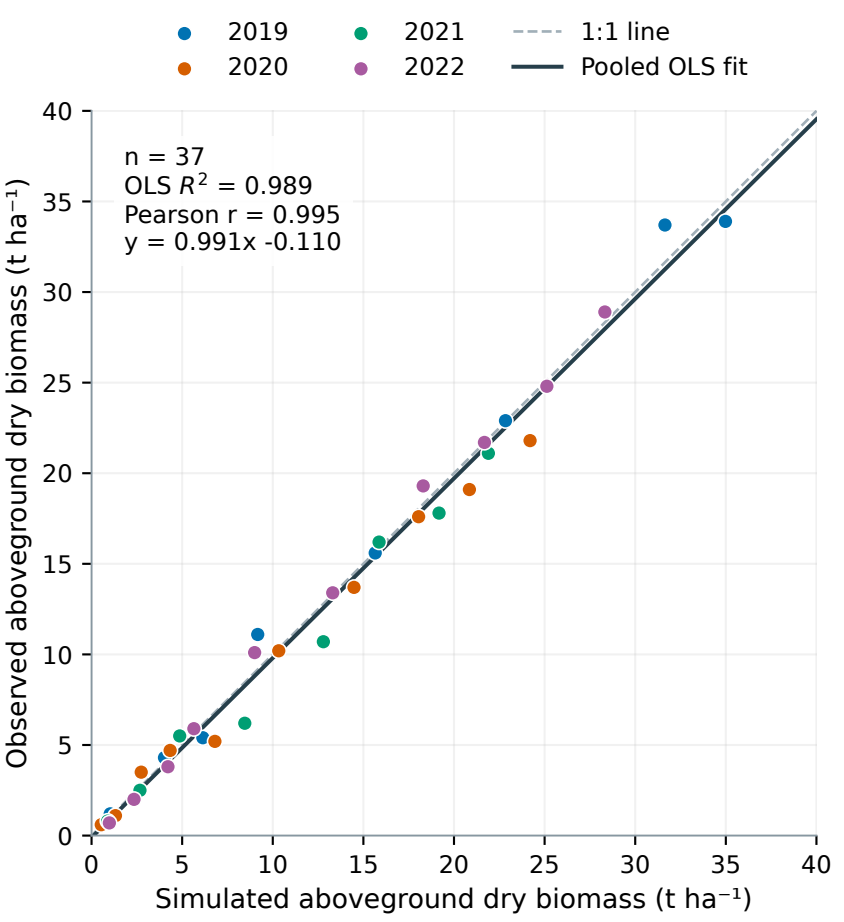

Maximum canopy cover
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.957
0.950
0.900
0.836
0.048
0.055
0.109
0.177
0.000
0.002
2019
2020
2021
2022
Year
Total-effect Sobol index (ST)
1.0
0.8
0.6
0.4
0.2
0.0

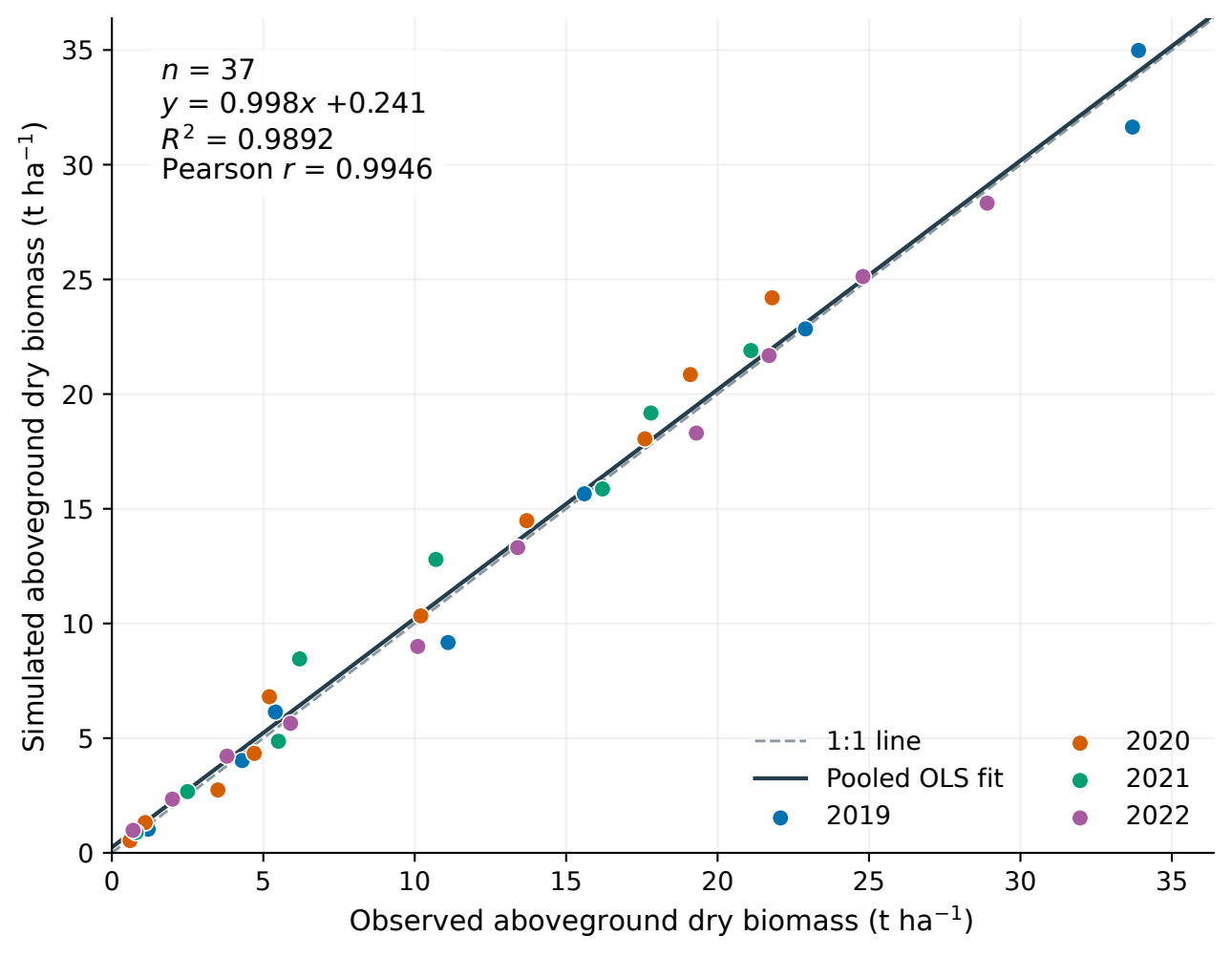

Original DI schedule
Selected candidate
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Irrigation per event (mm)
0
50
100
150
200
06-06
06-18
07-06
07-18
07-27
08-10
08-23
08-26
06-08
06-22
07-03
07-16
07-27
08-06
08-21
08-31
06-13
06-30
07-09
07-20
07-30
08-11
08-23
06-09
06-22
07-03
07-24
08-09

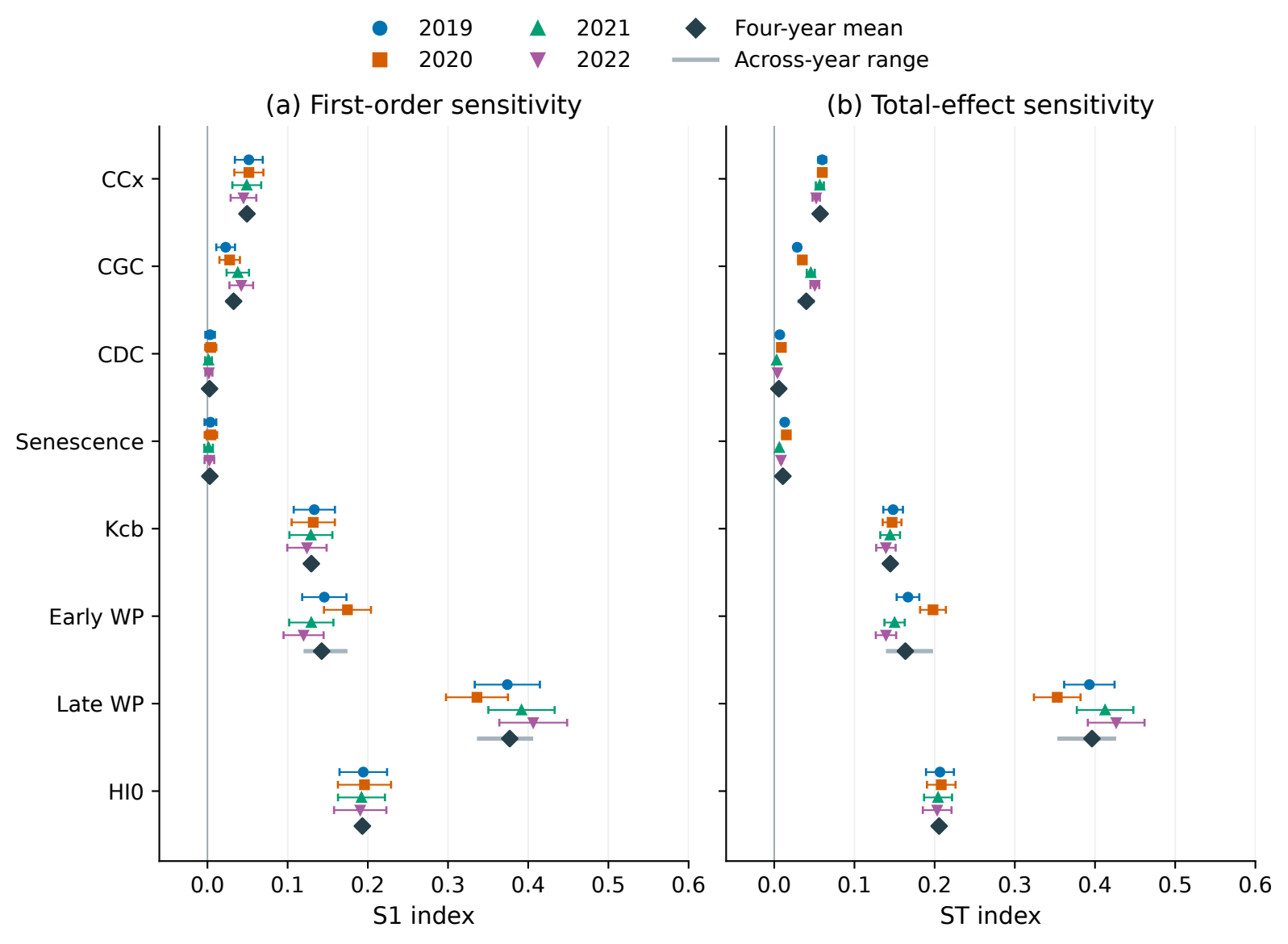

NSGA-III median and seed range
NSGA-II median and seed range
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Normalized hypervolume
Generation
0.52
0.51
0.50
0.49
0.48
0.47
0.46
0.44
0.42
0.40
0.38
0.36
0.56
0.54
0.52
0.50
0.36
0.34
0.32
0.30
0.28
20
40
60
80
100

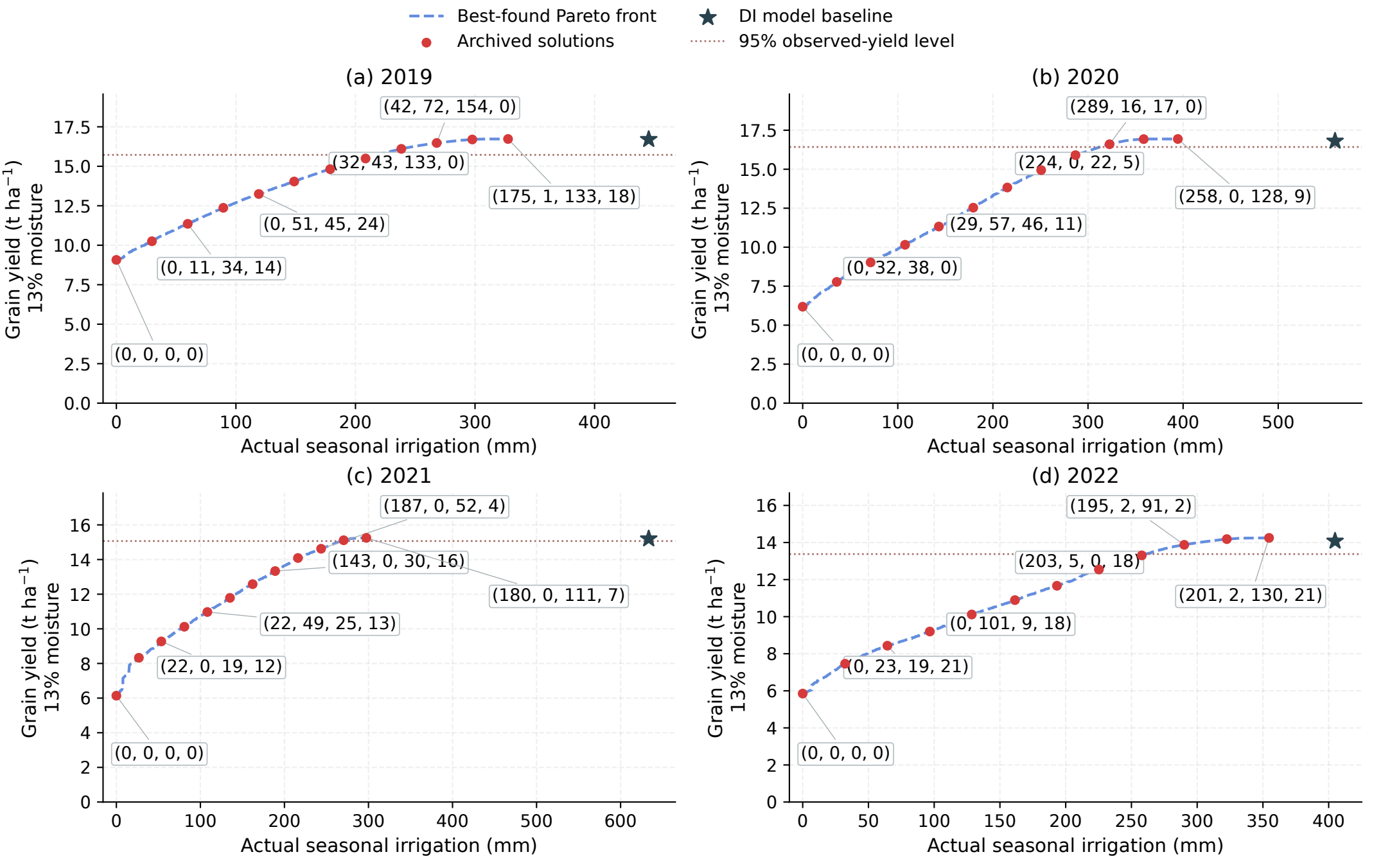

Vary stage 1; fix stage 2
Vary stage 2; fix stage 1
95% observed-yield gate
(a) Conditional gate exceedance
(b) Conditional lower-tail yield
2019
2020
2021
2022
0.0
0.2
0.4
0.6
0.8
1.0
Conditional fraction above yield gate
13
14
15
16
5th-percentile grain yield (t ha⁻¹)

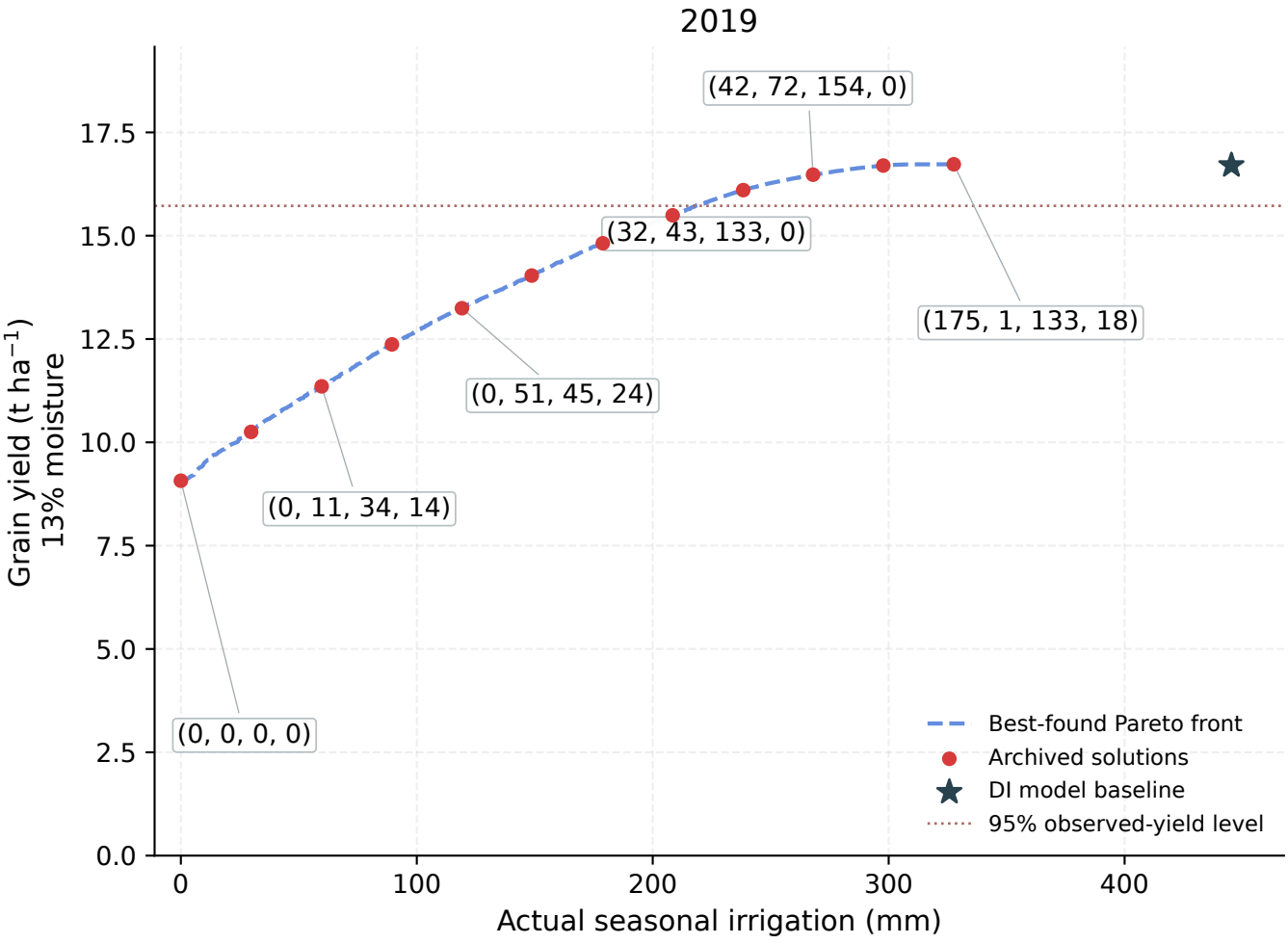

DI baseline
Original selected candidate
Existing point meeting baseline final-storage floor
Actual seasonal irrigation (mm)
0
100
200
300
400
500
600
700
2019
2020
2021
2022
Year

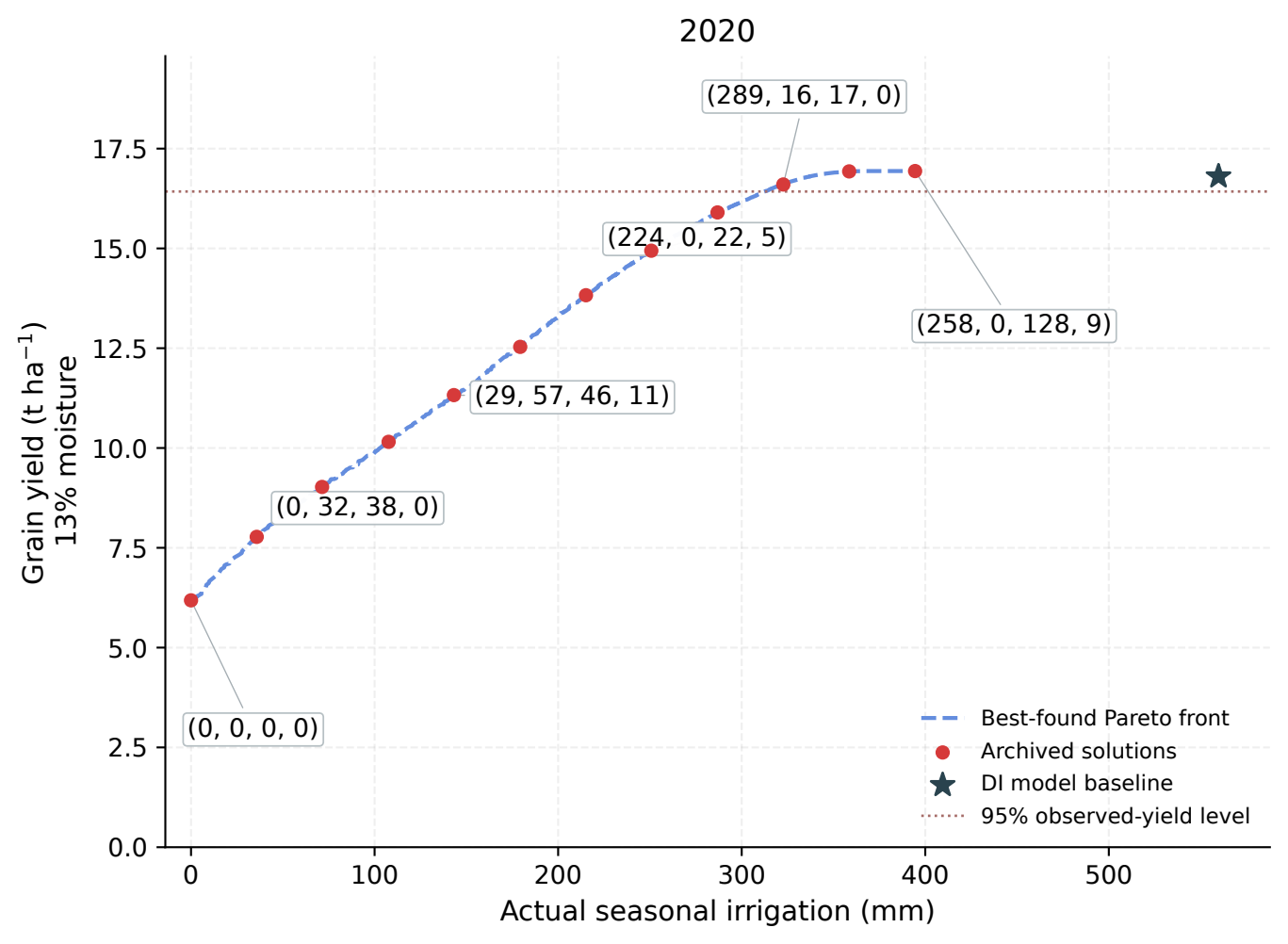

2019
2020
2021
2022
1:1 line
Pooled OLS fit
n = 37
OLS R^2 = 0.989
Pearson r = 0.995
y = 0.991x -0.110
Observed aboveground dry biomass (t ha^{-1})
Simulated aboveground dry biomass (t ha^{-1})
0
5
10
15
20
25
30
35
40

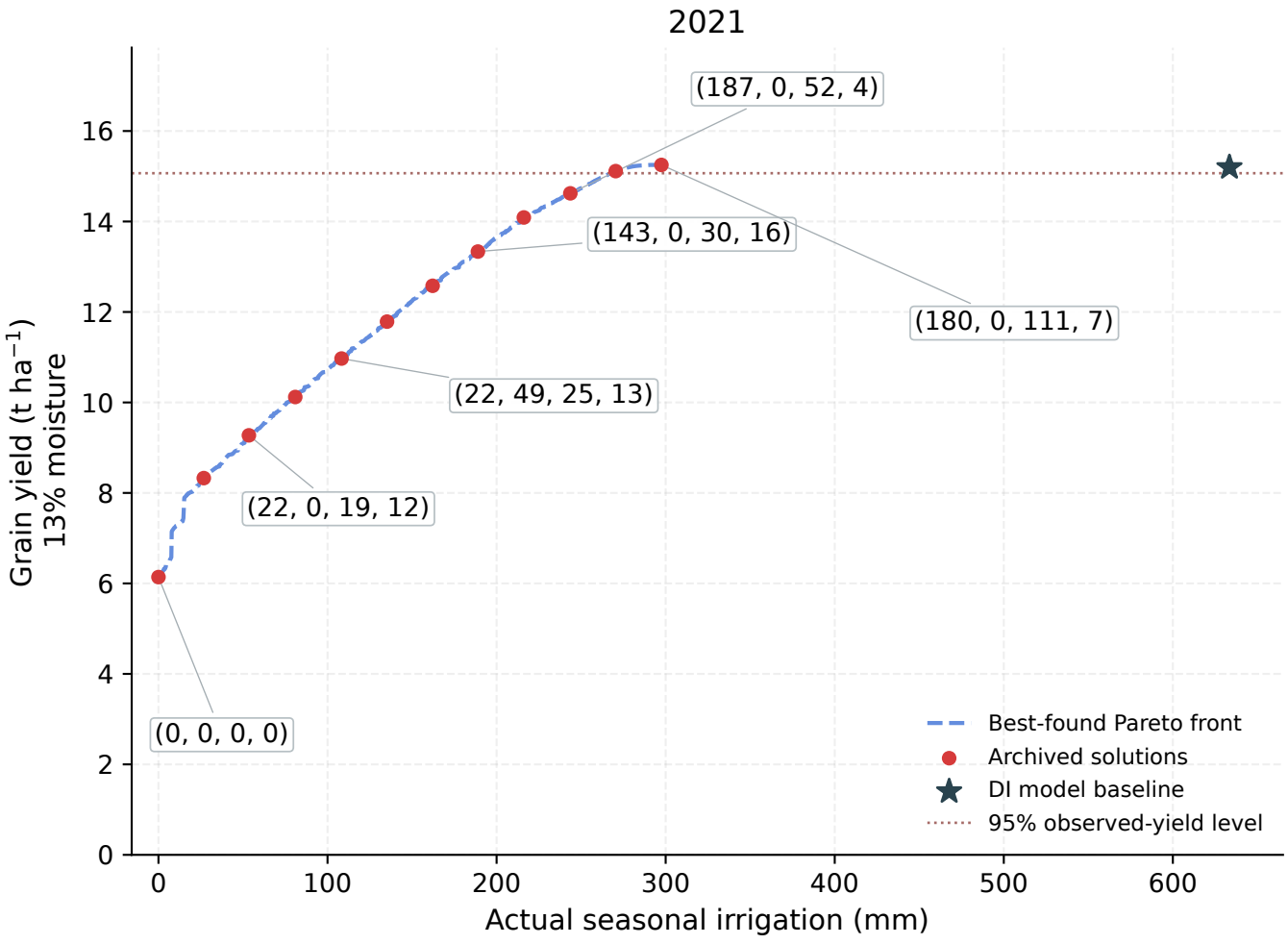

n = 37
y = 0.998x +0.241
R² = 0.9892
Pearson r = 0.9946
1:1 line
Pooled OLS fit
2019
2020
2021
2022
Simulated aboveground dry biomass (t ha⁻¹)
Observed aboveground dry biomass (t ha⁻¹)
0
5
10
15
20
25
30
35

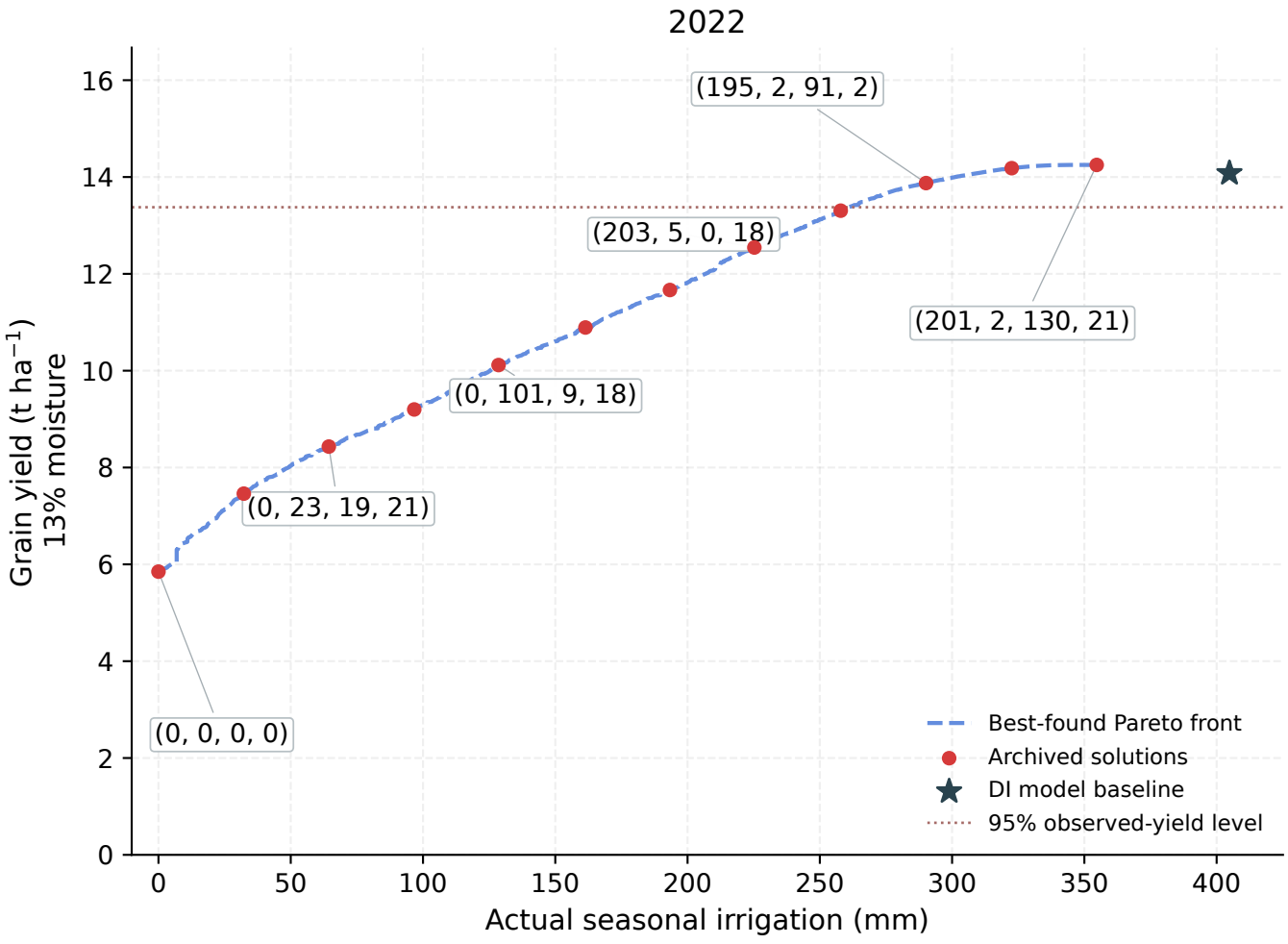

2019
2020
2021
2022
Four-year mean
Across-year range
(a) First-order sensitivity
(b) Total-effect sensitivity
CCx
CGC
CDC
Senescence
Kcb
Early WP
Late WP
HI0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
S1 index
ST index

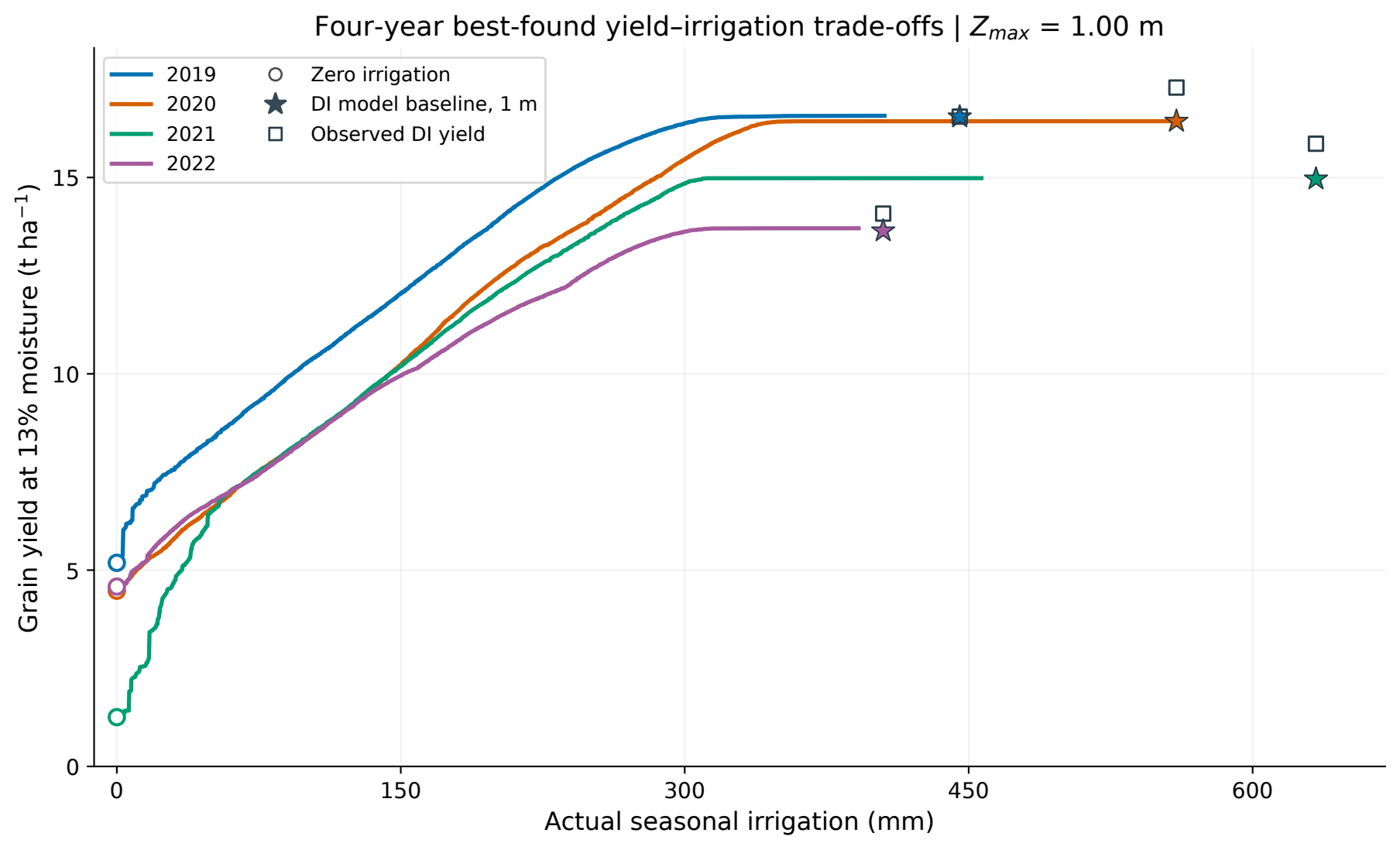

Best-found Pareto front
Archived solutions
DI model baseline
95% observed-yield level
(a) 2019
(b) 2020
(c) 2021
(d) 2022
Grain yield (t ha^{-1}) 13% moisture
Actual seasonal irrigation (mm)
(0, 0, 0, 0)
(0, 11, 34, 14)
(0, 51, 45, 24)
(32, 43, 133, 0)
(42, 72, 154, 0)
(175, 1, 133, 18)
(0, 32, 38, 0)
(29, 57, 46, 11)
(224, 0, 22, 5)
(289, 16, 17, 0)
(258, 0, 128, 9)
(22, 0, 19, 12)
(22, 49, 25, 13)
(143, 0, 30, 16)
(187, 0, 52, 4)
(180, 0, 111, 7)
(0, 23, 19, 21)
(0, 101, 9, 18)
(203, 5, 0, 18)
(195, 2, 91, 2)
(201, 2, 130, 21)

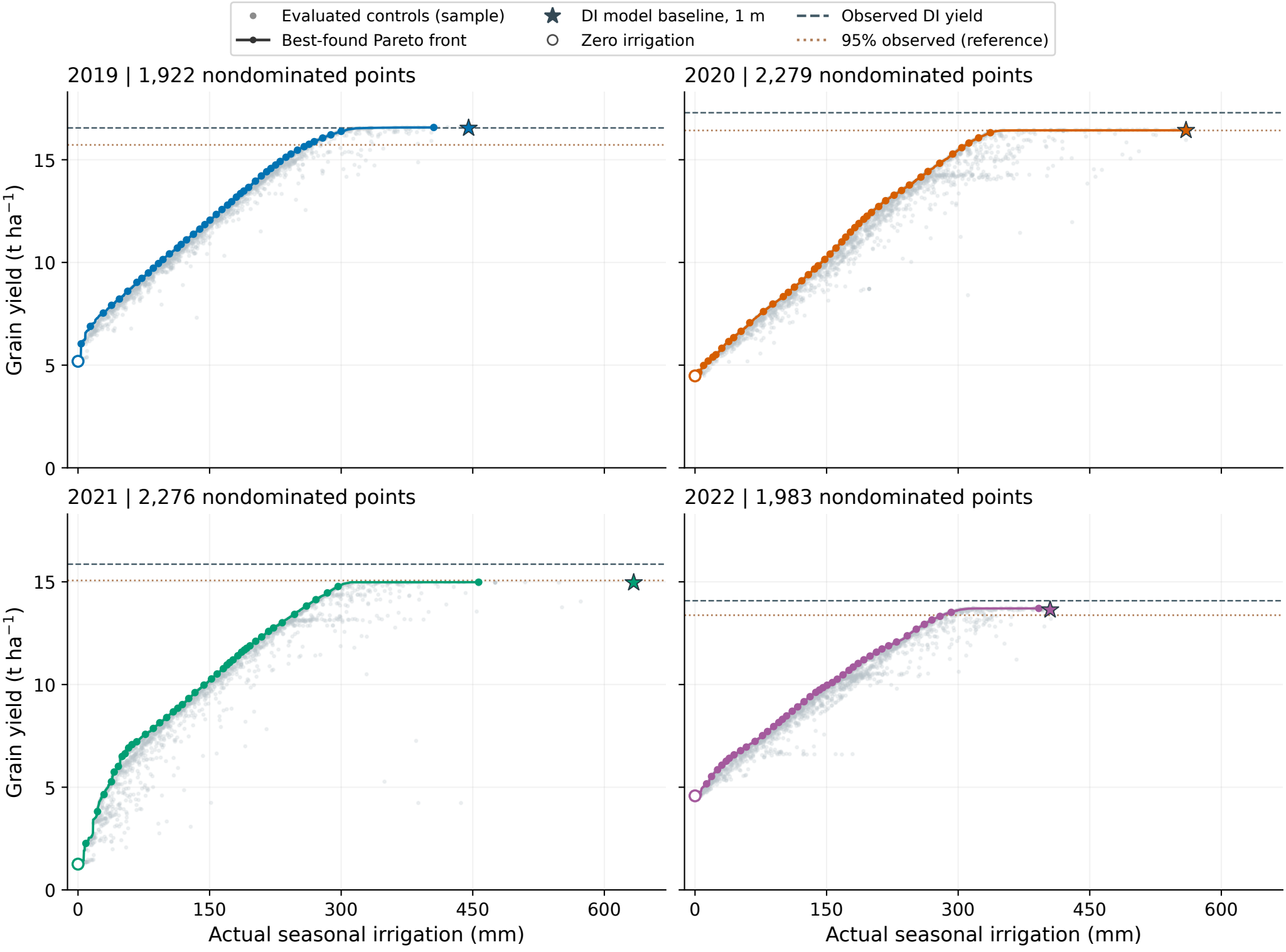

2019
(42, 72, 154, 0)
(32, 43, 133, 0)
(175, 1, 133, 18)
(0, 51, 45, 24)
(0, 11, 34, 14)
(0, 0, 0, 0)
Best-found Pareto front
Archived solutions
DI model baseline
95% observed-yield level
Grain yield (t ha^{-1})
13% moisture
Actual seasonal irrigation (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
100
200
300
400

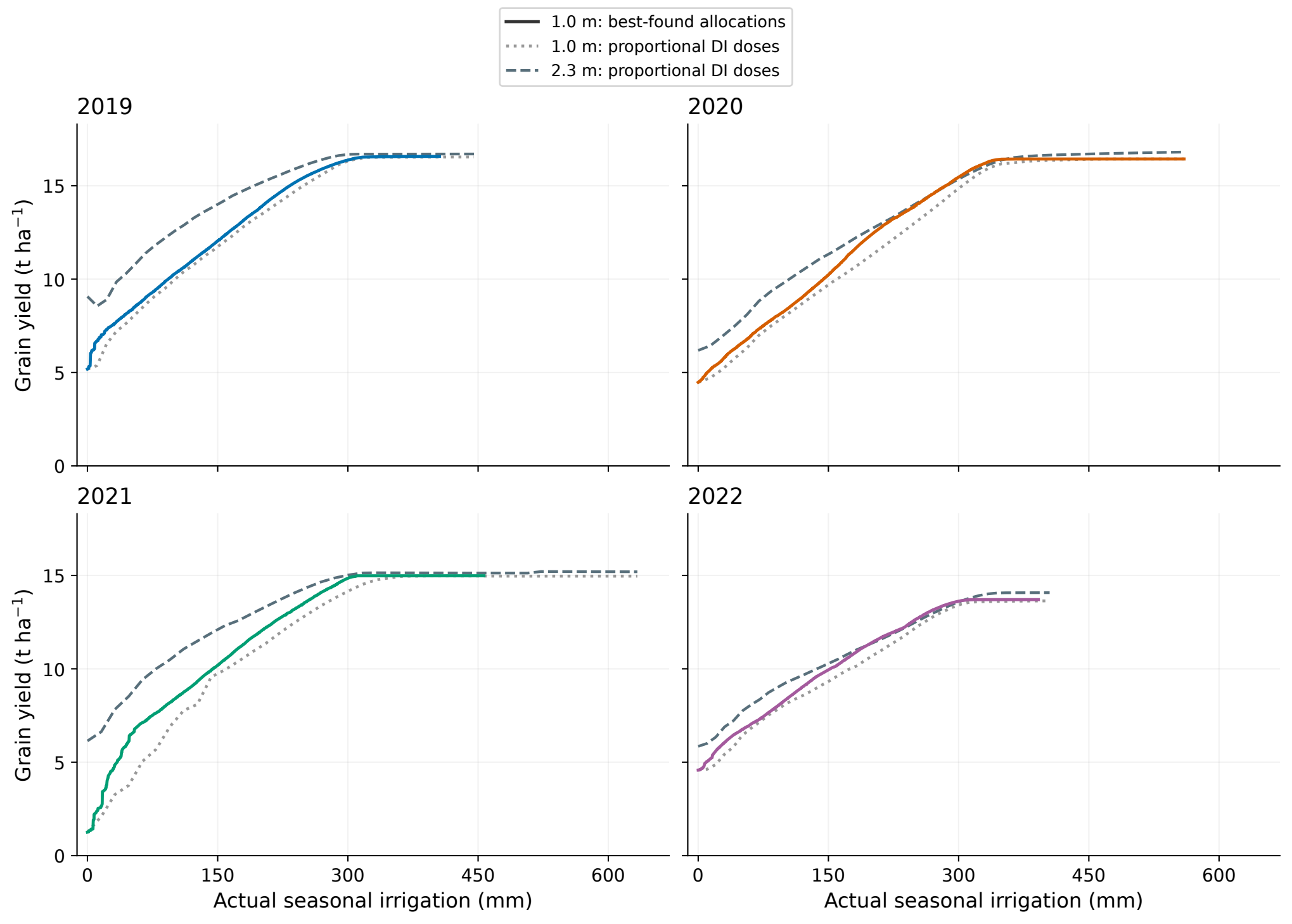

2020
(289, 16, 17, 0)
(224, 0, 22, 5)
(258, 0, 128, 9)
(29, 57, 46, 11)
(0, 32, 38, 0)
(0, 0, 0, 0)
Best-found Pareto front
Archived solutions
DI model baseline
95% observed-yield level
Grain yield (t ha^{-1})
13% moisture
Actual seasonal irrigation (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
100
200
300
400
500

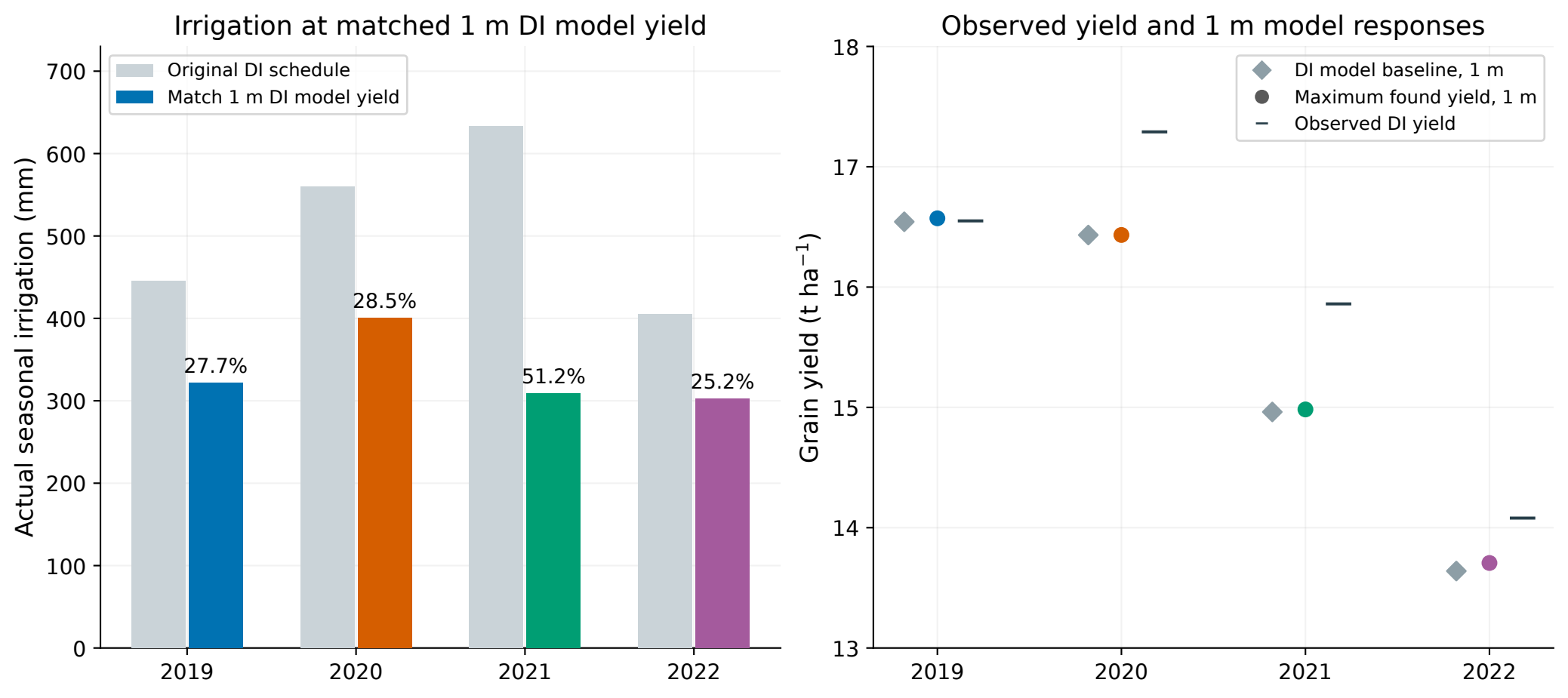

2021
(187, 0, 52, 4)
(143, 0, 30, 16)
(180, 0, 111, 7)
(22, 49, 25, 13)
(22, 0, 19, 12)
(0, 0, 0, 0)
Best-found Pareto front
Archived solutions
DI model baseline
95% observed-yield level
Grain yield (t ha^{-1})
13% moisture
Actual seasonal irrigation (mm)
0
2
4
6
8
10
12
14
16
0
100
200
300
400
500
600

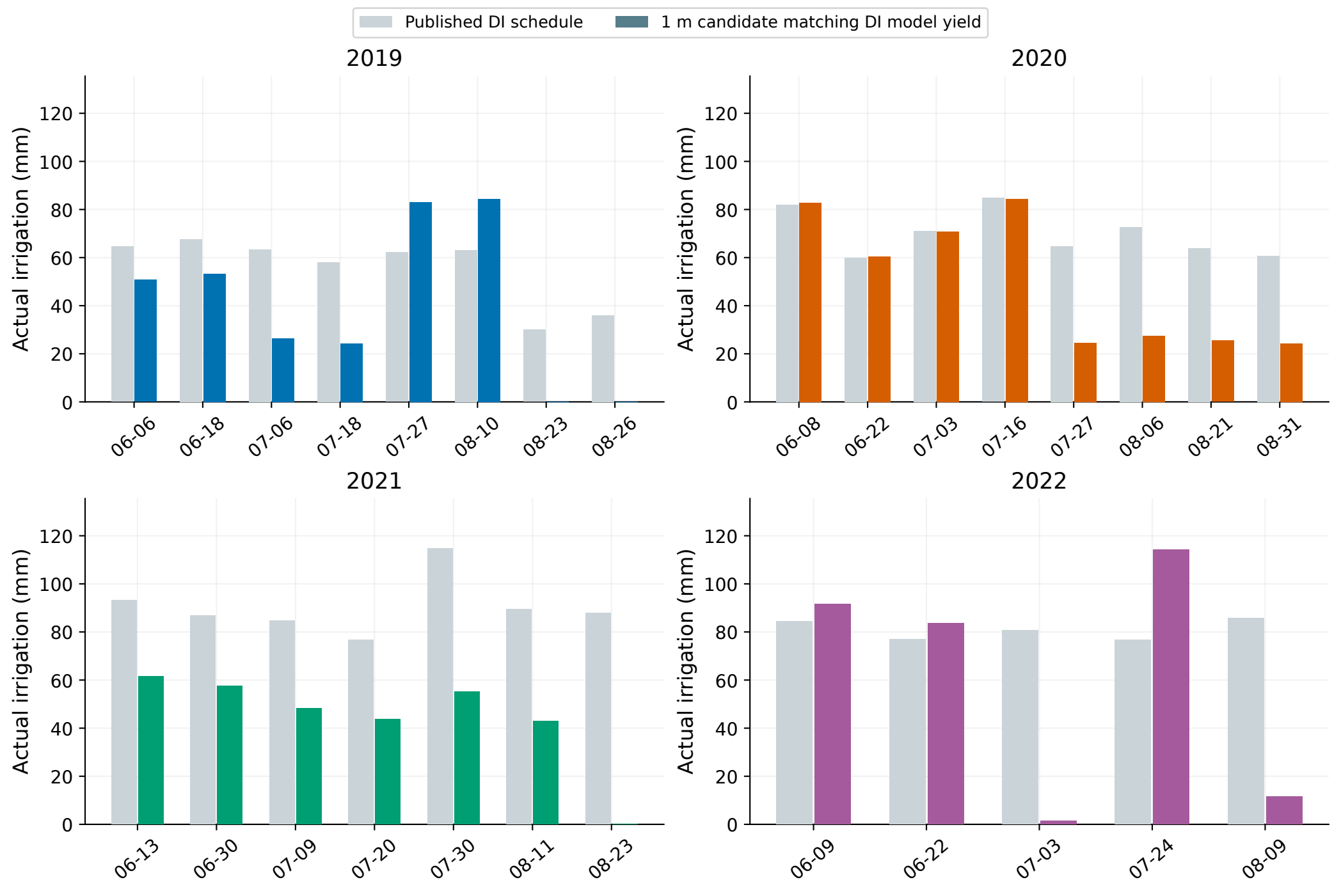

2022
Grain yield (t ha^{-1})
13% moisture
Actual seasonal irrigation (mm)
0
2
4
6
8
10
12
14
16
0
50
100
150
200
250
300
350
400
(195, 2, 91, 2)
(203, 5, 0, 18)
(201, 2, 130, 21)
(0, 101, 9, 18)
(0, 23, 19, 21)
(0, 0, 0, 0)
Best-found Pareto front
Archived solutions
DI model baseline
95% observed-yield level

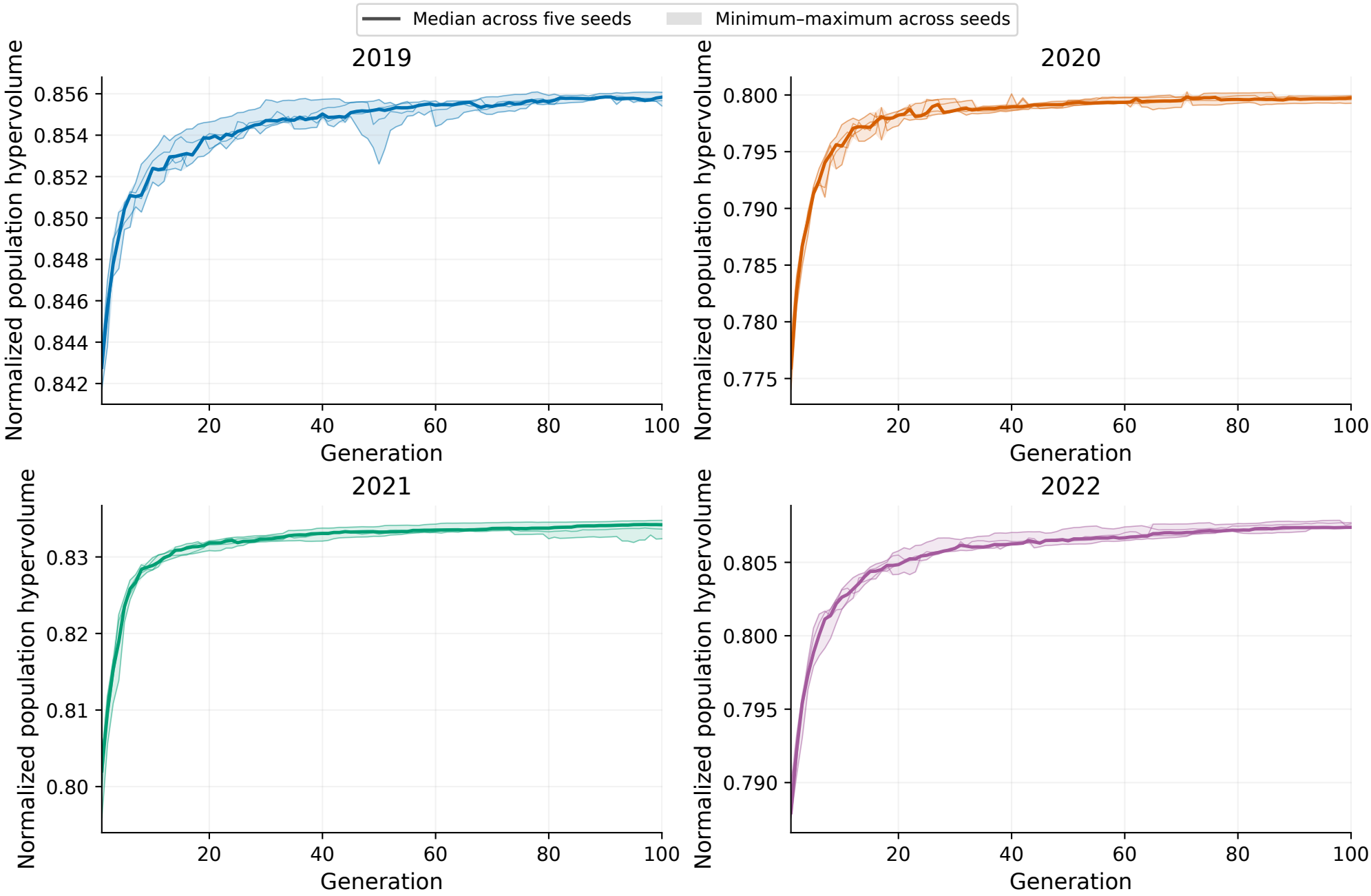

Four-year best-found yield–irrigation trade-offs | Z_{max} = 1.00 m
2019
2020
2021
2022
Zero irrigation
DI model baseline, 1 m
Observed DI yield
Grain yield at 13% moisture (t ha^{-1})
Actual seasonal irrigation (mm)
0
5
10
15
0
150
300
450
600

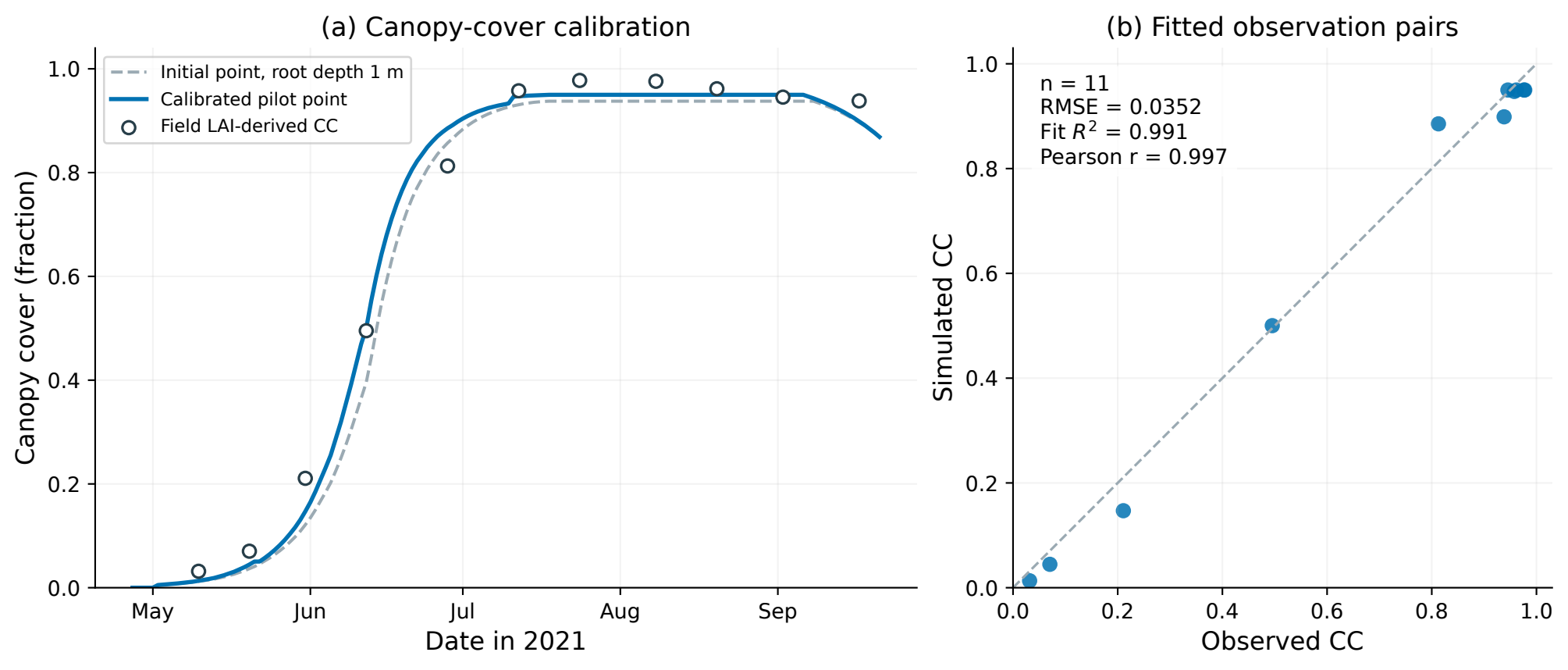

Evaluated controls (sample)
Best-found Pareto front
DI model baseline, 1 m
Zero irrigation
Observed DI yield
95% observed (reference)
2019 | 1,922 nondominated points
2020 | 2,279 nondominated points
2021 | 2,276 nondominated points
2022 | 1,983 nondominated points
Grain yield (t ha⁻¹)
Actual seasonal irrigation (mm)
0
5
10
15
0
150
300
450
600

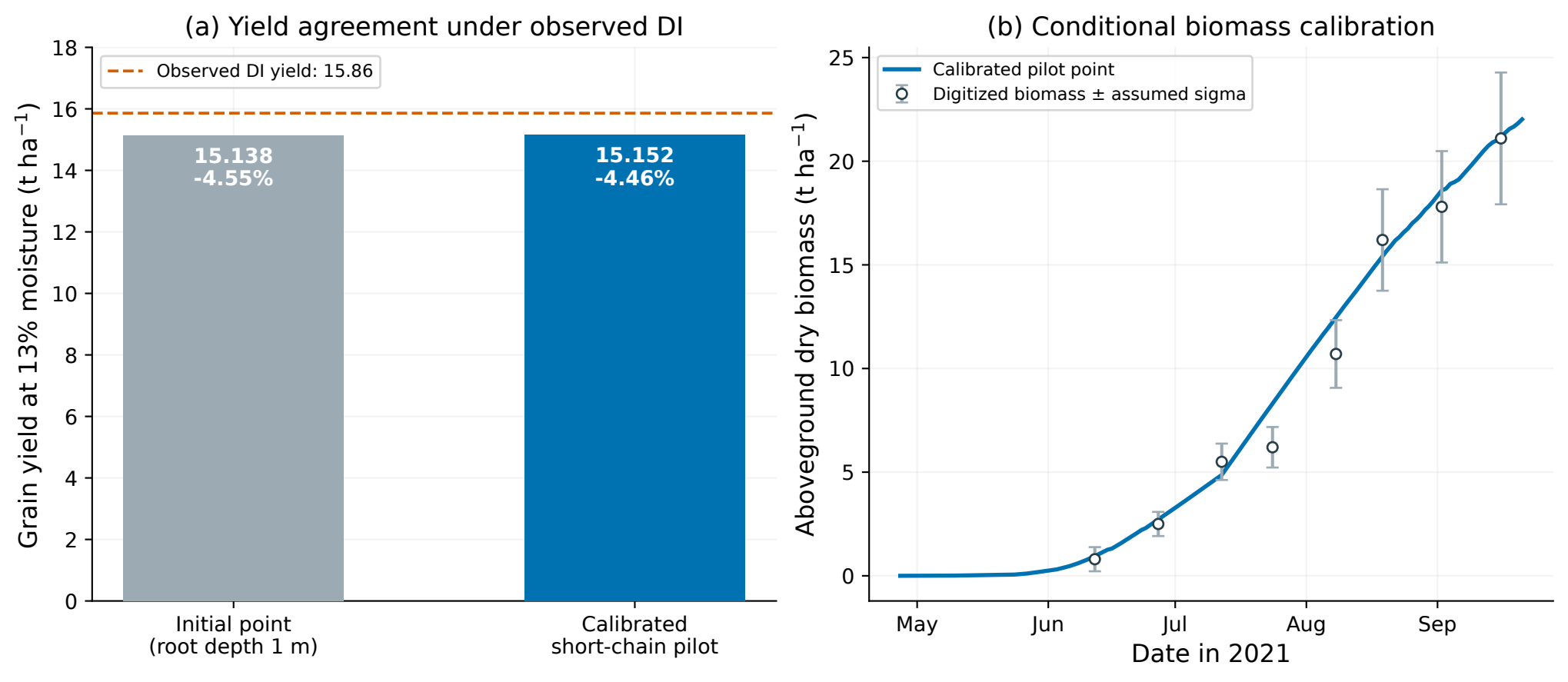

1.0 m: best-found allocations
1.0 m: proportional DI doses
2.3 m: proportional DI doses
2019
2020
2021
2022
Grain yield (t ha^{-1})
Grain yield (t ha^{-1})
0
5
10
15
0
150
300
450
600
Actual seasonal irrigation (mm)
Actual seasonal irrigation (mm)

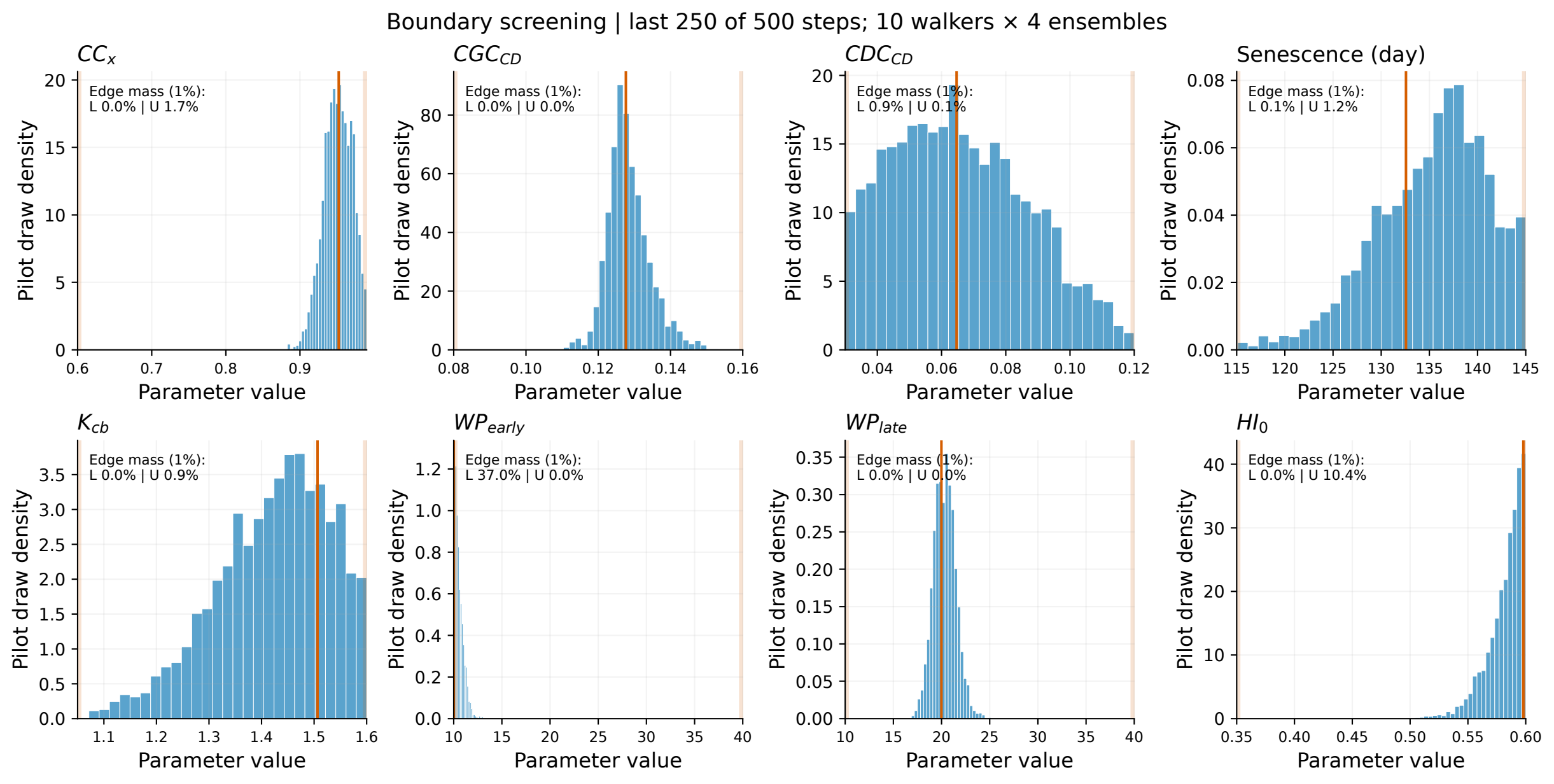

Irrigation at matched 1 m DI model yield
Original DI schedule
Match 1 m DI model yield
Actual seasonal irrigation (mm)
0
100
200
300
400
500
600
700
27.7%
28.5%
51.2%
25.2%
2019
2020
2021
2022
Observed yield and 1 m model responses
DI model baseline, 1 m
Maximum found yield, 1 m
Observed DI yield
Grain yield (t ha^{-1})
13
14
15
16
17
18
2019
2020
2021
2022

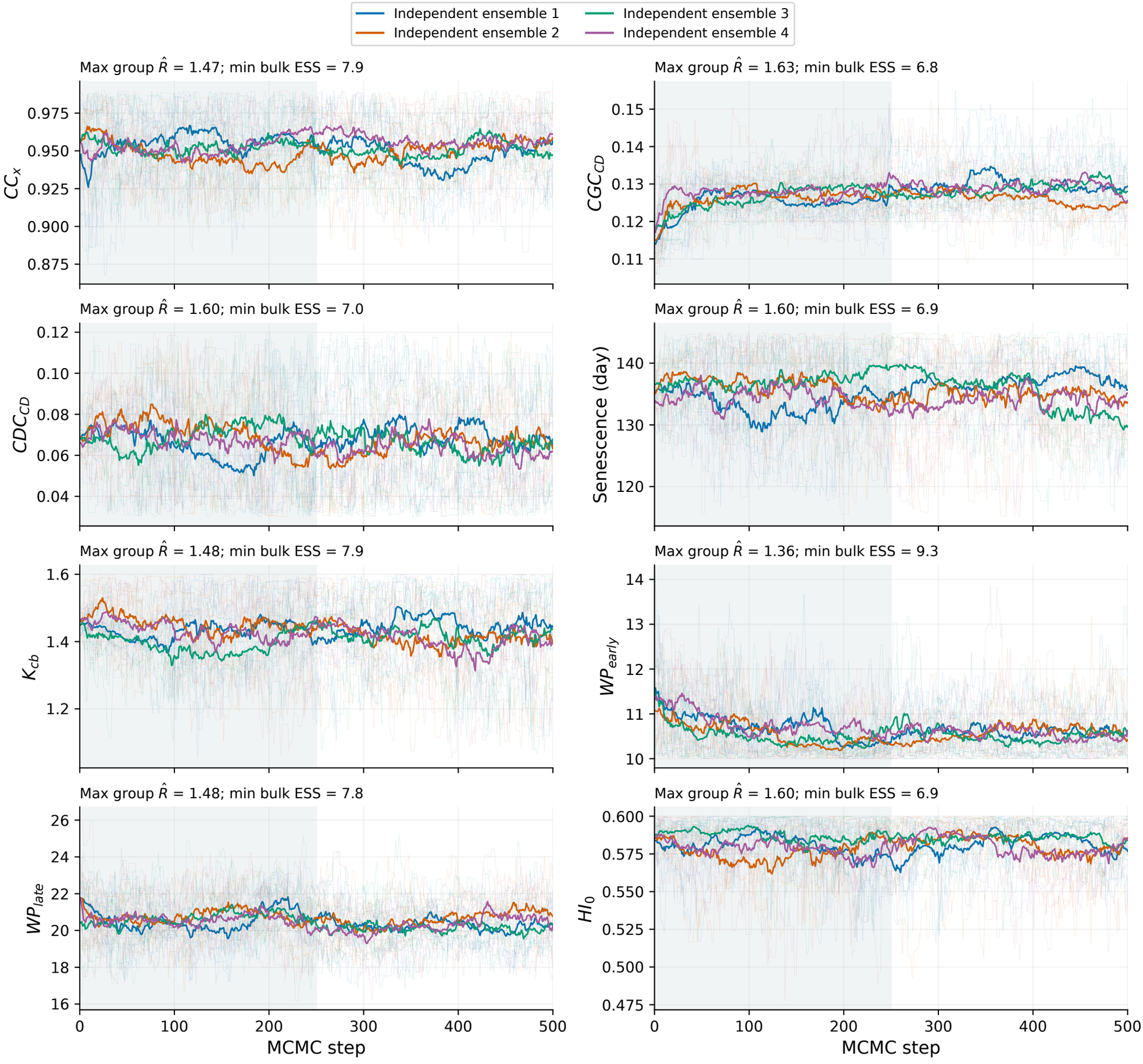

Published DI schedule
1 m candidate matching DI model yield
2019
2020
2021
2022
Actual irrigation (mm)
0
20
40
60
80
100
120
06-06
06-18
07-06
07-18
07-27
08-10
08-23
08-26
06-08
06-22
07-03
07-16
07-27
08-06
08-21
08-31
06-13
06-30
07-09
07-20
07-30
08-11
08-23
06-09
06-22
07-03
07-24
08-09

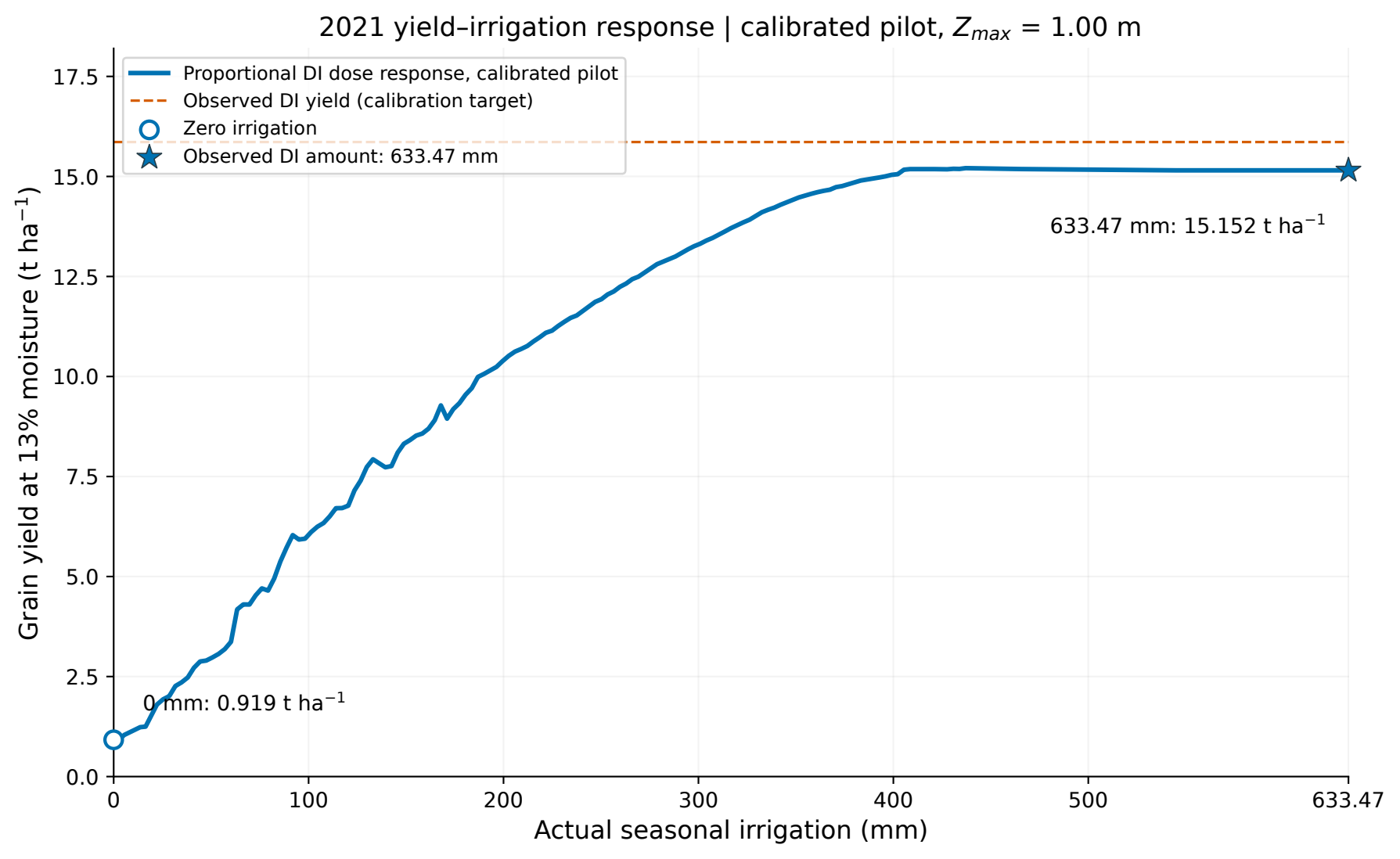

Median across five seeds
Minimum–maximum across seeds
2019
2020
2021
2022
Normalized population hypervolume
Generation
0.856
0.854
0.852
0.850
0.848
0.846
0.844
0.842
0.800
0.795
0.790
0.785
0.780
0.775
0.83
0.82
0.81
0.80
0.805
0.800
0.795
0.790
20
40
60
80
100

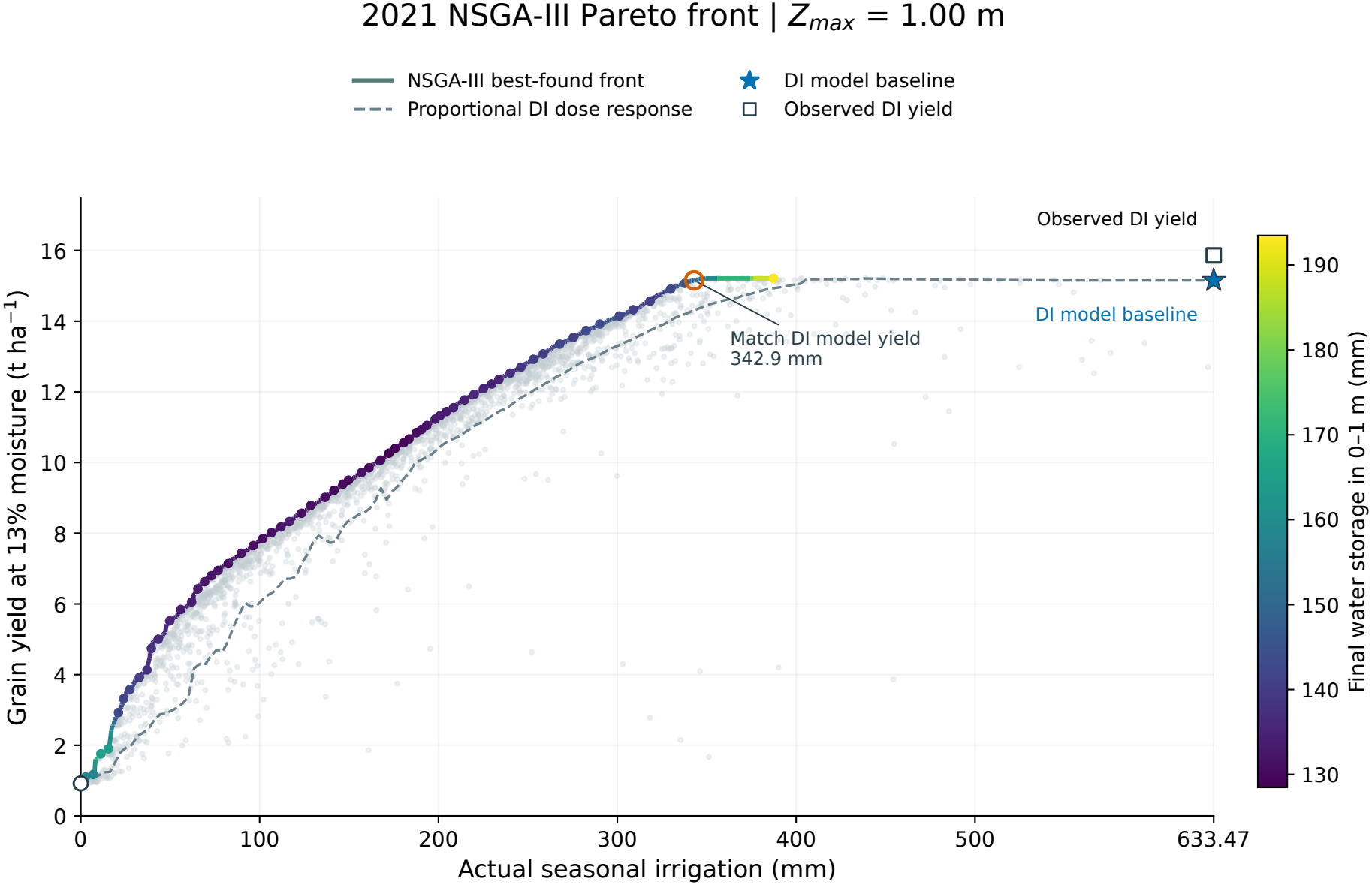

(a) Canopy-cover calibration
Initial point, root depth 1 m
Calibrated pilot point
Field LAI-derived CC
Canopy cover (fraction)
Date in 2021
May
Jun
Jul
Aug
Sep
(b) Fitted observation pairs
n = 11
RMSE = 0.0352
Fit R^2 = 0.991
Pearson r = 0.997
Simulated CC
Observed CC


(a) Yield agreement under observed DI
Observed DI yield: 15.86
15.138
-4.55%
15.152
-4.46%
Grain yield at 13% moisture (t ha^{-1})
Initial point
(root depth 1 m)
Calibrated
short-chain pilot
(b) Conditional biomass calibration
Calibrated pilot point
Digitized biomass ± assumed sigma
Aboveground dry biomass (t ha^{-1})
May
Jun
Jul
Aug
Sep
Date in 2021

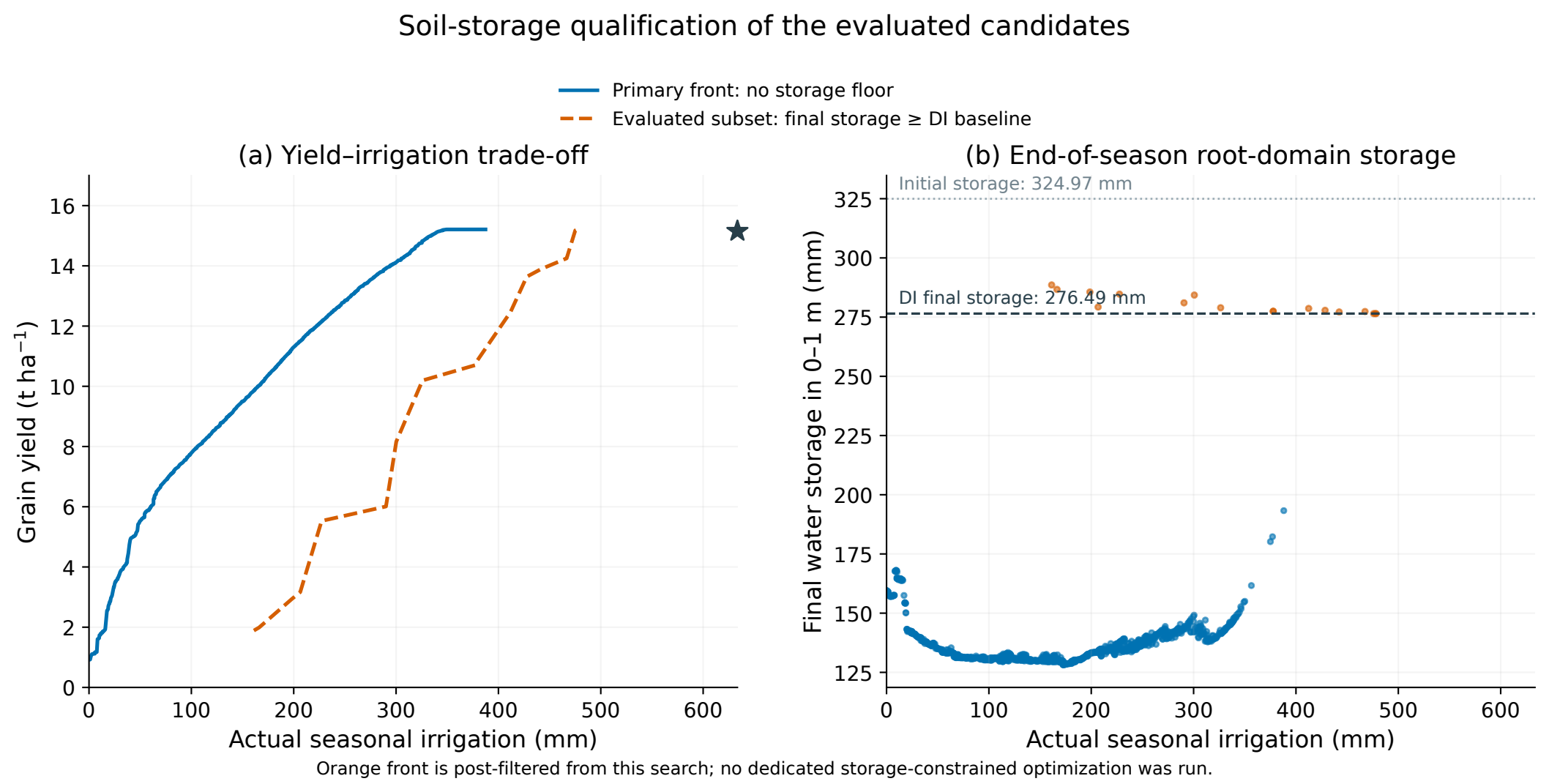

Boundary screening | last 250 of 500 steps; 10 walkers × 4 ensembles
CC_x
Edge mass (1%):
L 0.0% | U 1.7%
CGC_{CD}
Edge mass (1%):
L 0.0% | U 0.0%
CDC_{CD}
Edge mass (1%):
L 0.9% | U 0.1%
Senescence (day)
Edge mass (1%):
L 0.1% | U 1.2%
K_{cb}
Edge mass (1%):
L 0.0% | U 0.9%
WP_{early}
Edge mass (1%):
L 37.0% | U 0.0%
WP_{late}
Edge mass (1%):
L 0.0% | U 0.0%
HI_0
Edge mass (1%):
L 0.0% | U 10.4%
Pilot draw density
Parameter value

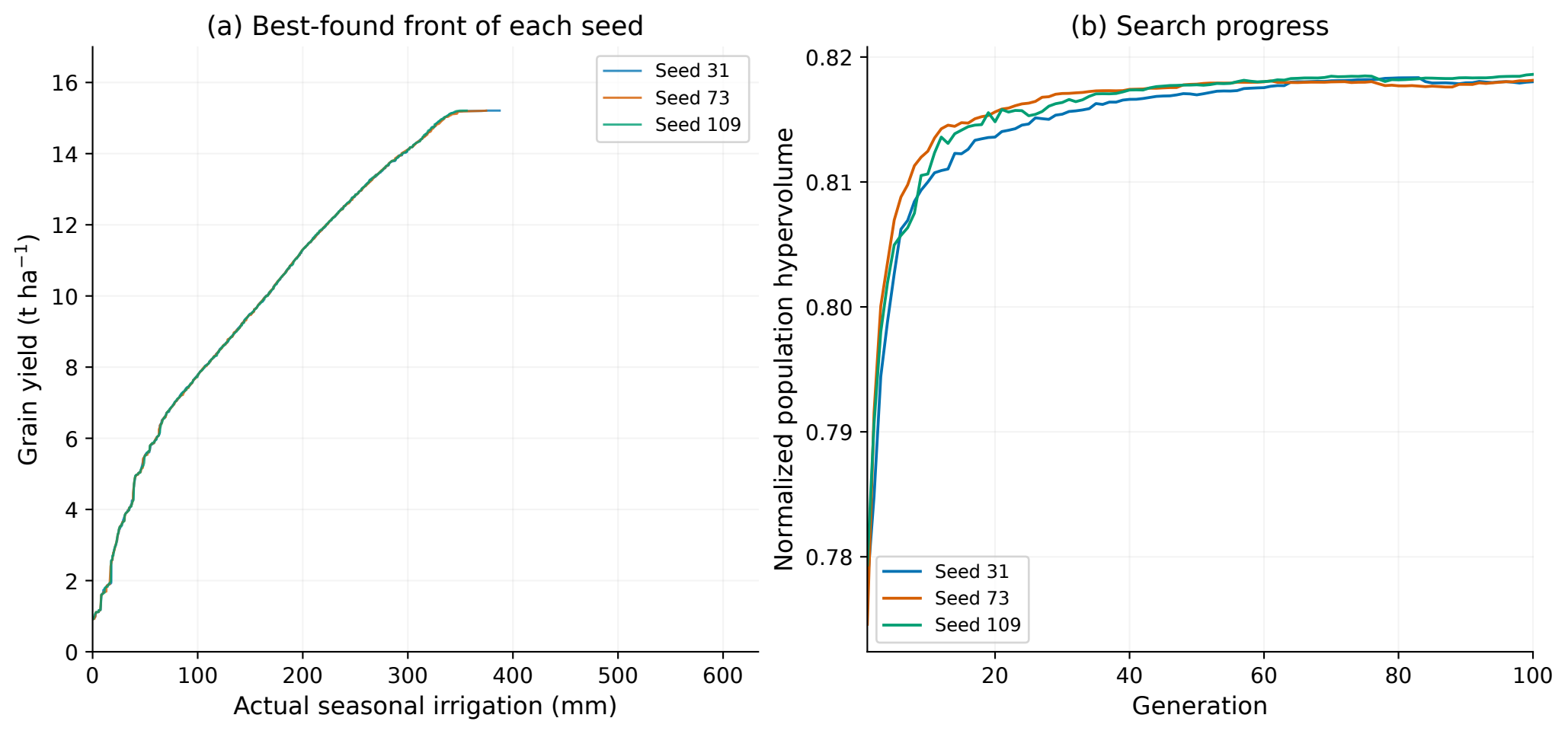

Independent ensemble 1
Independent ensemble 2
Independent ensemble 3
Independent ensemble 4
Max group $\hat{R}$ = 1.47; min bulk ESS = 7.9
CC_x
Max group $\hat{R}$ = 1.63; min bulk ESS = 6.8
CGC_{CD}
Max group $\hat{R}$ = 1.60; min bulk ESS = 7.0
CDC_{CD}
Max group $\hat{R}$ = 1.60; min bulk ESS = 6.9
Senescence (day)
Max group $\hat{R}$ = 1.48; min bulk ESS = 7.9
K_{cb}
Max group $\hat{R}$ = 1.36; min bulk ESS = 9.3
WP_{early}
Max group $\hat{R}$ = 1.48; min bulk ESS = 7.8
WP_{late}
Max group $\hat{R}$ = 1.60; min bulk ESS = 6.9
HI_0
MCMC step
MCMC step

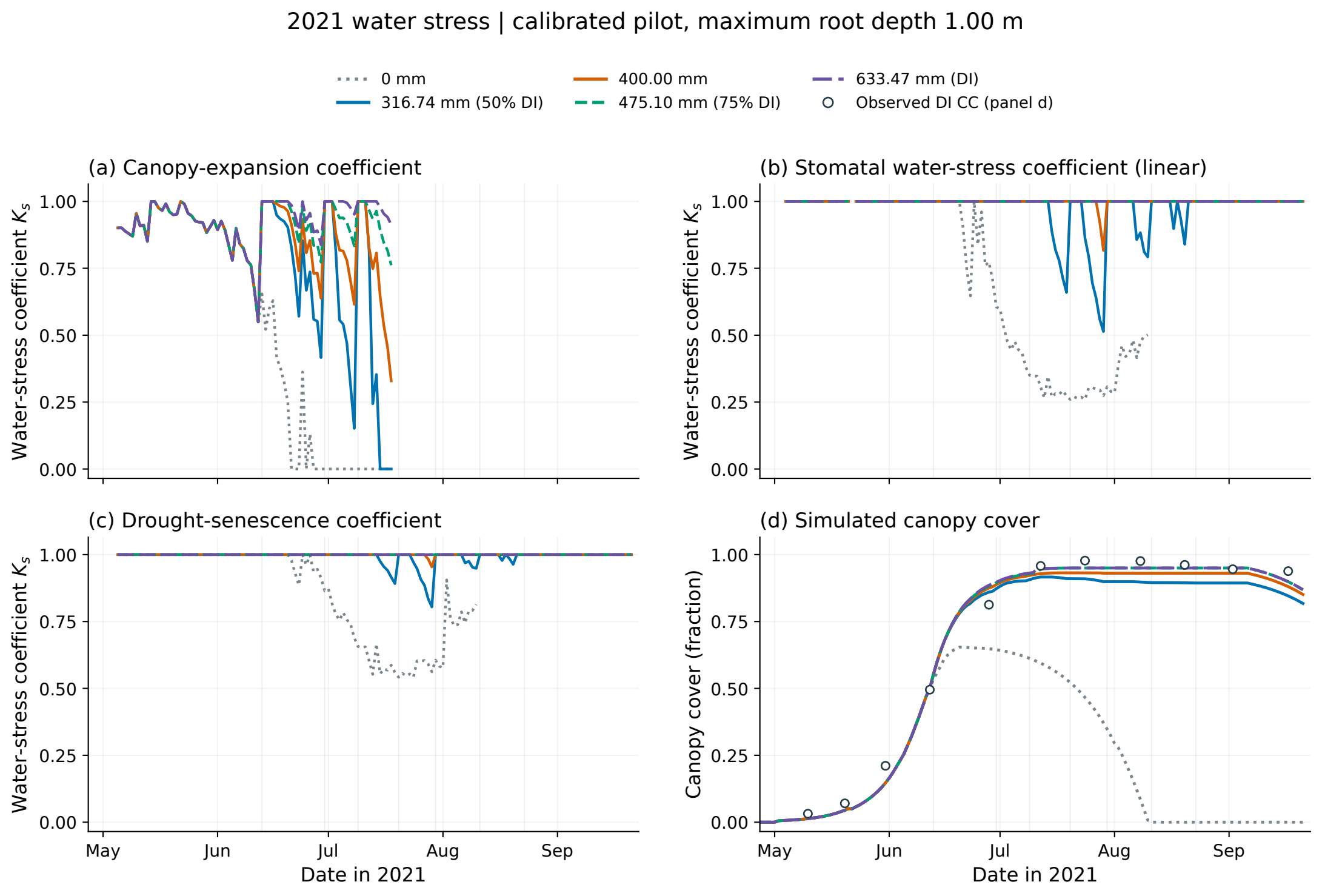

2021 yield–irrigation response | calibrated pilot, Z_{max} = 1.00 m
Proportional DI dose response, calibrated pilot
Observed DI yield (calibration target)
Zero irrigation
Observed DI amount: 633.47 mm
633.47 mm: 15.152 t ha^{-1}
0 mm: 0.919 t ha^{-1}
Grain yield at 13% moisture (t ha^{-1})
Actual seasonal irrigation (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
100
200
300
400
500
633.47



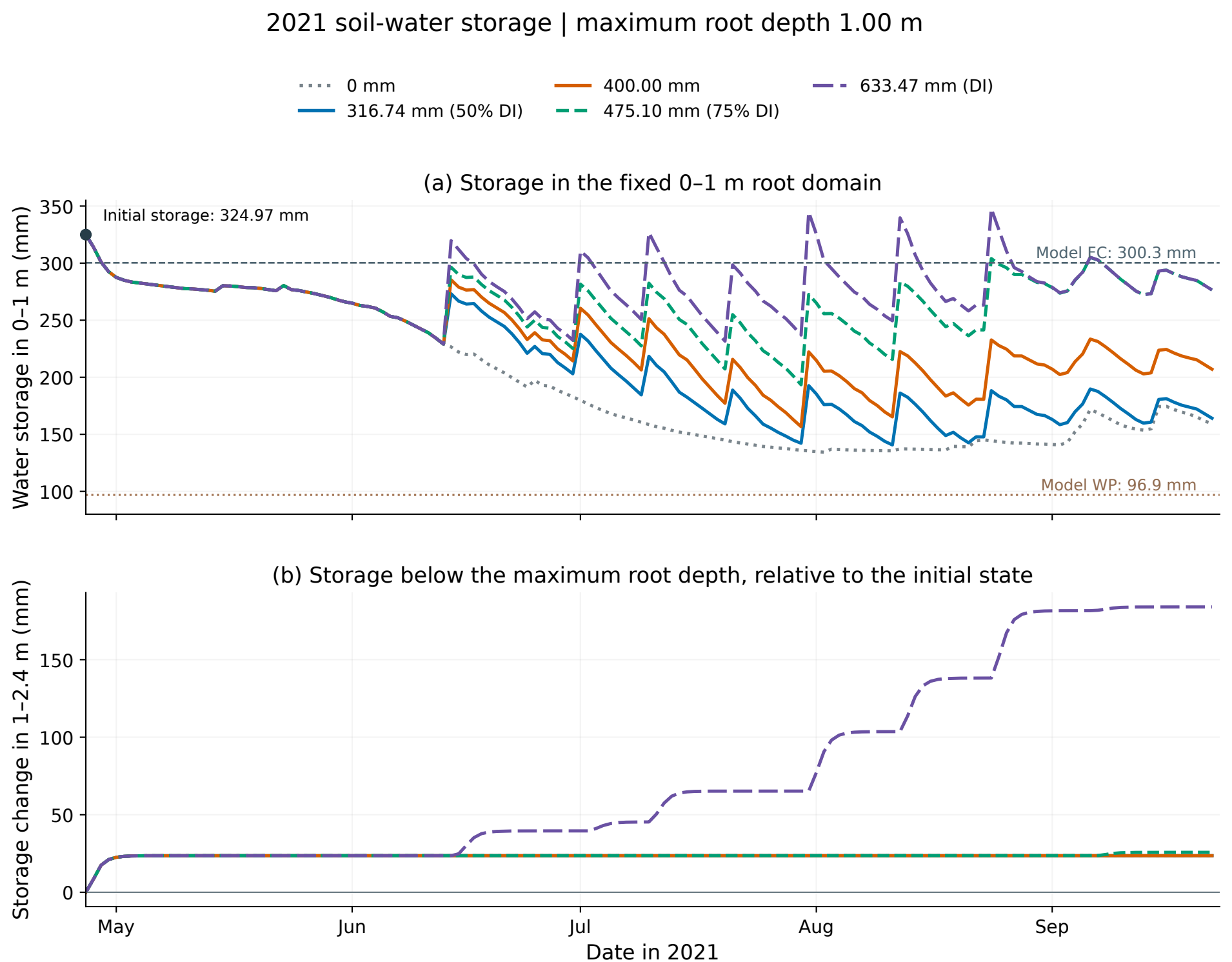
3 seeds × 100 individuals × 100 generations; fixed pilot-calibrated parameters. Finite best-found front, not field validation.


Soil-storage qualification of the evaluated candidates
Primary front: no storage floor
Evaluated subset: final storage ≥ DI baseline
(a) Yield–irrigation trade-off
Grain yield (t ha^{-1})
Actual seasonal irrigation (mm)
(b) End-of-season root-domain storage
Initial storage: 324.97 mm
DI final storage: 276.49 mm
Final water storage in 0–1 m (mm)
Actual seasonal irrigation (mm)
Orange front is post-filtered from this search; no dedicated storage-constrained optimization was run.

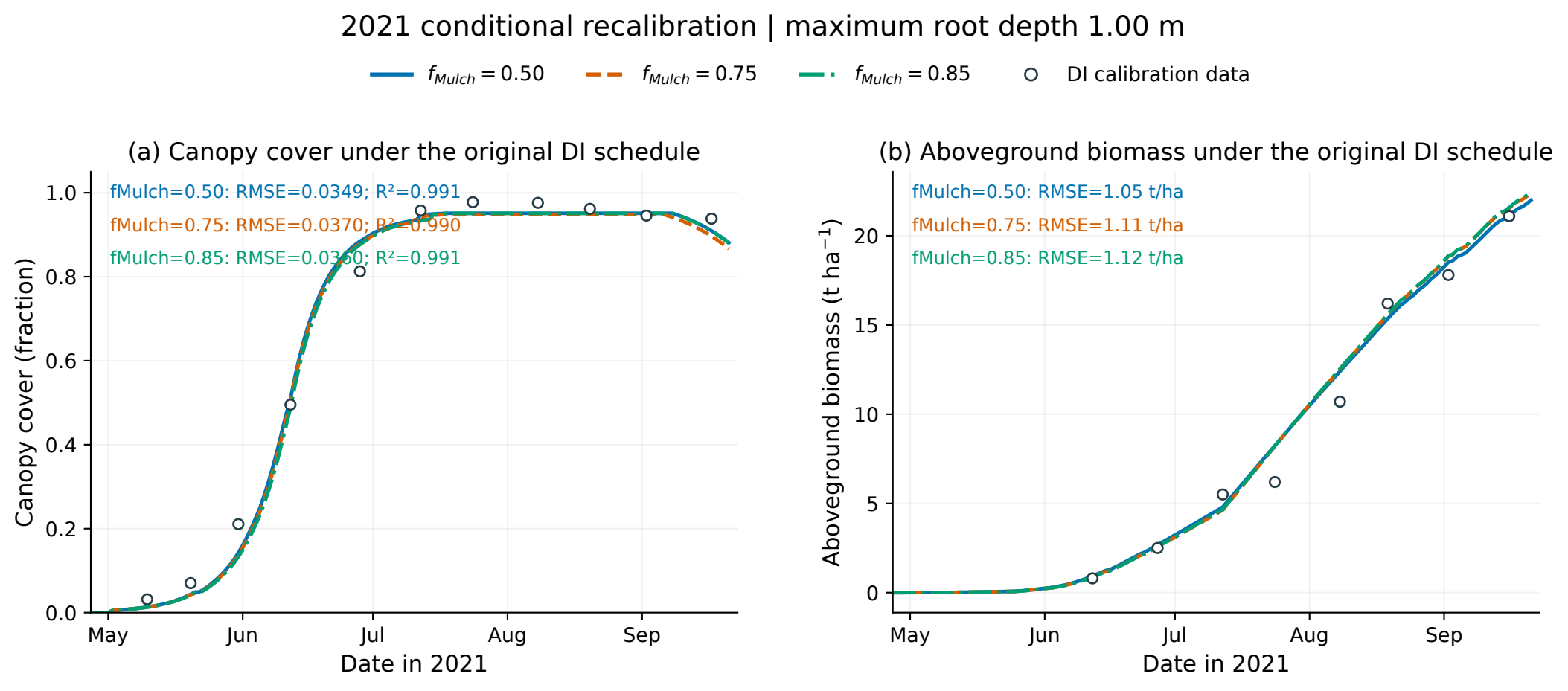

(a) Best-found front of each seed
Seed 31
Seed 73
Seed 109
Grain yield (t ha^{-1})
Actual seasonal irrigation (mm)
0
2
4
6
8
10
12
14
16
0
100
200
300
400
500
600
(b) Search progress
Normalized population hypervolume
Generation
0.78
0.79
0.80
0.81
0.82
20
40
60
80
100
Seed 31
Seed 73
Seed 109

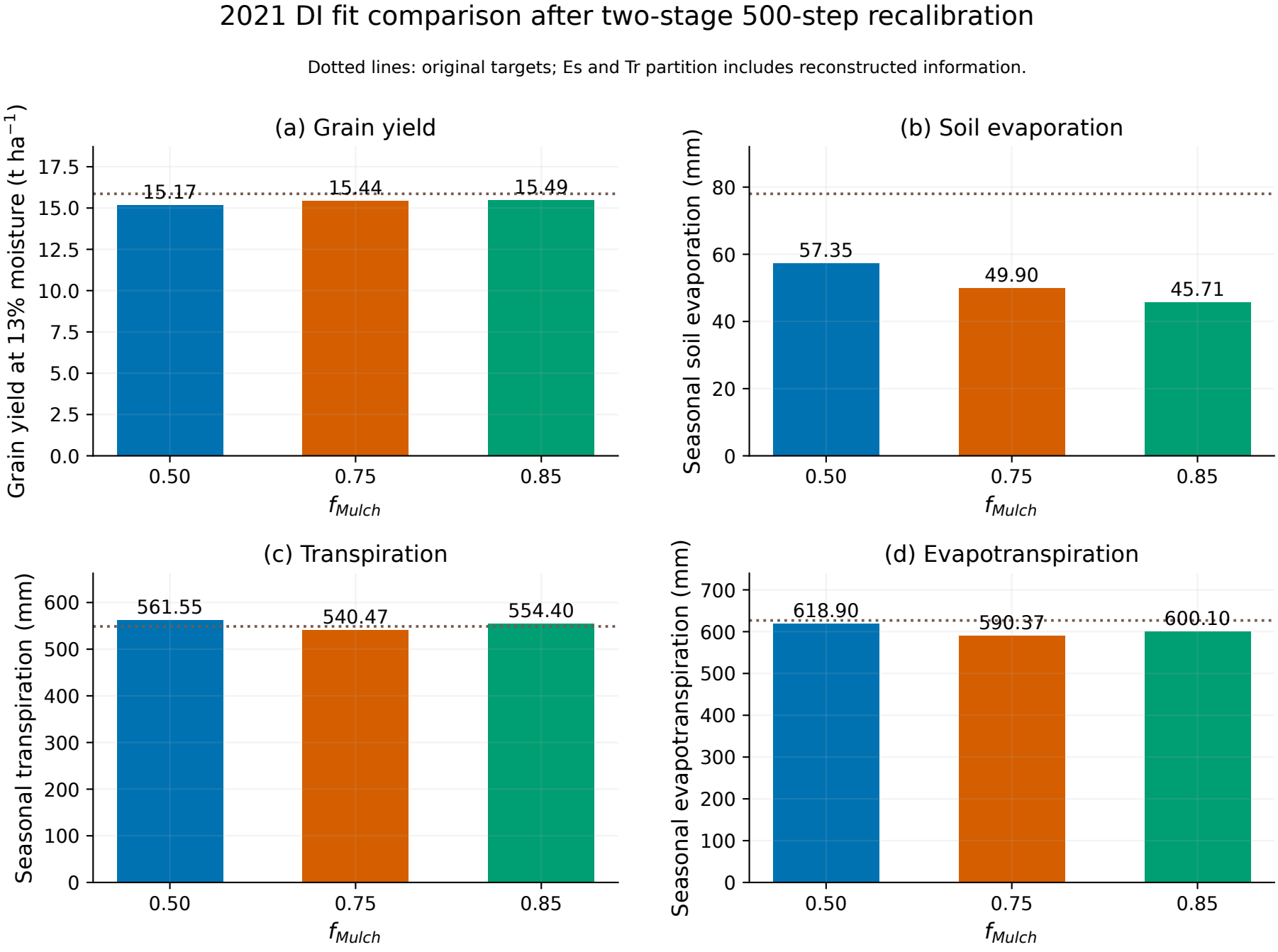


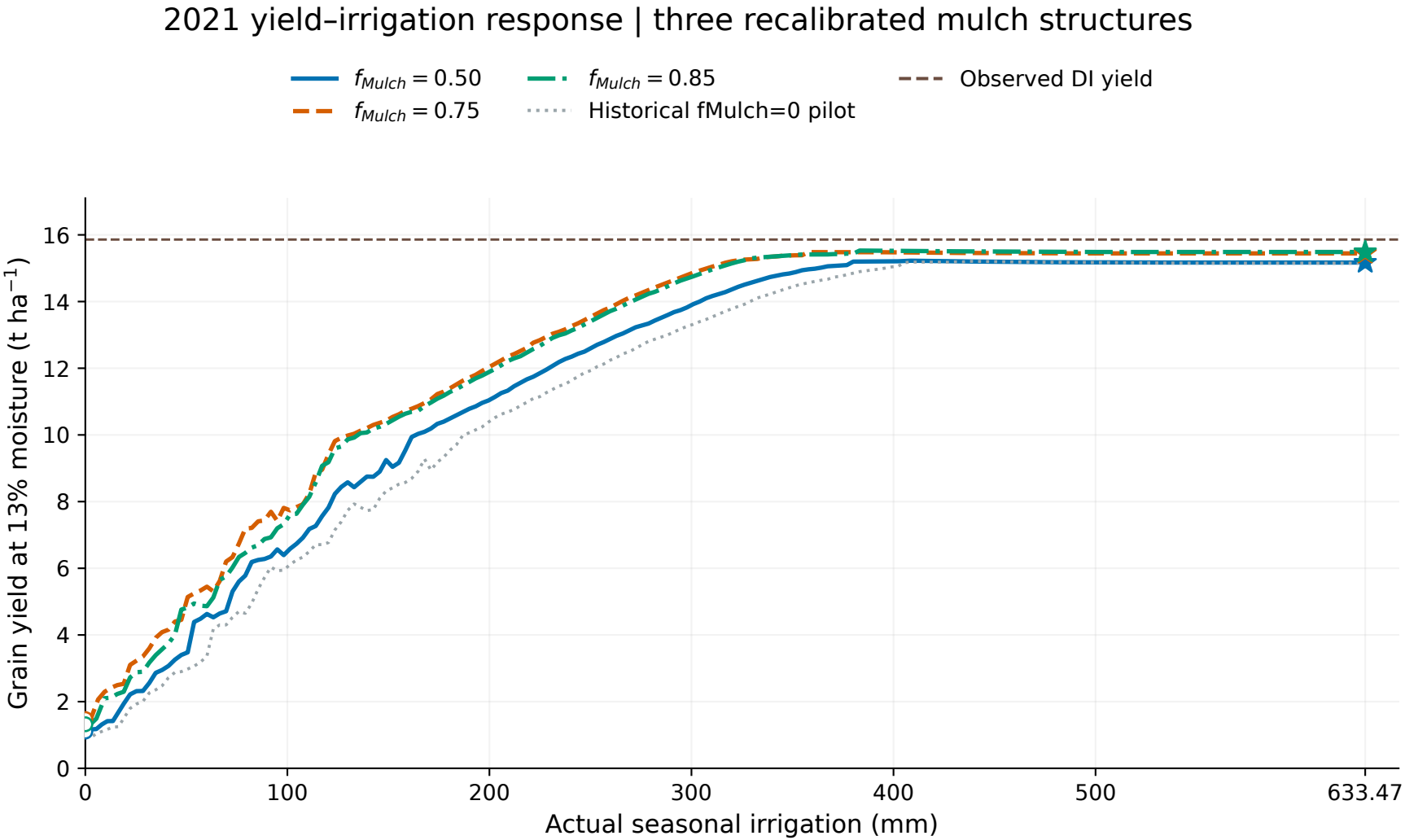
$K_s = 1$: no drought limitation; lower K_s: stronger limitation. Coefficients shown only in stated process windows.



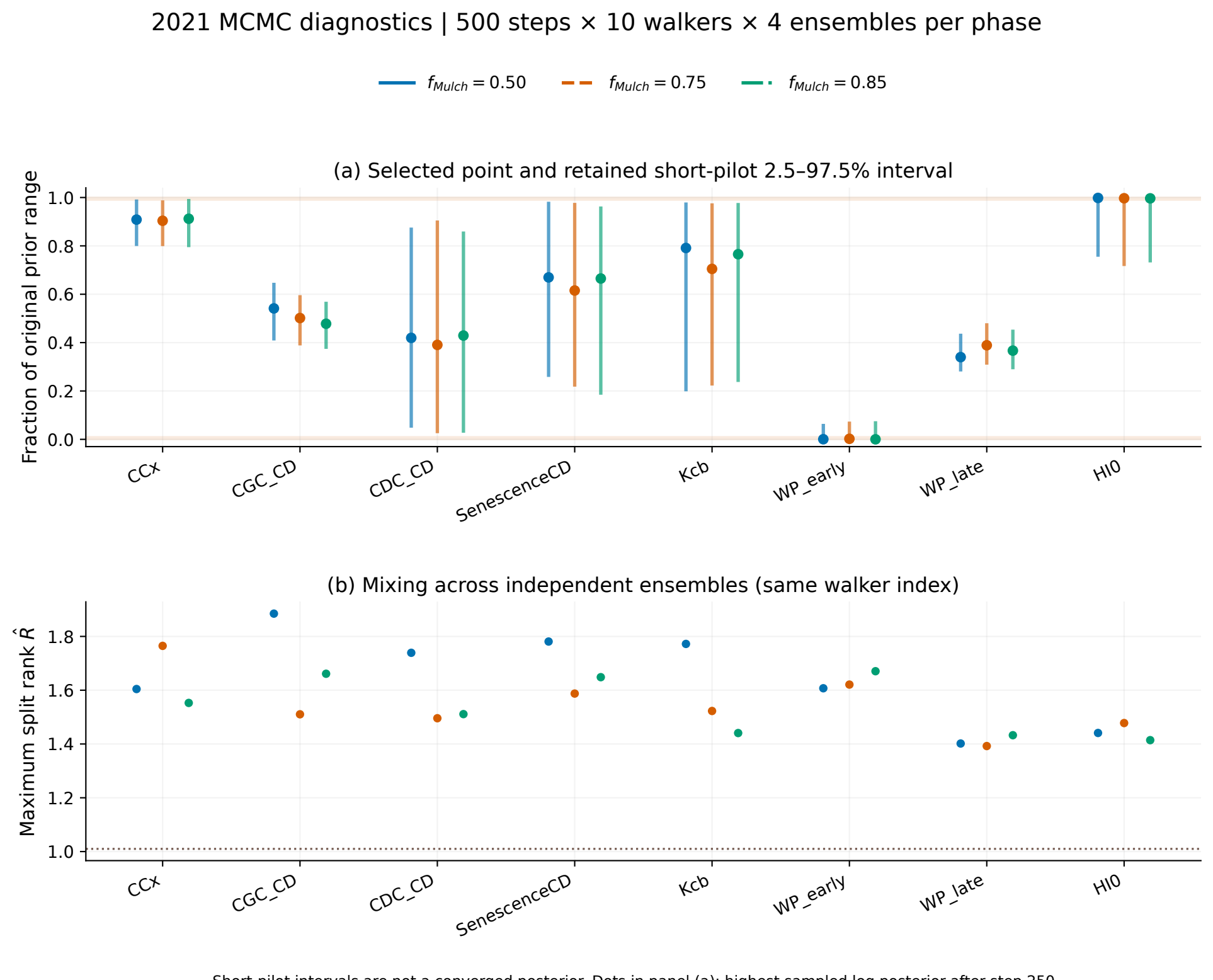
Panel (a) uses a fixed depth for comparison; growing roots do not access the full 1 m from planting. FC/WP are model-derived.


2021 conditional recalibration | maximum root depth 1.00 m
$f_{Mulch} = 0.50$
$f_{Mulch} = 0.75$
$f_{Mulch} = 0.85$
DI calibration data
(a) Canopy cover under the original DI schedule
fMulch=0.50: RMSE=0.0349; R²=0.991
fMulch=0.75: RMSE=0.0370; R²=0.990
fMulch=0.85: RMSE=0.0360; R²=0.991
Canopy cover (fraction)
1.0
0.8
0.6
0.4
0.2
0.0
May
Jun
Jul
Aug
Sep
Date in 2021
(b) Aboveground biomass under the original DI schedule
fMulch=0.50: RMSE=1.05 t/ha
fMulch=0.75: RMSE=1.11 t/ha
fMulch=0.85: RMSE=1.12 t/ha
Aboveground biomass (t ha^{-1})
20
15
10
5
0
May
Jun
Jul
Aug
Sep
Date in 2021

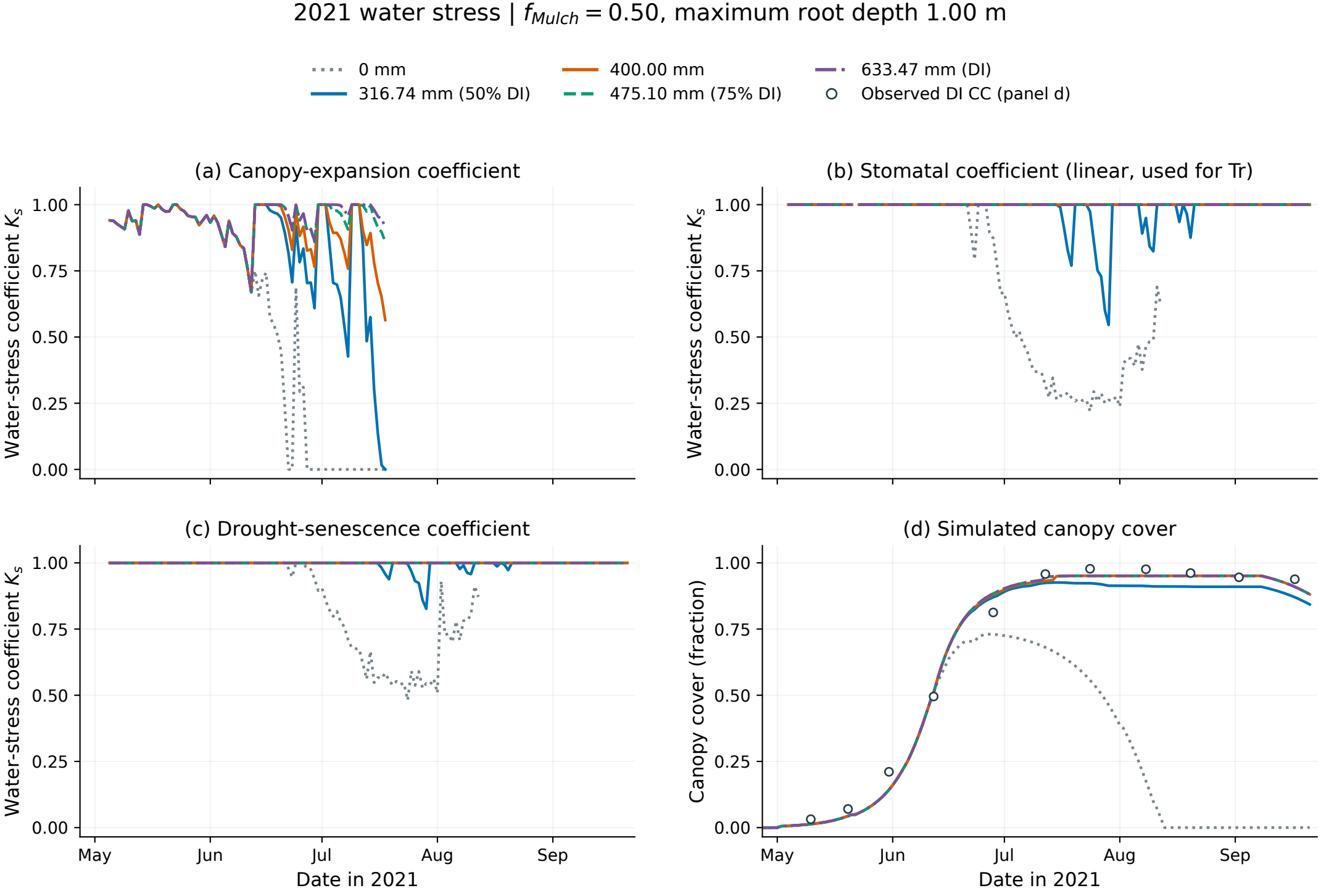

2021 DI fit comparison after two-stage 500-step recalibration
Dotted lines: original targets; Es and Tr partition includes reconstructed information.
(a) Grain yield
Grain yield at 13% moisture (t ha^{-1})
15.17
15.44
15.49
(b) Soil evaporation
Seasonal soil evaporation (mm)
57.35
49.90
45.71
(c) Transpiration
Seasonal transpiration (mm)
561.55
540.47
554.40
(d) Evapotranspiration
Seasonal evapotranspiration (mm)
618.90
590.37
600.10
0.50
0.75
0.85
f_{Mulch}

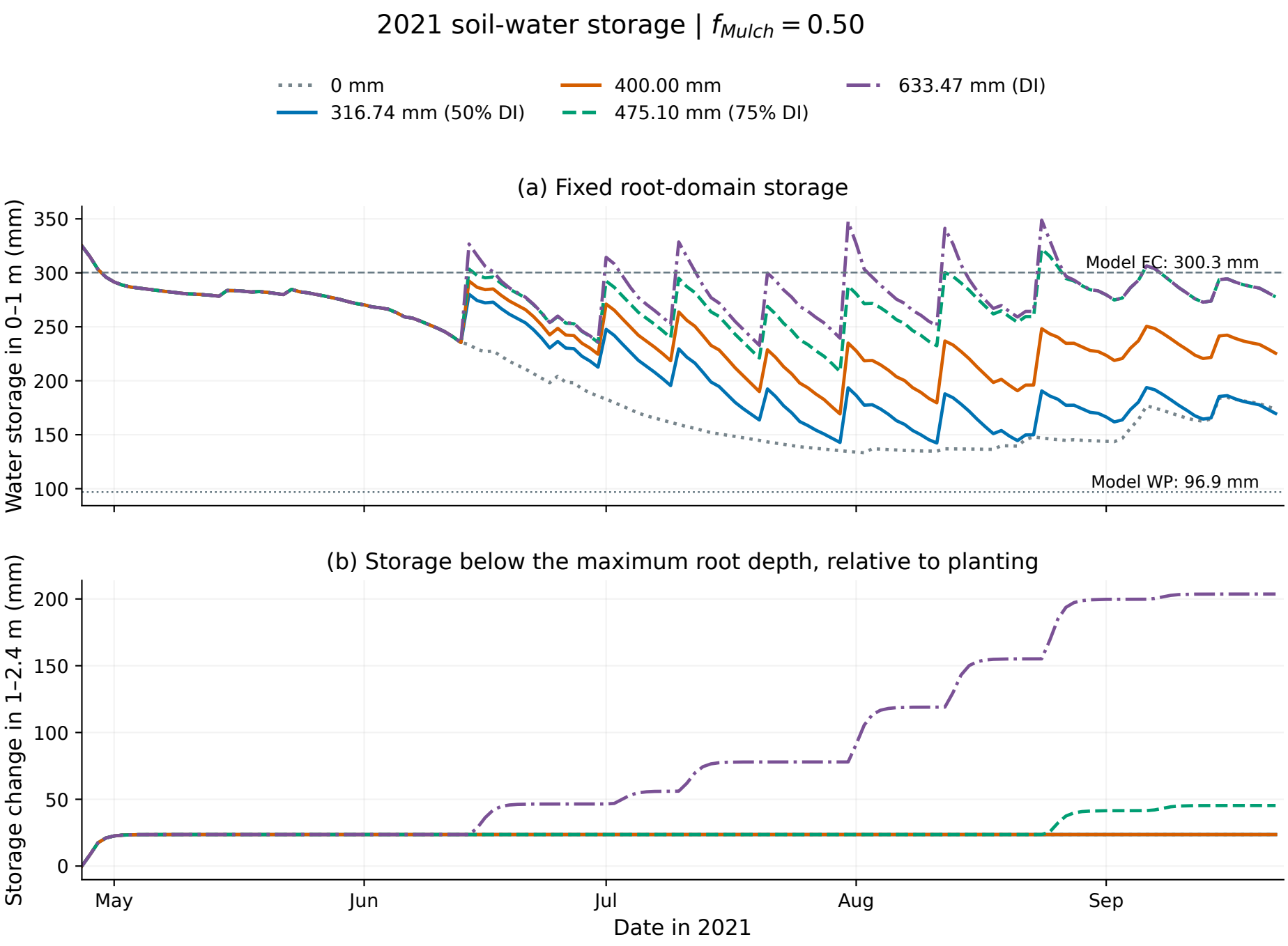


Root depth 1.00 m; original dates and proportions retained; open circles: zero irrigation; stars: DI model baselines.

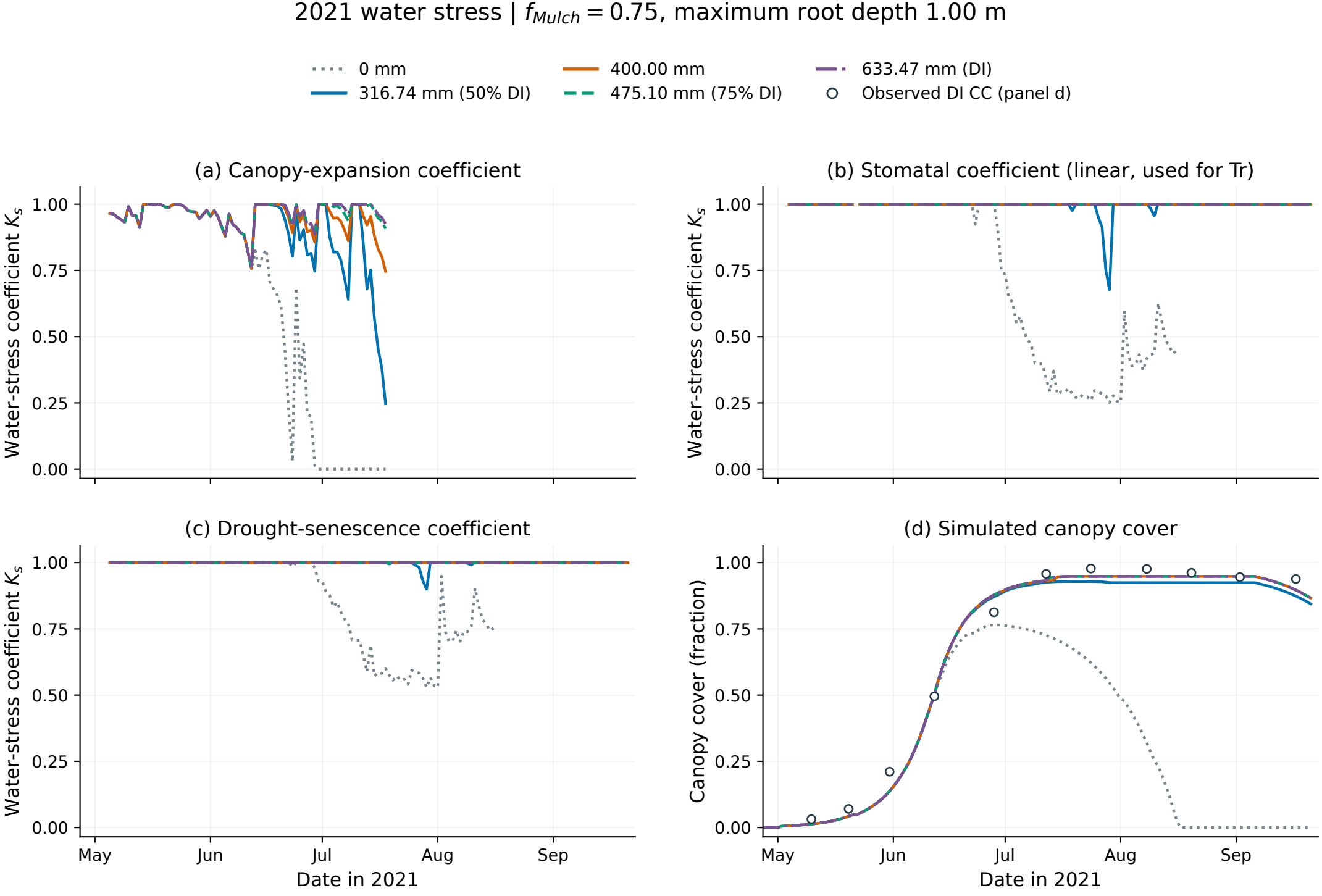


Short-pilot intervals are not a converged posterior. Dots in panel (a): highest sampled log posterior after step 250.

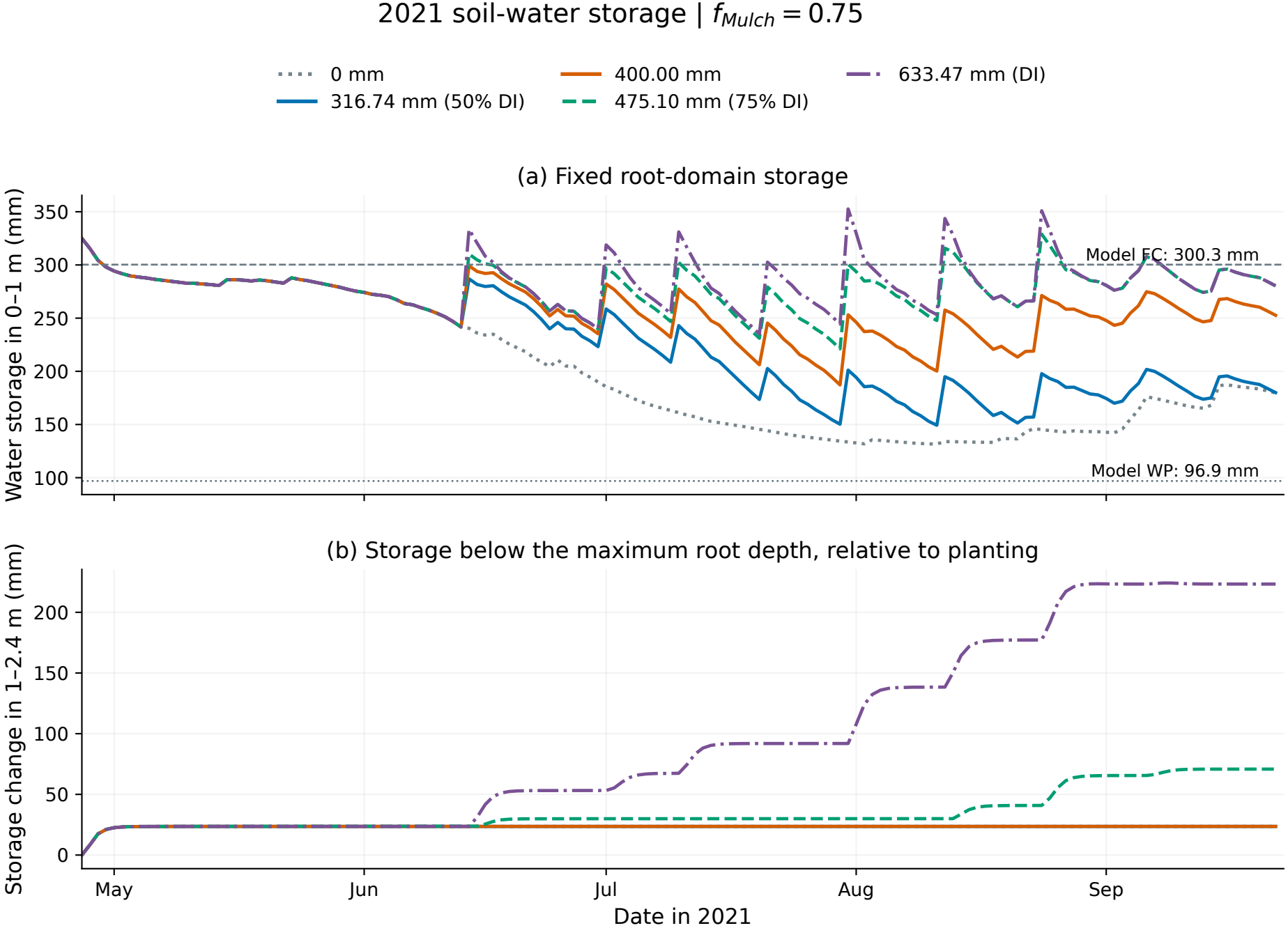

2021 water stress | $f_{Mulch} = 0.50$, maximum root depth 1.00 m
0 mm
316.74 mm (50% DI)
400.00 mm
475.10 mm (75% DI)
633.47 mm (DI)
Observed DI CC (panel d)
(a) Canopy-expansion coefficient
(b) Stomatal coefficient (linear, used for Tr)
(c) Drought-senescence coefficient
(d) Simulated canopy cover
Water-stress coefficient K_s
Canopy cover (fraction)
1.00
0.75
0.50
0.25
0.00
May
Jun
Jul
Aug
Sep
Date in 2021
$K_s = 1$: no drought limitation; lower K_s: stronger limitation. Missing process windows remain blank.



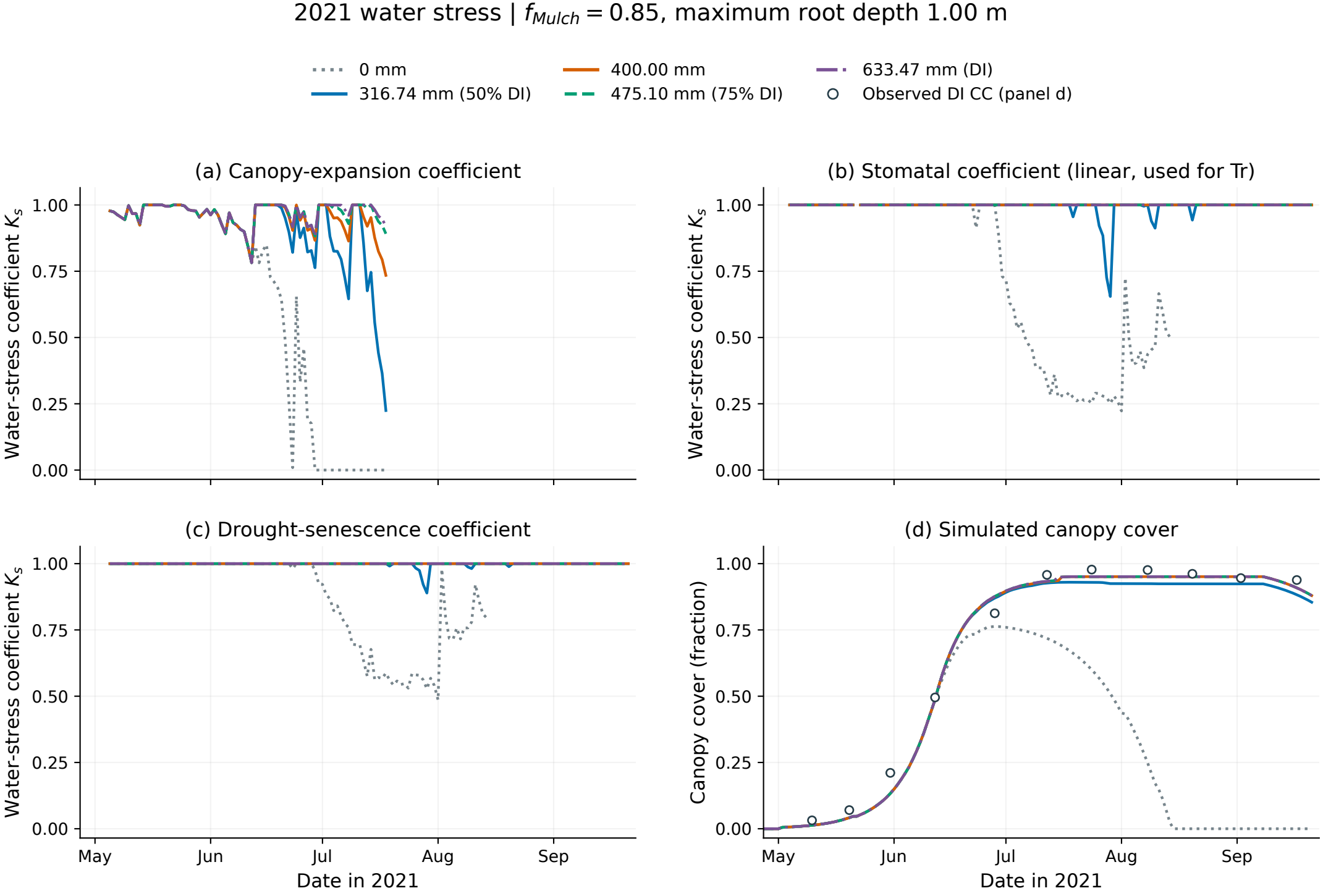
Fixed 0–1 m storage is not fully accessible to growing roots from planting. FC/WP and the initial state are model inputs.



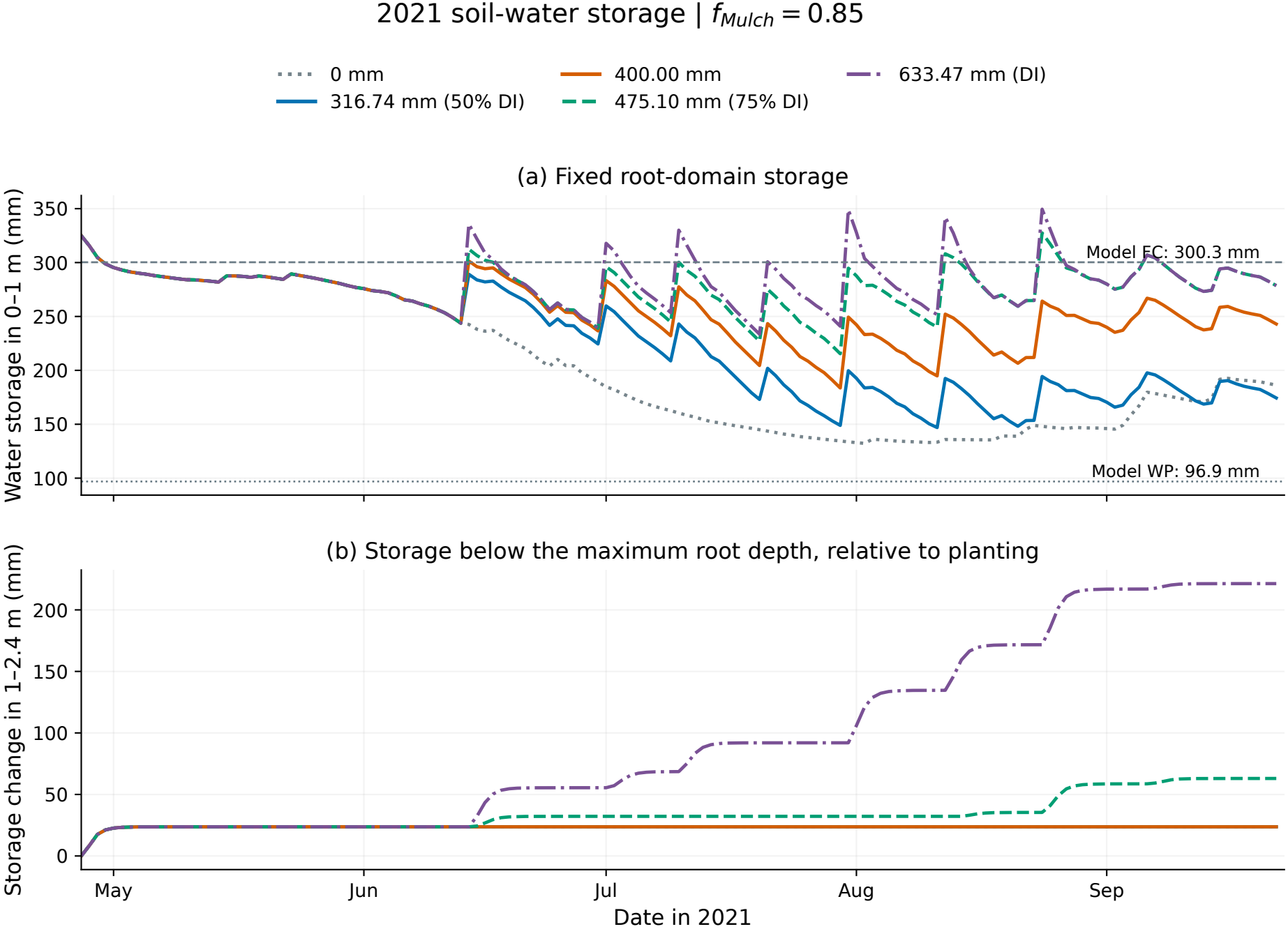
$K_s = 1$: no drought limitation; lower K_s: stronger limitation. Missing process windows remain blank.



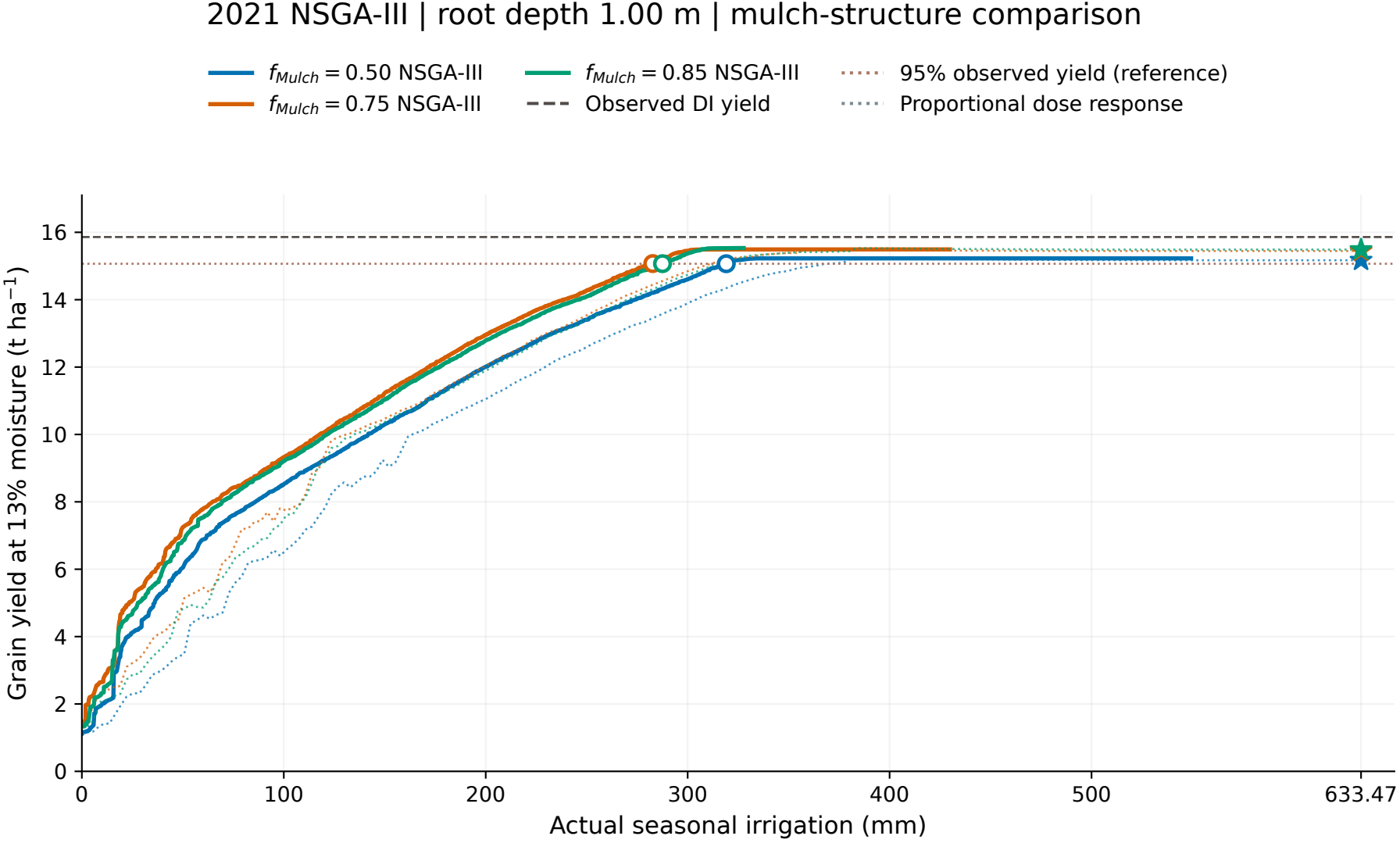
Fixed 0–1 m storage is not fully accessible to growing roots from planting. FC/WP and the initial state are model inputs.



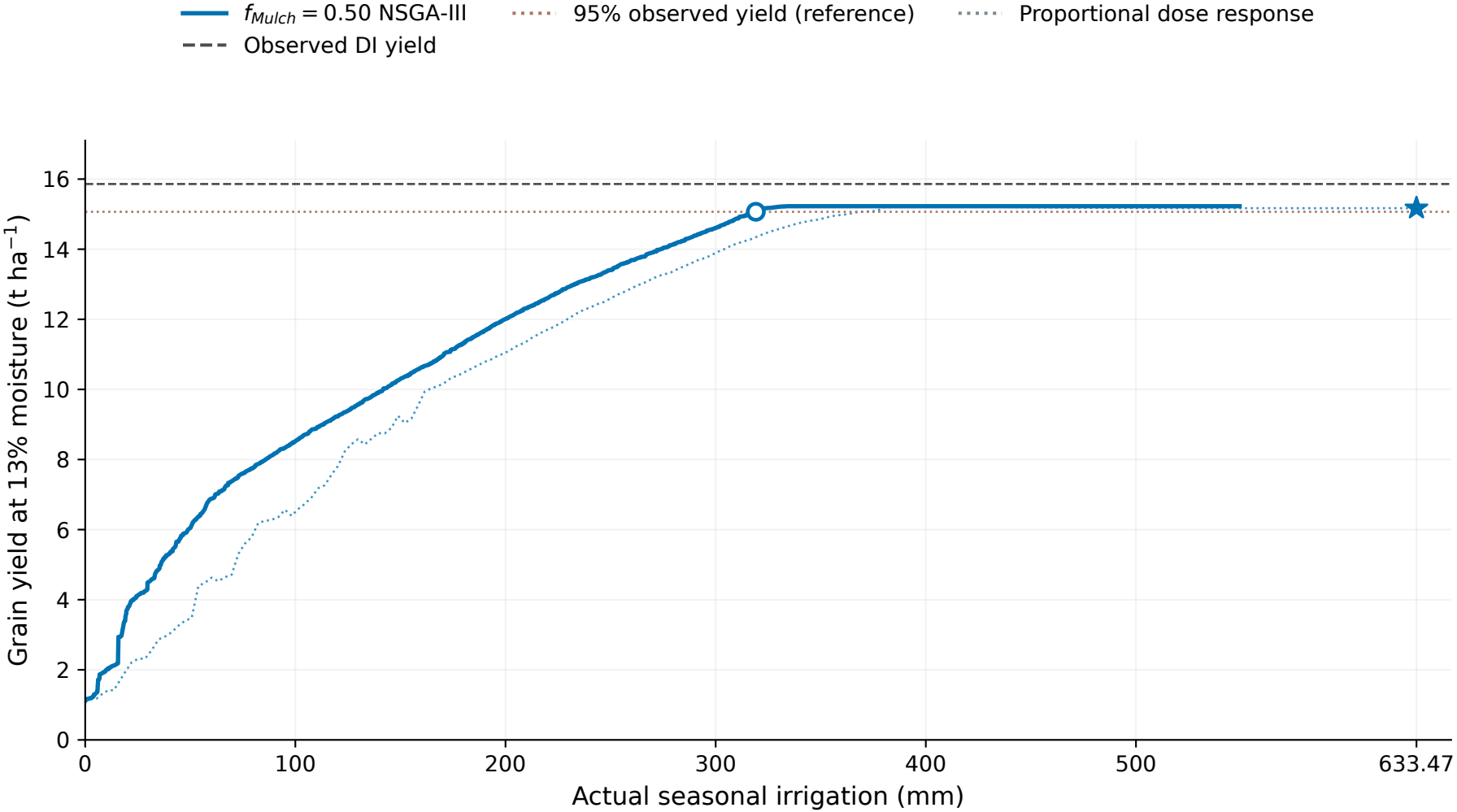
$K_s = 1$: no drought limitation; lower K_s: stronger limitation. Missing process windows remain blank.



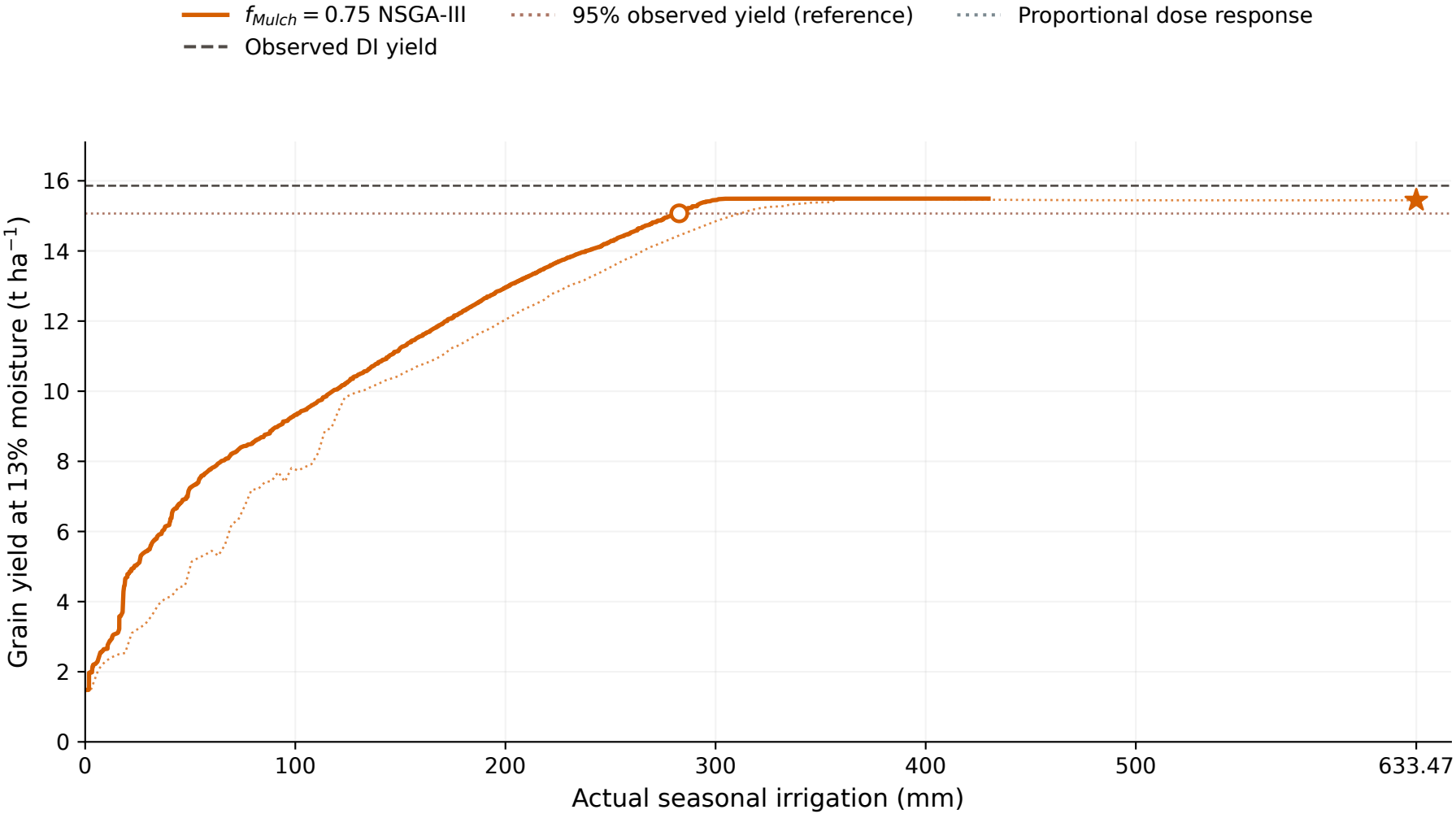
Fixed 0–1 m storage is not fully accessible to growing roots from planting. FC/WP and the initial state are model inputs.



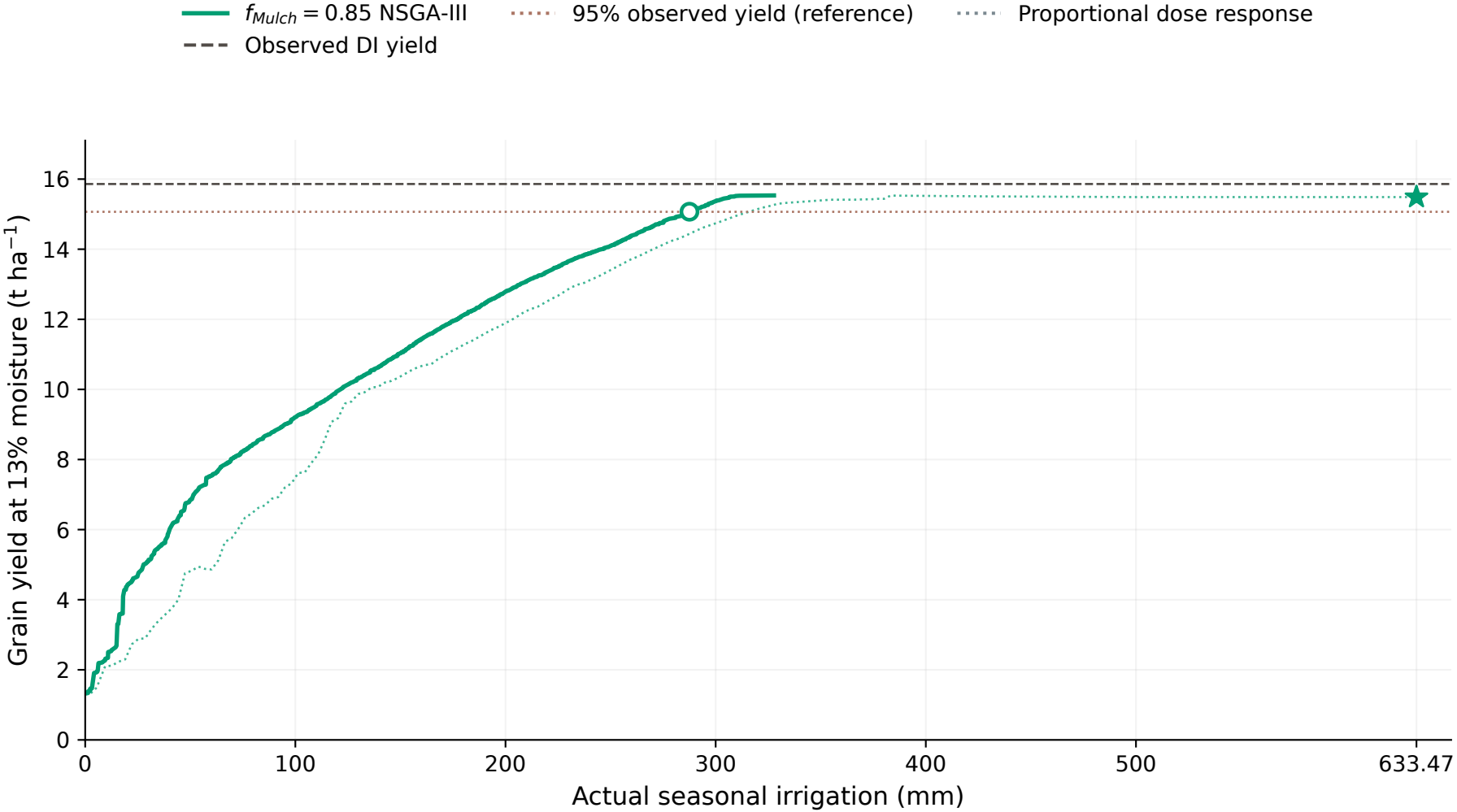
Each structure: 3 seeds × 100 individuals × 100 generations; fixed pilot parameters. Stars: own DI model baselines.



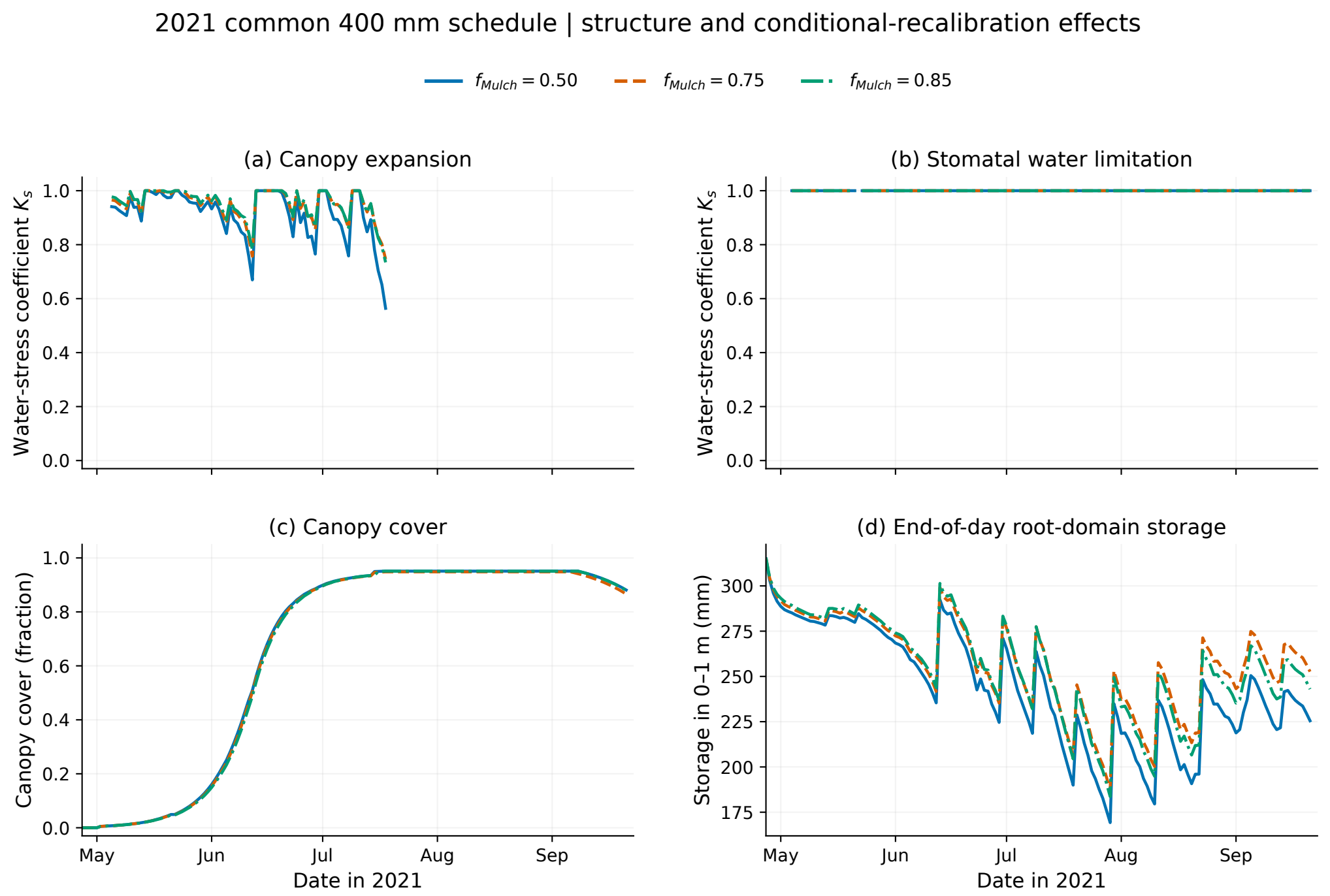
Each structure: 3 seeds × 100 individuals × 100 generations; fixed pilot parameters. Stars: own DI model baselines.

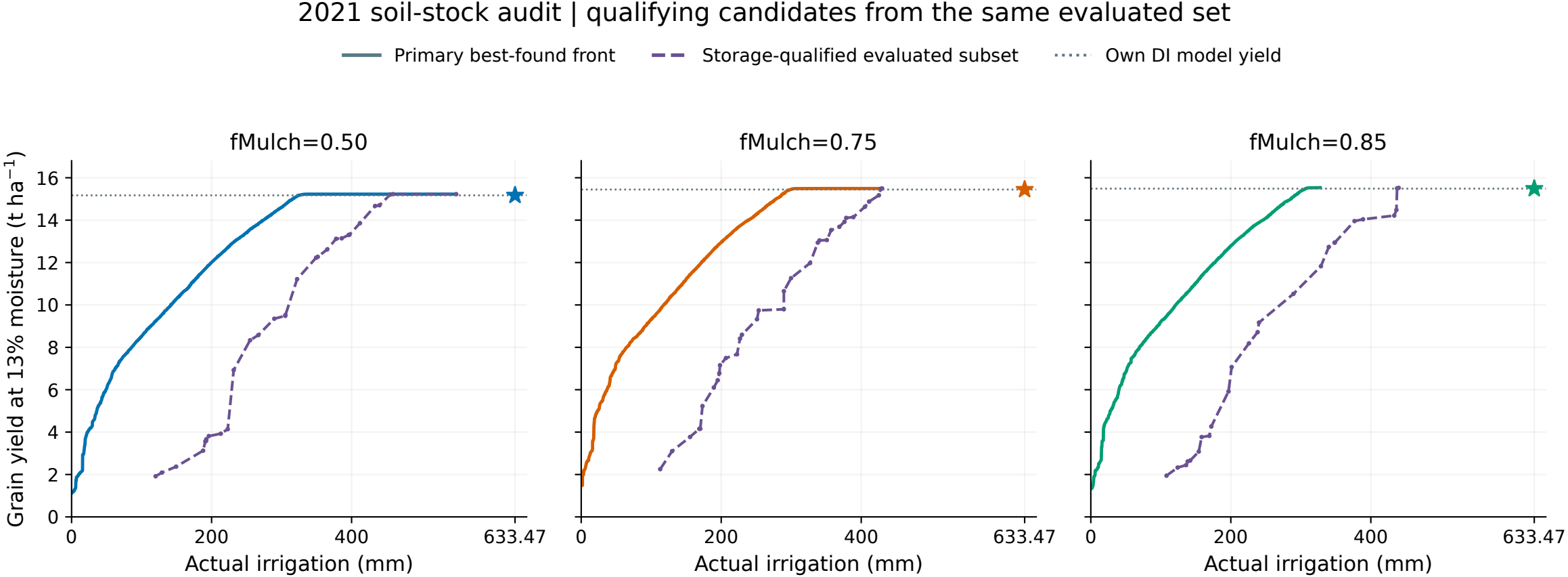


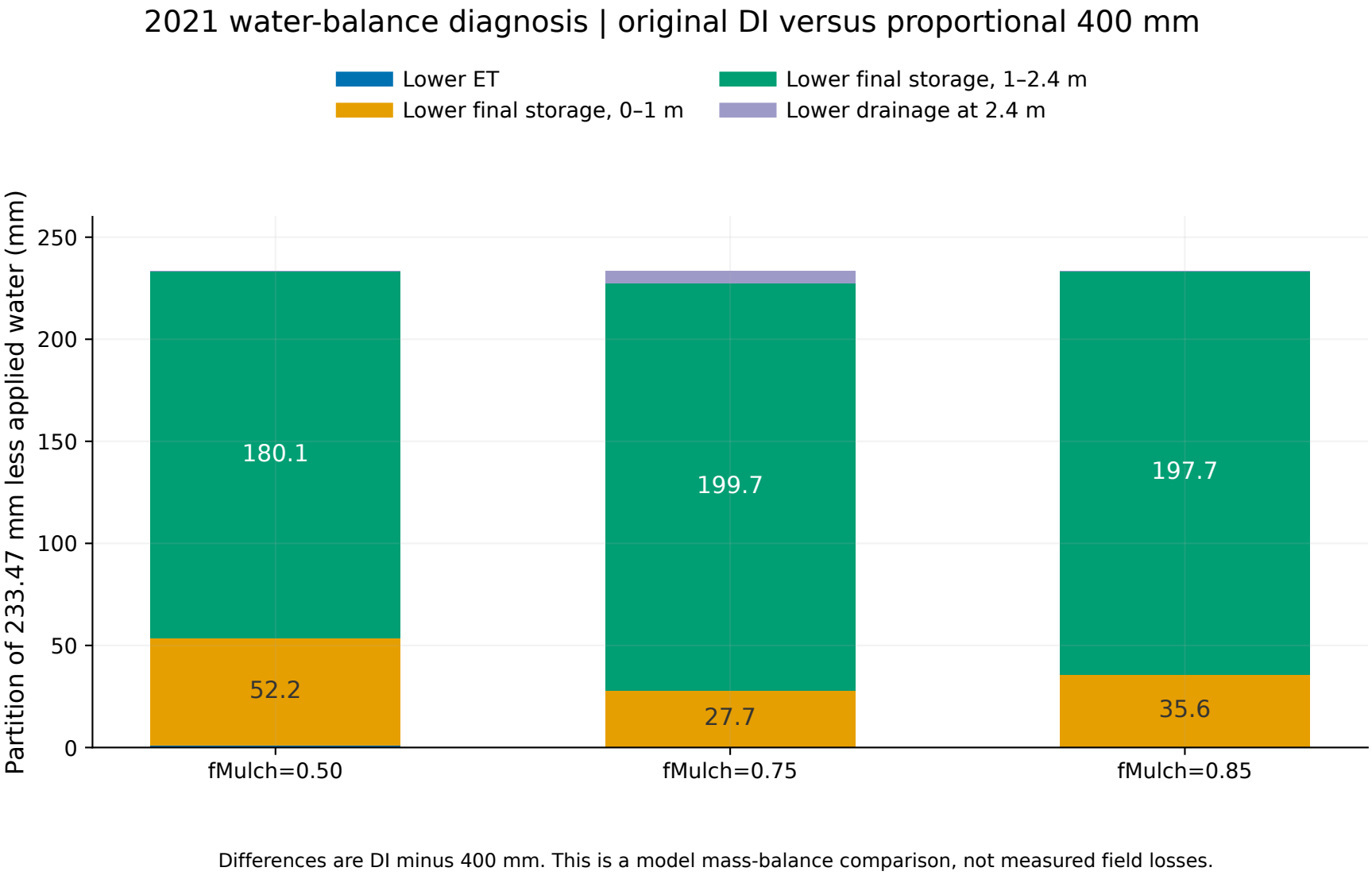
Each structure: 3 seeds × 100 individuals × 100 generations; fixed pilot parameters. Stars: own DI model baselines.

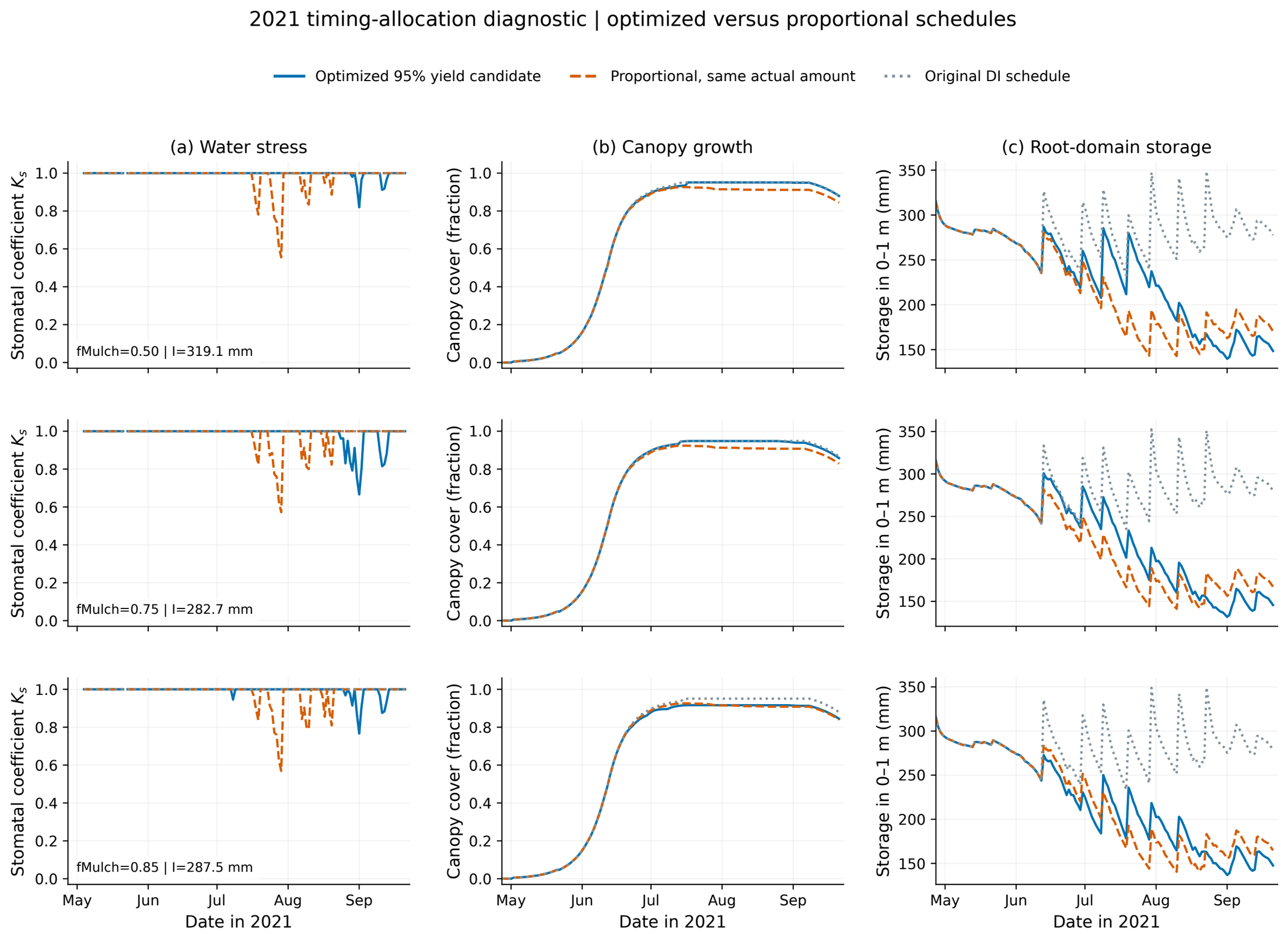


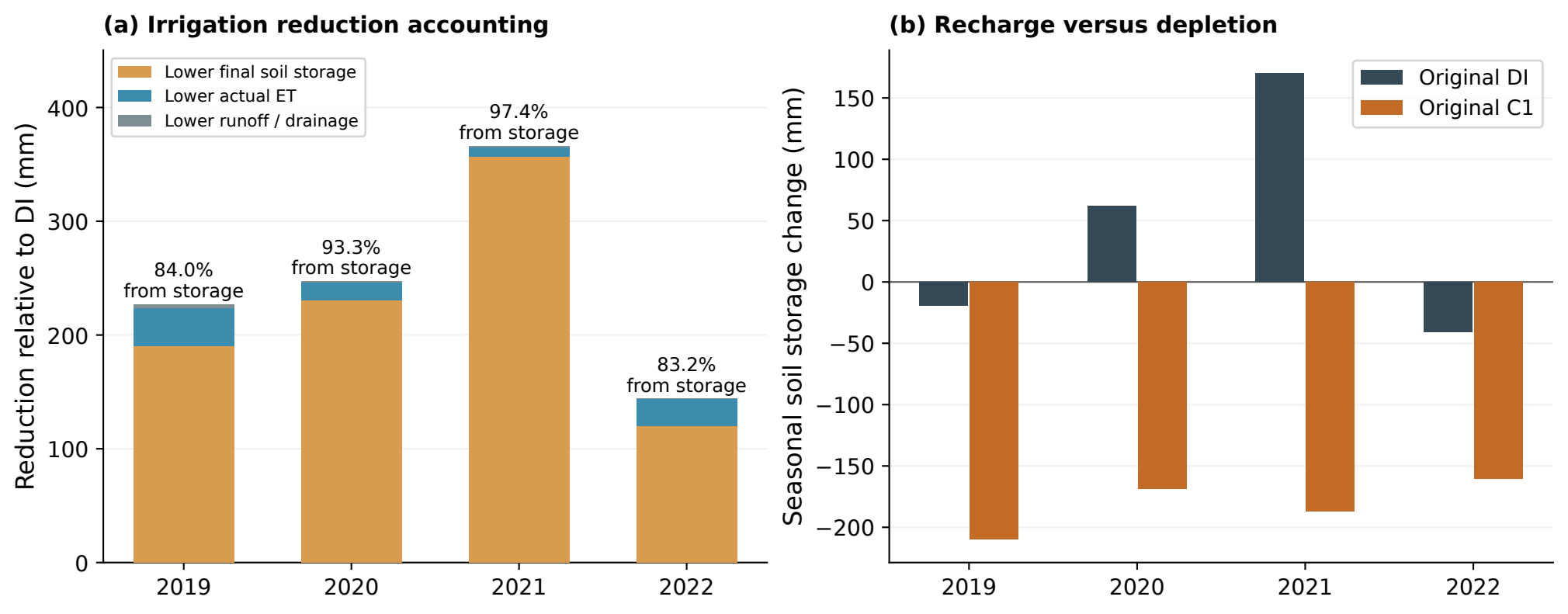
Each structure: 3 seeds × 100 individuals × 100 generations; fixed pilot parameters. Stars: own DI model baselines.

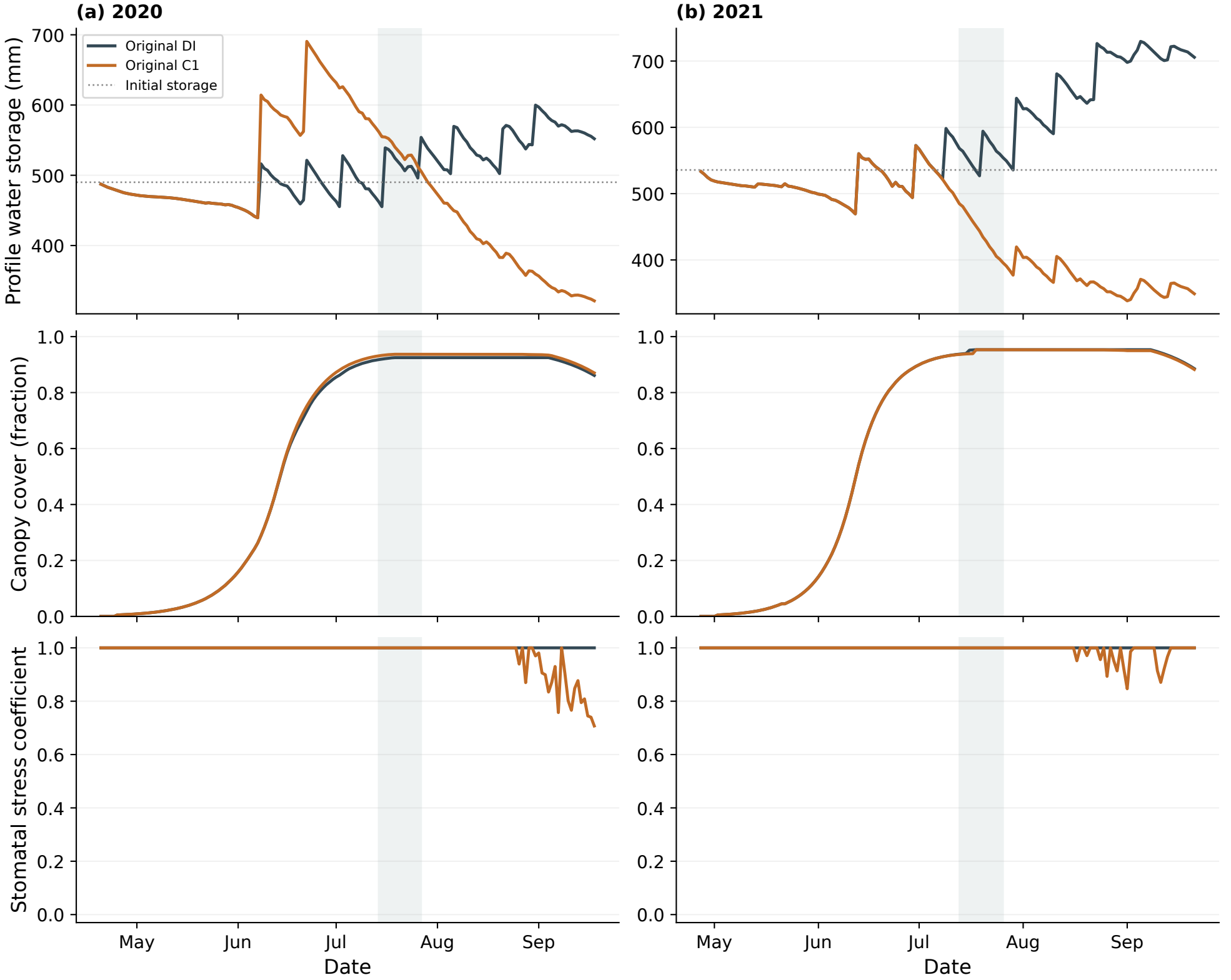

2021 common 400 mm schedule | structure and conditional-recalibration effects
$f_{Mulch} = 0.50$
$f_{Mulch} = 0.75$
$f_{Mulch} = 0.85$
(a) Canopy expansion
(b) Stomatal water limitation
(c) Canopy cover
(d) End-of-day root-domain storage
Water-stress coefficient K_s
Canopy cover (fraction)
Storage in 0–1 m (mm)
May
Jun
Jul
Aug
Sep
Date in 2021
No deficit-treatment observations are available here; these are model diagnostic comparisons.

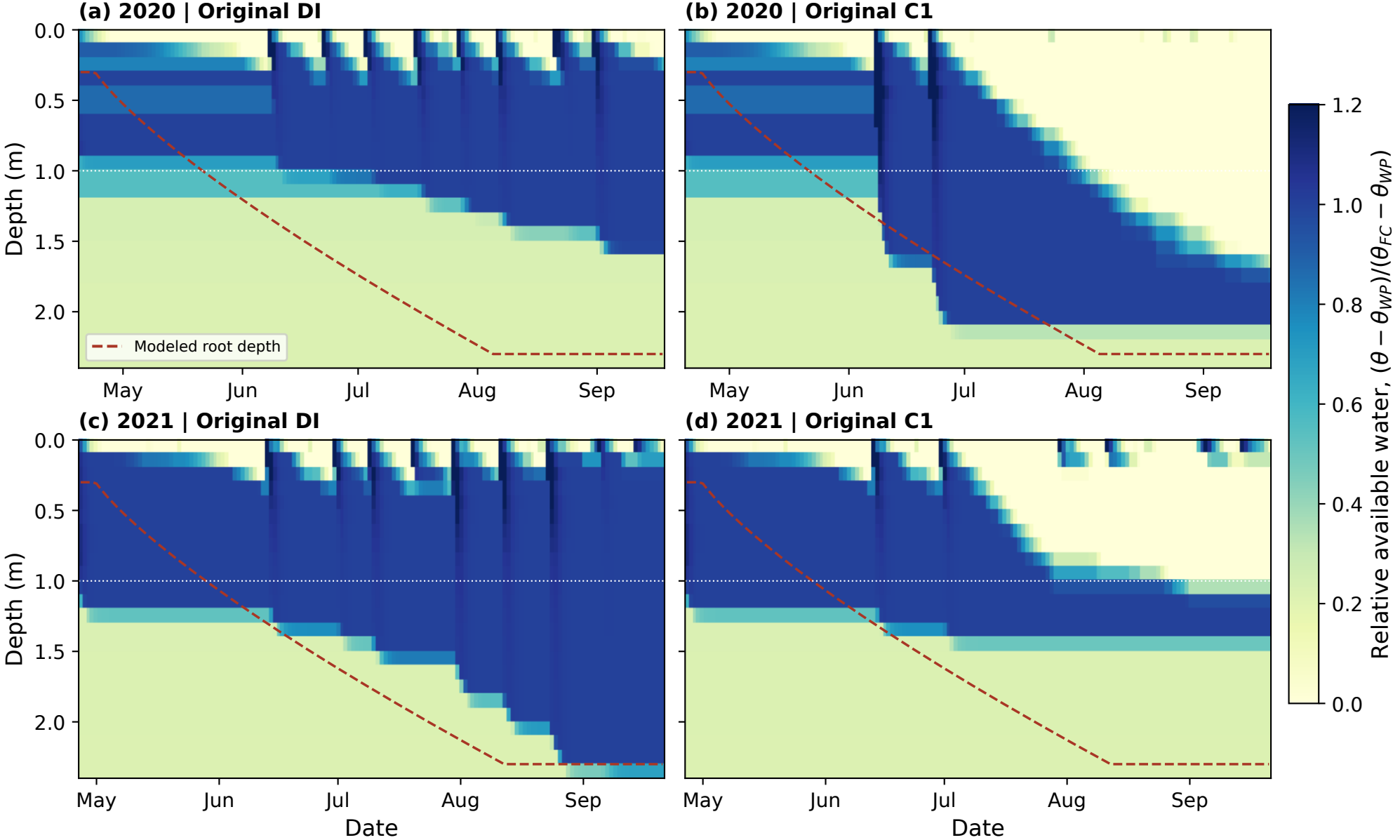

2021 soil-stock audit | qualifying candidates from the same evaluated set
Primary best-found front
Storage-qualified evaluated subset
Own DI model yield
fMulch=0.50
fMulch=0.75
fMulch=0.85
Grain yield at 13% moisture (t ha^{-1})
0
2
4
6
8
10
12
14
16
0
200
400
633.47
Actual irrigation (mm)
Subset: final 0–1 m storage >= own DI minus 0.01 mm; not a dedicated constrained optimization.

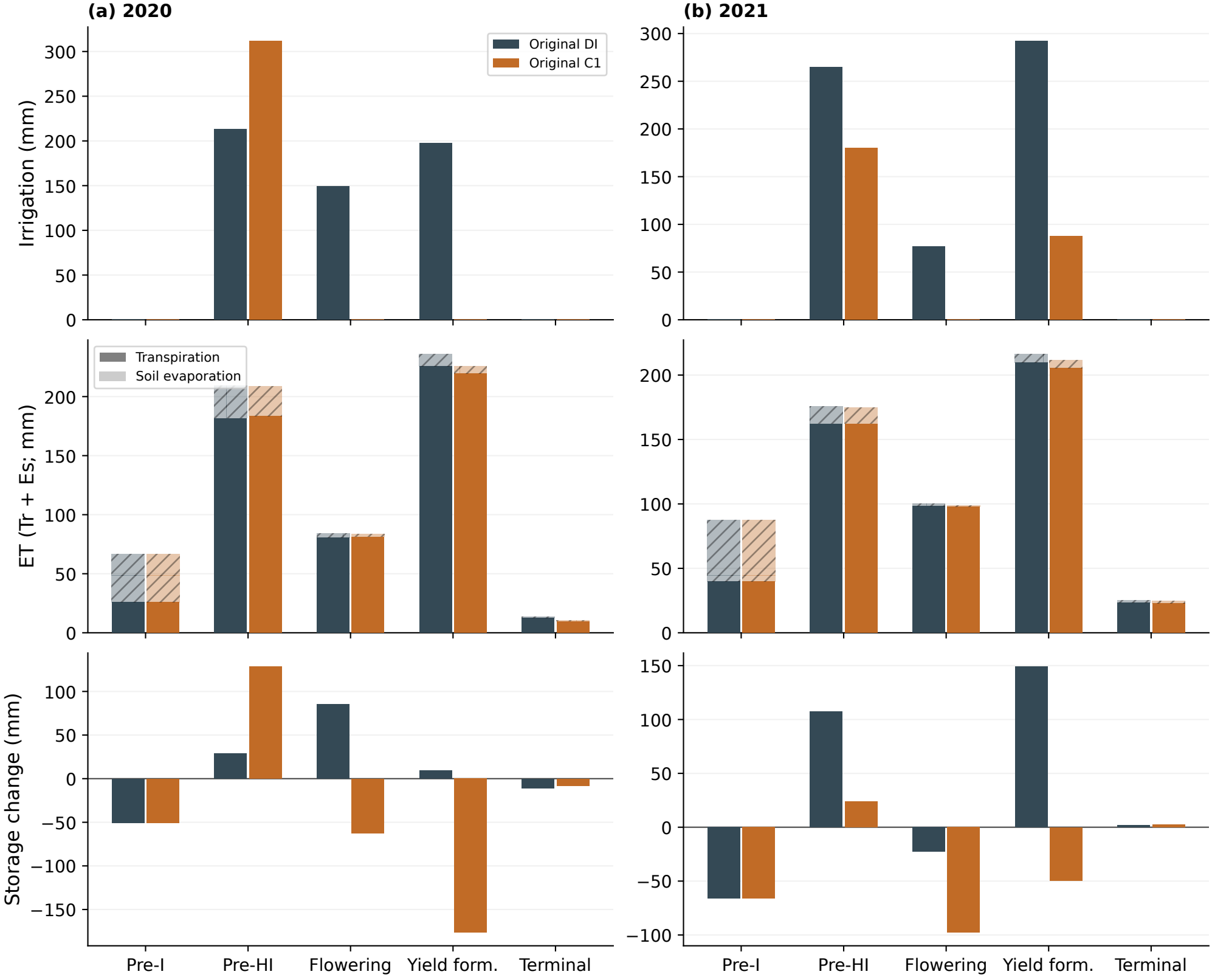

2021 water-balance diagnosis | original DI versus proportional 400 mm
Lower ET
Lower final storage, 0–1 m
Lower final storage, 1–2.4 m
Lower drainage at 2.4 m
Partition of 233.47 mm less applied water (mm)
250
200
150
100
50
0
180.1
52.2
199.7
27.7
197.7
35.6
fMulch=0.50
fMulch=0.75
fMulch=0.85
Differences are DI minus 400 mm. This is a model mass-balance comparison, not measured field losses.

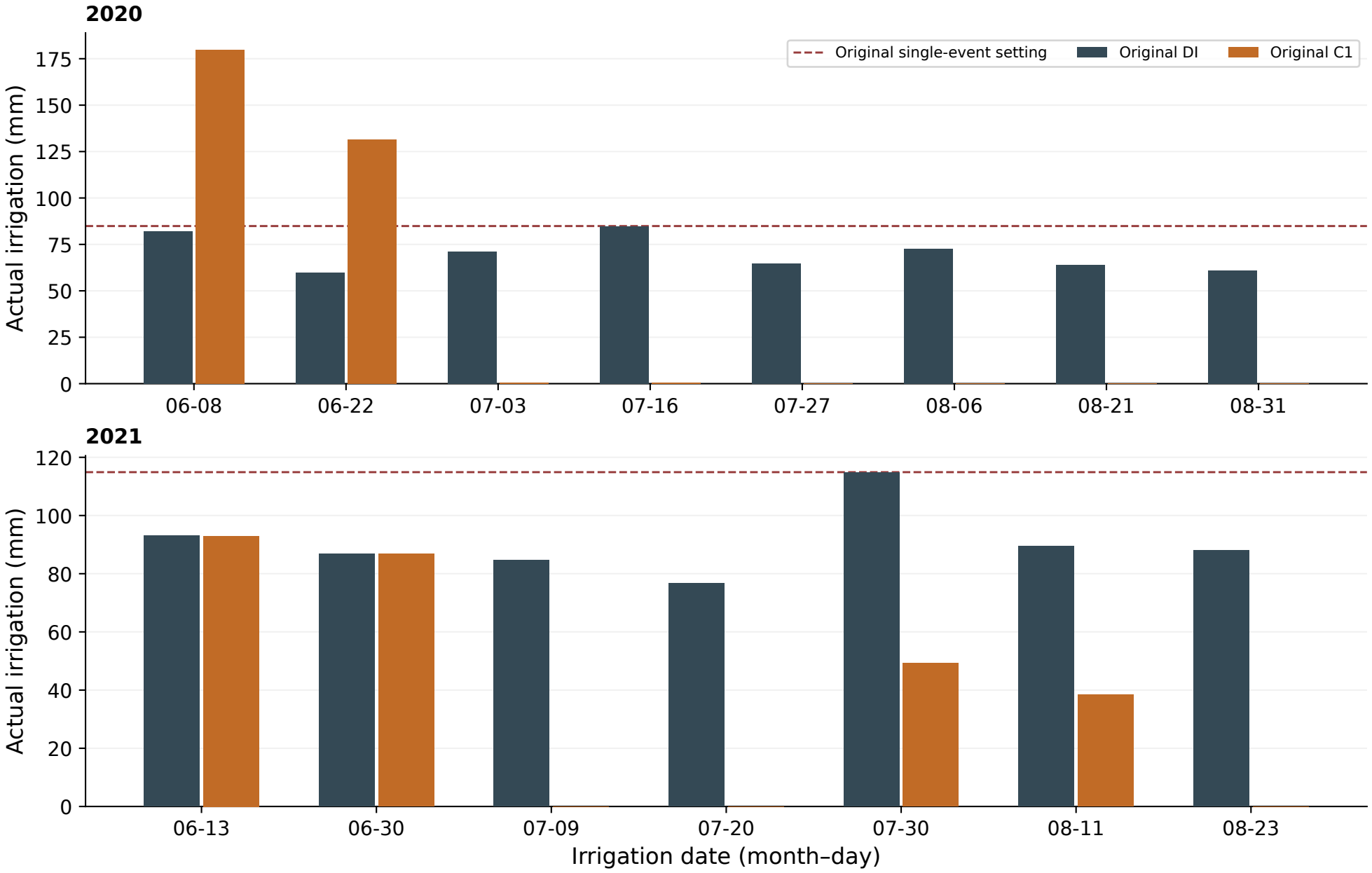

2021 timing-allocation diagnostic | optimized versus proportional schedules
Optimized 95% yield candidate
Proportional, same actual amount
Original DI schedule
(a) Water stress
(b) Canopy growth
(c) Root-domain storage
Stomatal coefficient K_s
Canopy cover (fraction)
Storage in 0–1 m (mm)
fMulch=0.50 | I=319.1 mm
fMulch=0.75 | I=282.7 mm
fMulch=0.85 | I=287.5 mm
May
Jun
Jul
Aug
Sep
Date in 2021
Dates remain fixed. Only the amounts in the four original irrigation windows change. No deficit-treatment validation is implied.

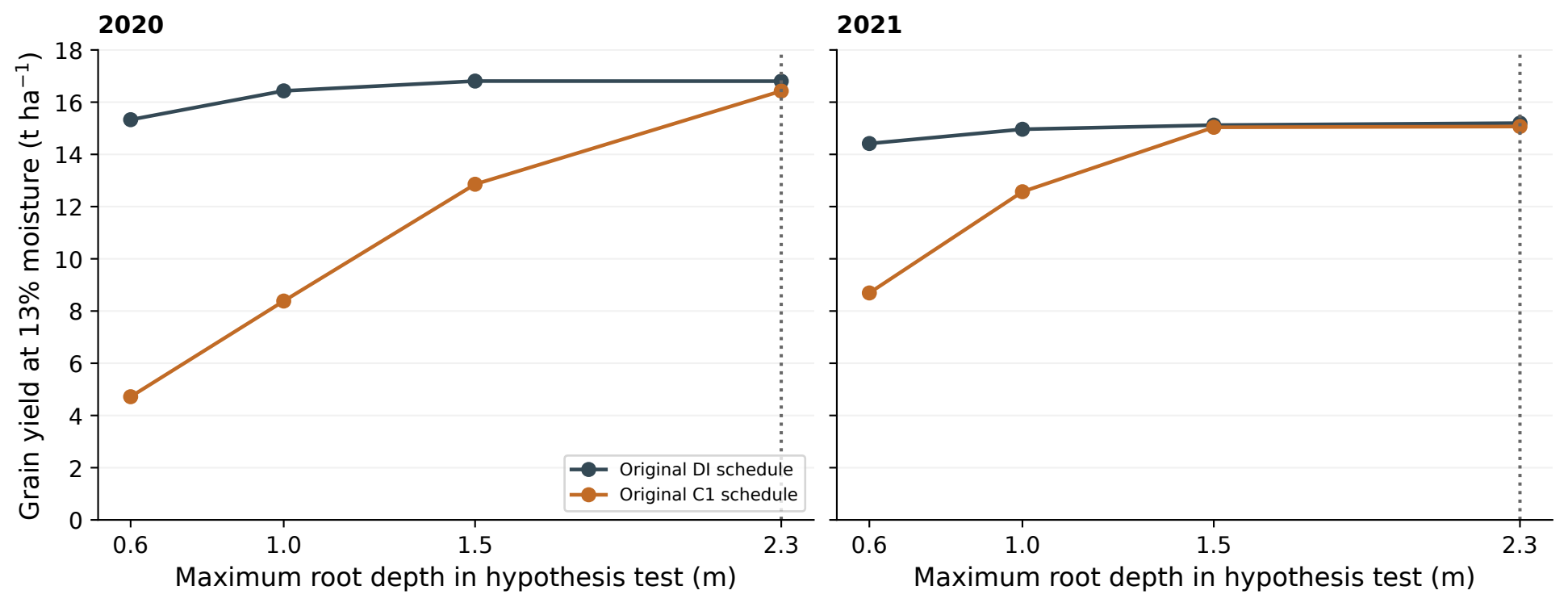

(a) Irrigation reduction accounting
Lower final soil storage
Lower actual ET
Lower runoff / drainage
Reduction relative to DI (mm)
0
100
200
300
400
84.0%
from storage
93.3%
from storage
97.4%
from storage
83.2%
from storage
2019
2020
2021
2022
(b) Recharge versus depletion
Original DI
Original C1
Seasonal soil storage change (mm)
150
100
50
0
−50
−100
−150
−200
2019
2020
2021
2022

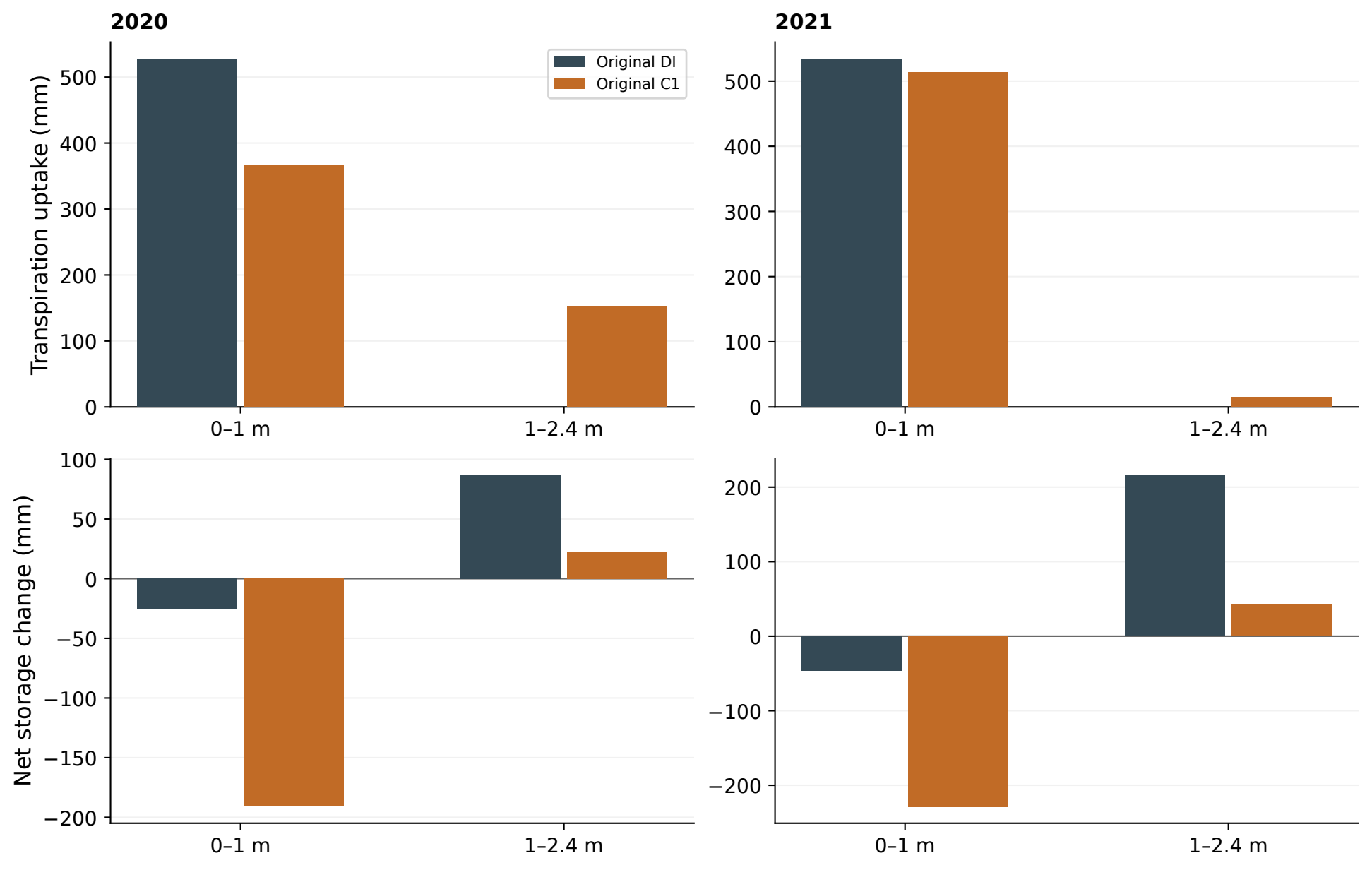

(a) 2020
(b) 2021
Original DI
Original C1
Initial storage
Profile water storage (mm)
Canopy cover (fraction)
Stomatal stress coefficient
May
Jun
Jul
Aug
Sep
Date

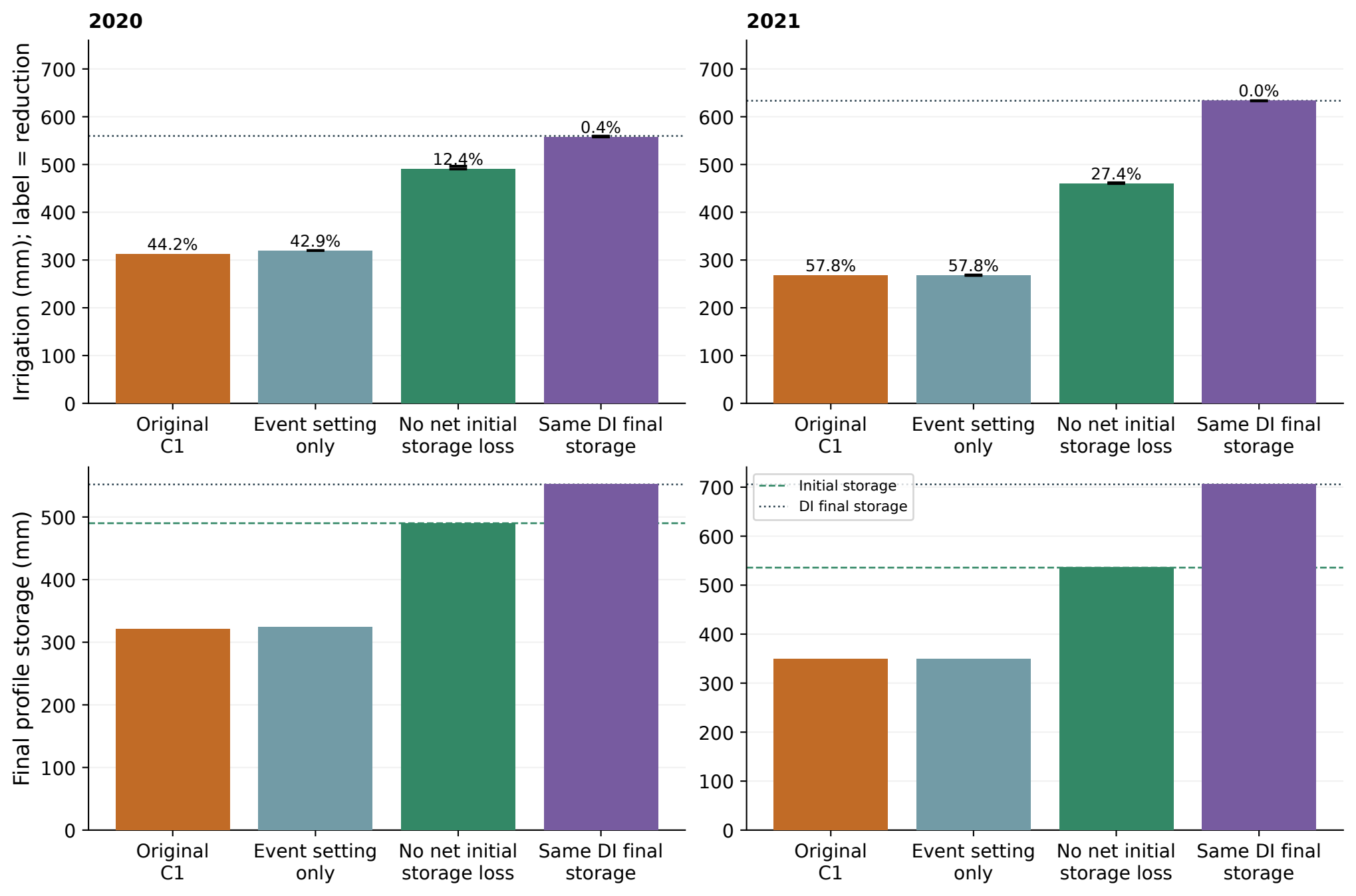

(a) 2020 | Original DI
(b) 2020 | Original C1
(c) 2021 | Original DI
(d) 2021 | Original C1
Depth (m)
0.0
0.5
1.0
1.5
2.0
Modeled root depth
May
Jun
Jul
Aug
Sep
Date
Relative available water, $(\theta - \theta_{WP})/(\theta_{FC} - \theta_{WP})$
1.2
1.0
0.8
0.6
0.4
0.2
0.0

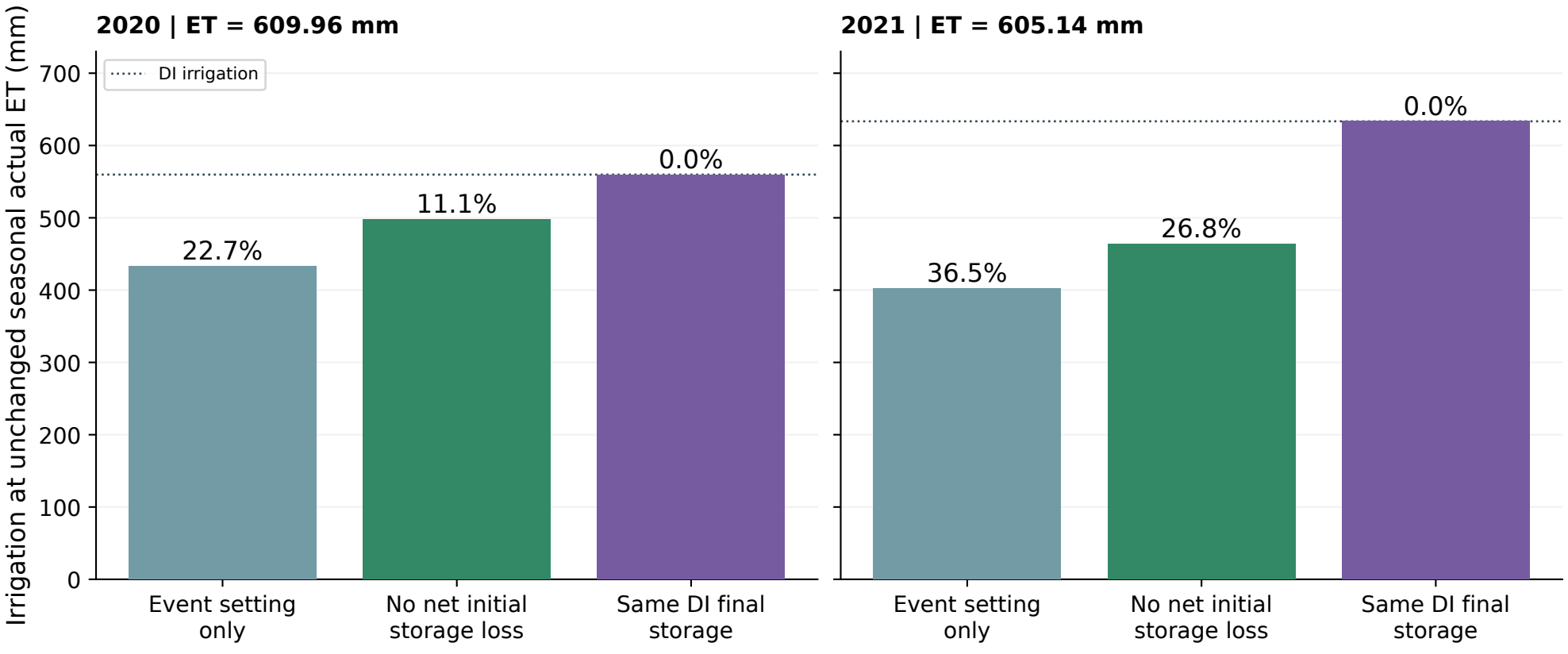

(a) 2020
(b) 2021
Original DI
Original C1
Transpiration
Soil evaporation
Irrigation (mm)
ET (Tr + Es; mm)
Storage change (mm)
Pre-I
Pre-HI
Flowering
Yield form.
Terminal

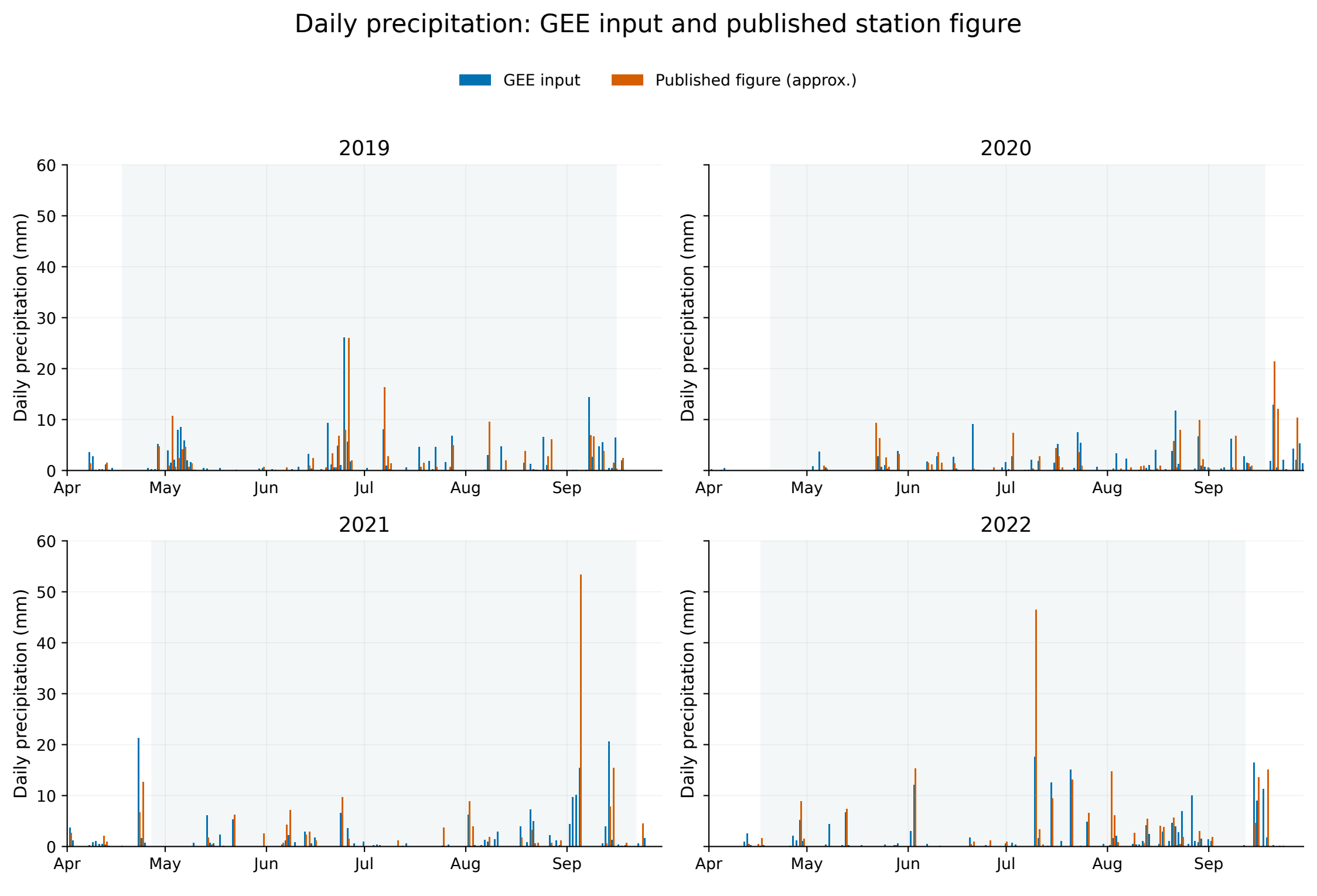

2020
Original single-event setting
Original DI
Original C1
Actual irrigation (mm)
0
25
50
75
100
125
150
175
06-08
06-22
07-03
07-16
07-27
08-06
08-21
08-31
2021
Actual irrigation (mm)
0
20
40
60
80
100
120
06-13
06-30
07-09
07-20
07-30
08-11
08-23
Irrigation date (month–day)

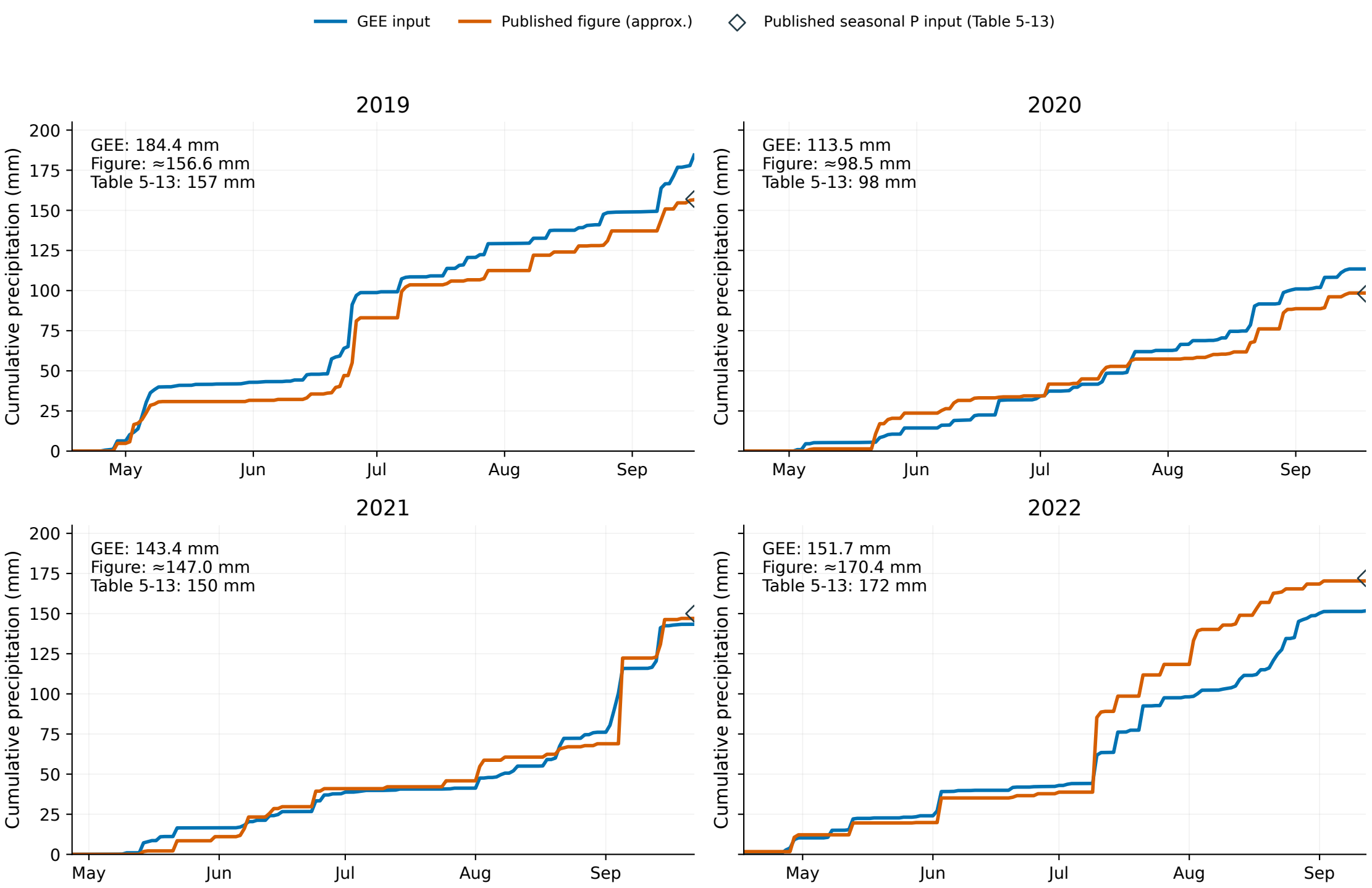

2020
2021
Grain yield at 13% moisture (t ha^{-1})
0
2
4
6
8
10
12
14
16
18
0.6
1.0
1.5
2.3
Maximum root depth in hypothesis test (m)
Original DI schedule
Original C1 schedule

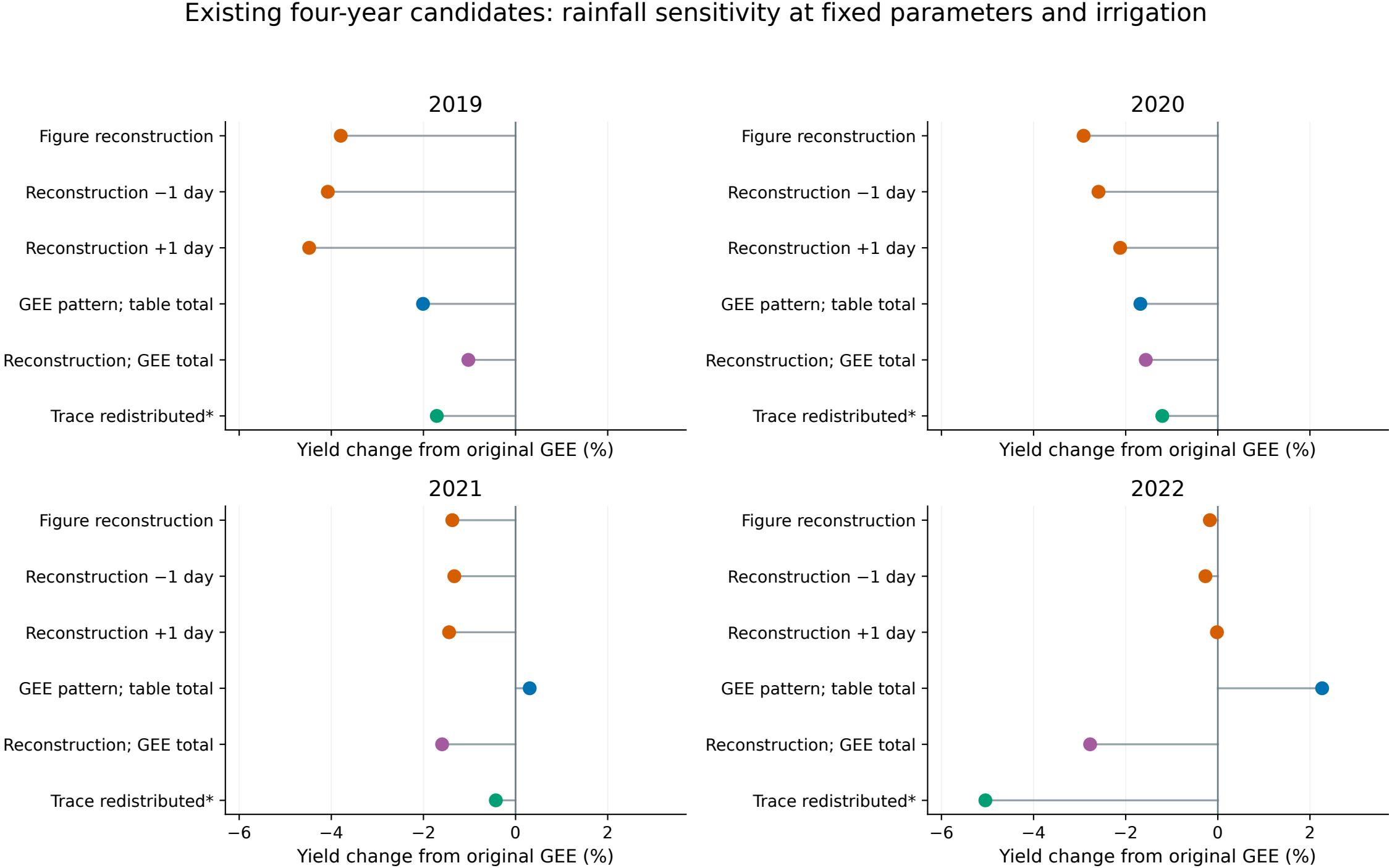

2020
2021
Original DI
Original C1
Transpiration uptake (mm)
Net storage change (mm)
0
100
200
300
400
500
−50
−100
−150
−200
50
0–1 m
1–2.4 m

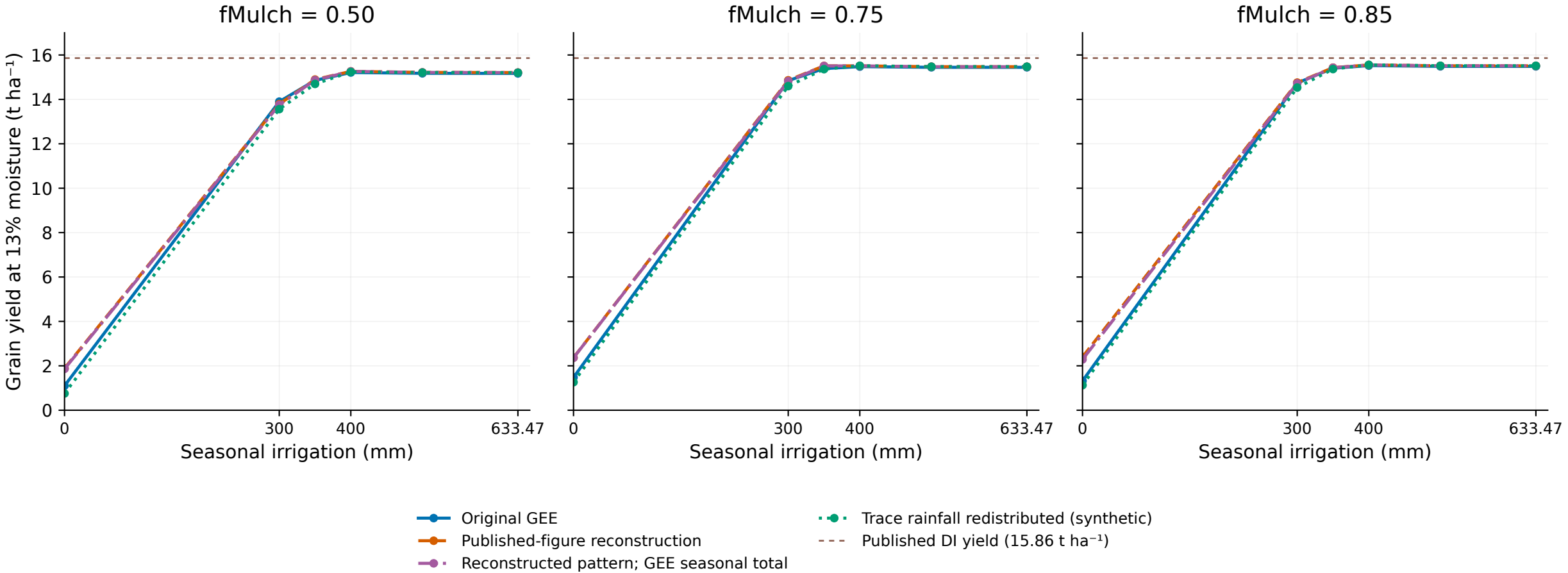

2020
2021
Irrigation (mm); label = reduction
Final profile storage (mm)
0
100
200
300
400
500
600
700
44.2%
42.9%
12.4%
0.4%
57.8%
57.8%
27.4%
0.0%
Initial storage
DI final storage
Original C1
Event setting only
No net initial storage loss
Same DI final storage

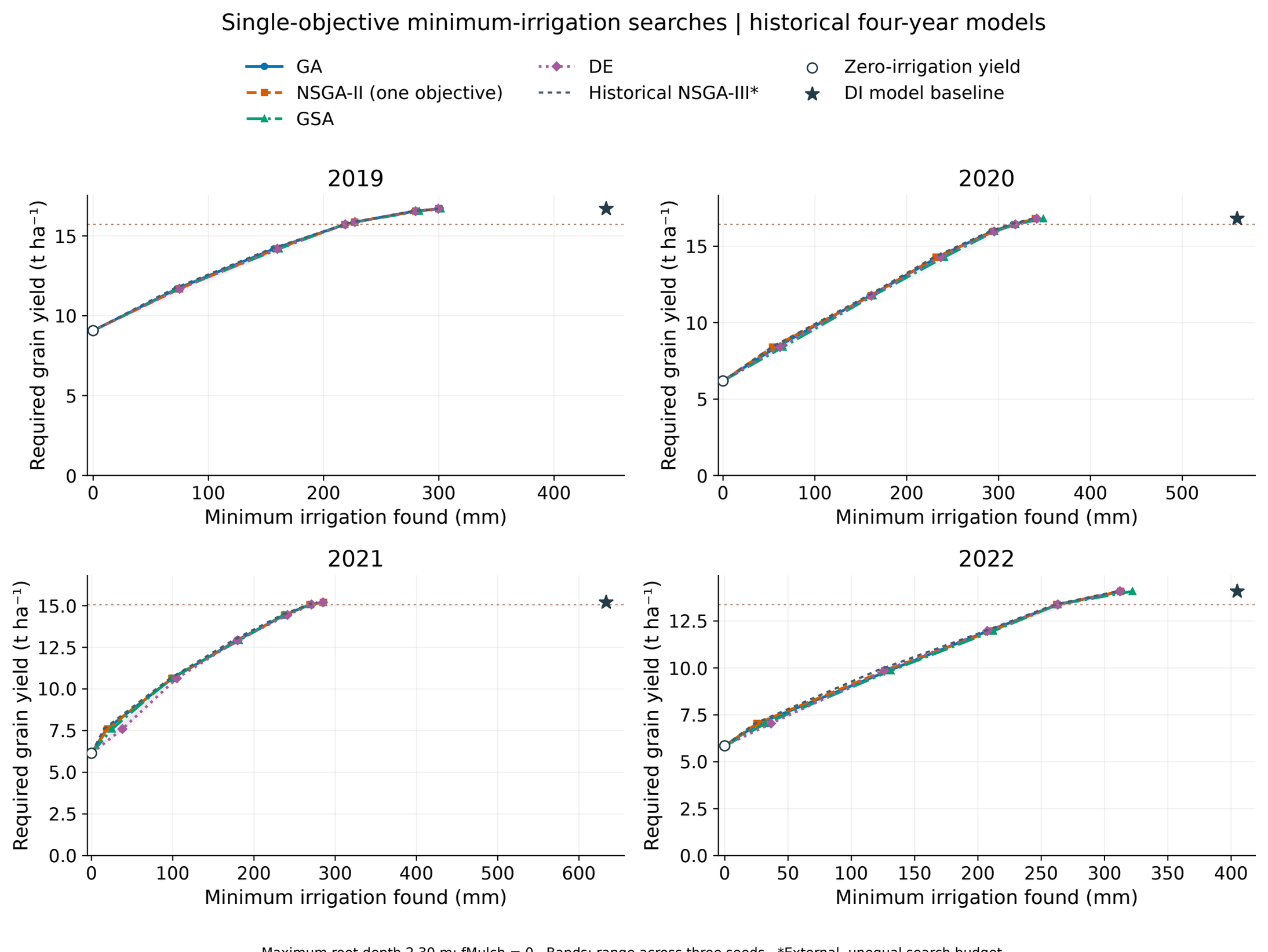

2020 | ET = 609.96 mm
2021 | ET = 605.14 mm
Irrigation at unchanged seasonal actual ET (mm)
DI irrigation
0
100
200
300
400
500
600
700
22.7%
11.1%
0.0%
36.5%
26.8%
0.0%
Event setting only
No net initial storage loss
Same DI final storage
Event setting only
No net initial storage loss
Same DI final storage

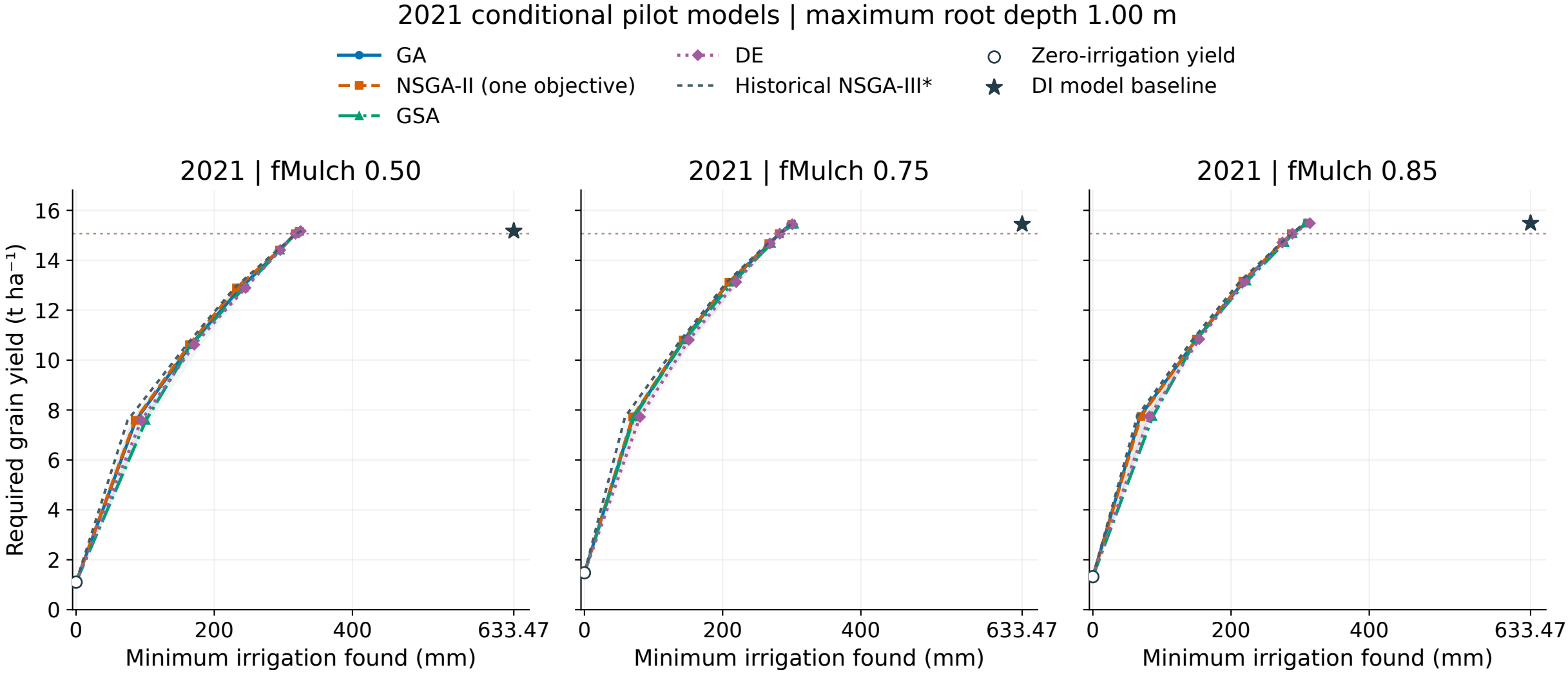

Daily precipitation: GEE input and published station figure
GEE input
Published figure (approx.)
2019
2020
2021
2022
Daily precipitation (mm)
0
10
20
30
40
50
60
Apr
May
Jun
Jul
Aug
Sep
Shading: published DI crop period. Graphic dates are approximate (±1 day); small bars may be unresolved.


Matched-period cumulative precipitation
GEE input
Published figure (approx.)
Published seasonal P input (Table 5-13)
2019
GEE: 184.4 mm
Figure: ≈156.6 mm
Table 5-13: 157 mm
2020
GEE: 113.5 mm
Figure: ≈98.5 mm
Table 5-13: 98 mm
2021
GEE: 143.4 mm
Figure: ≈147.0 mm
Table 5-13: 150 mm
2022
GEE: 151.7 mm
Figure: ≈170.4 mm
Table 5-13: 172 mm
Cumulative precipitation (mm)
0
25
50
75
100
125
150
175
200
May
Jun
Jul
Aug
Sep
All three values use the published DI sowing–harvest period, including the harvest day.

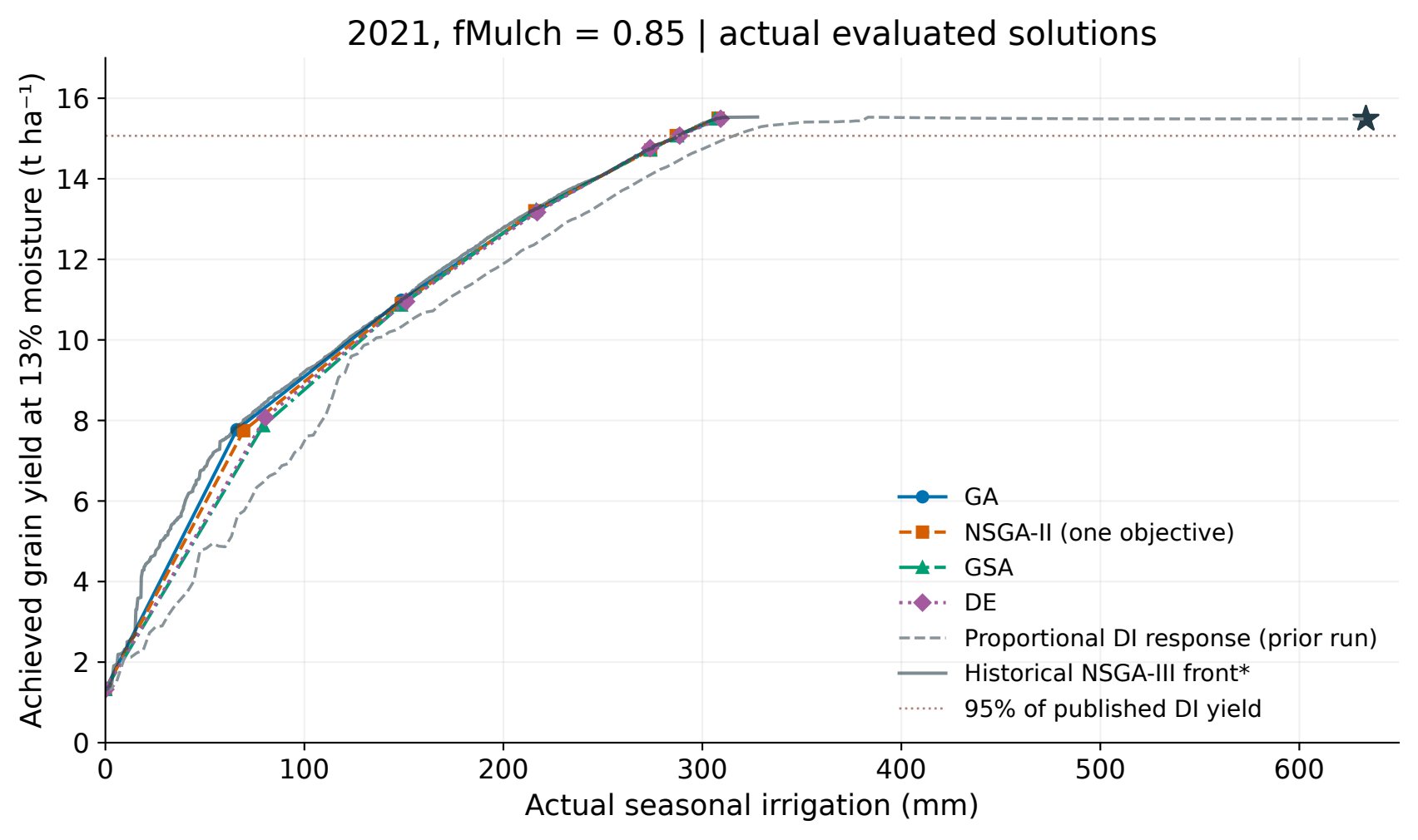

Existing four-year candidates: rainfall sensitivity at fixed parameters and irrigation
2019
2020
2021
2022
Figure reconstruction
Reconstruction −1 day
Reconstruction +1 day
GEE pattern; table total
Reconstruction; GEE total
Trace redistributed*
−6
−4
−2
0
2
Yield change from original GEE (%)
Historical model: maximum root depth 2.30 m, fMulch = 0. *Synthetic redistribution; not a station-data correction.

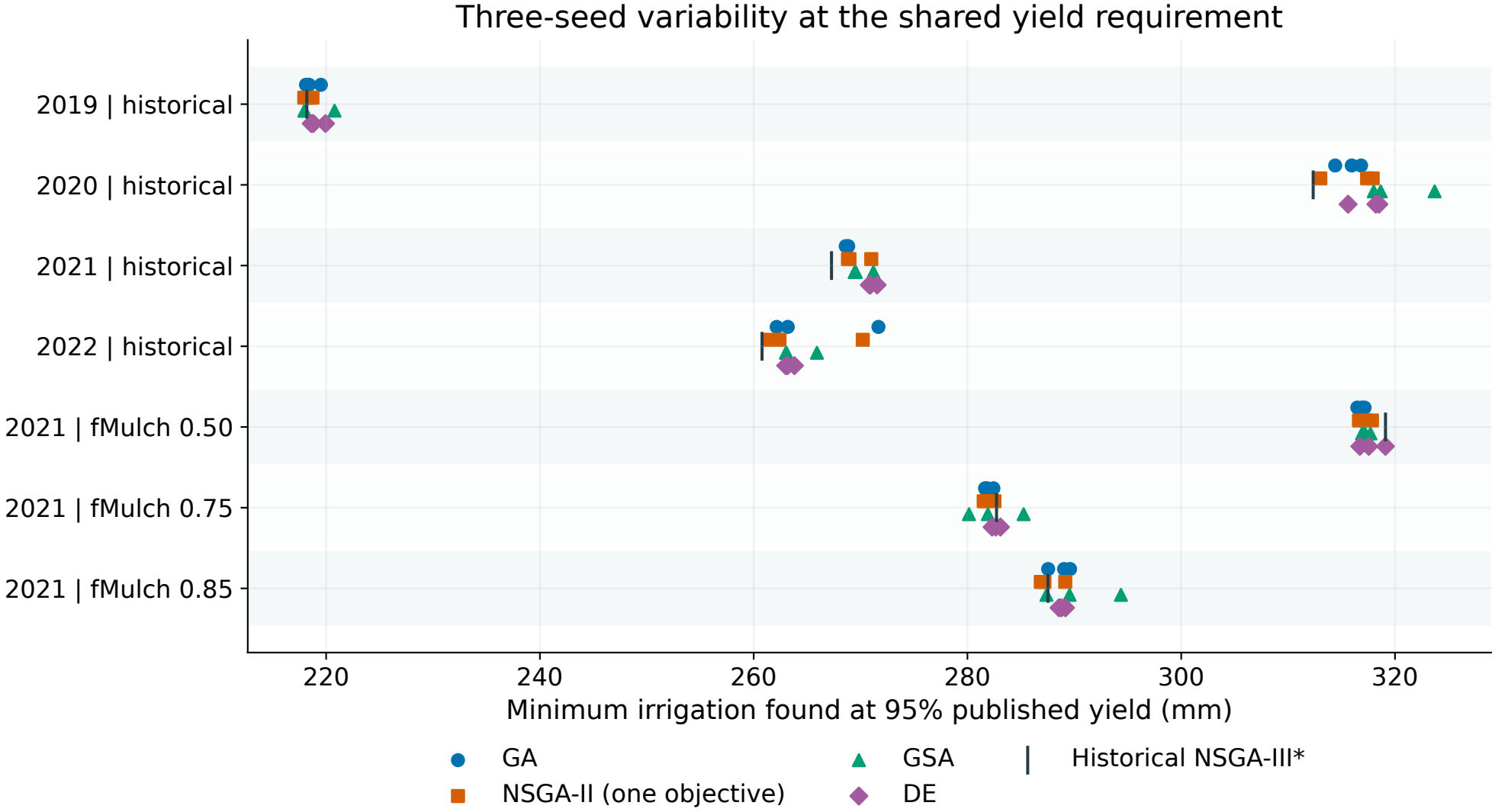

2021 rainfall checks at six fixed proportional-irrigation levels
fMulch = 0.50
fMulch = 0.75
fMulch = 0.85
Grain yield at 13% moisture (t ha⁻¹)
0
2
4
6
8
10
12
14
16
0
300
400
633.47
Seasonal irrigation (mm)
Original GEE
Published-figure reconstruction
Reconstructed pattern; GEE seasonal total
Trace rainfall redistributed (synthetic)
Published DI yield (15.86 t ha⁻¹)

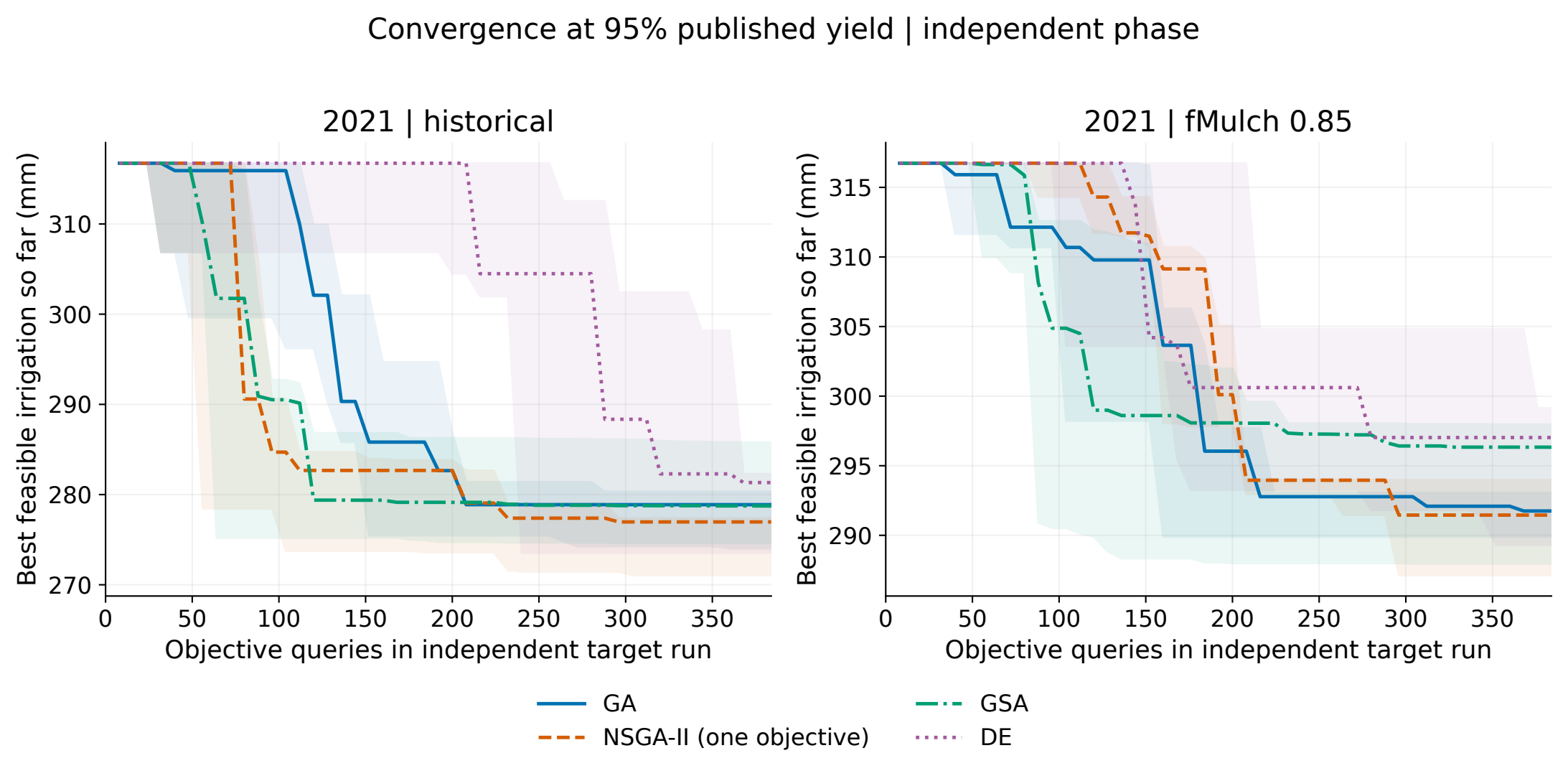

Single-objective minimum-irrigation searches | historical four-year models
GA
NSGA-II (one objective)
GSA
DE
Historical NSGA-III*
Zero-irrigation yield
DI model baseline
2019
2020
2021
2022
Required grain yield (t ha⁻¹)
Minimum irrigation found (mm)
Maximum root depth 2.30 m; fMulch = 0. Bands: range across three seeds. *External, unequal search budget.

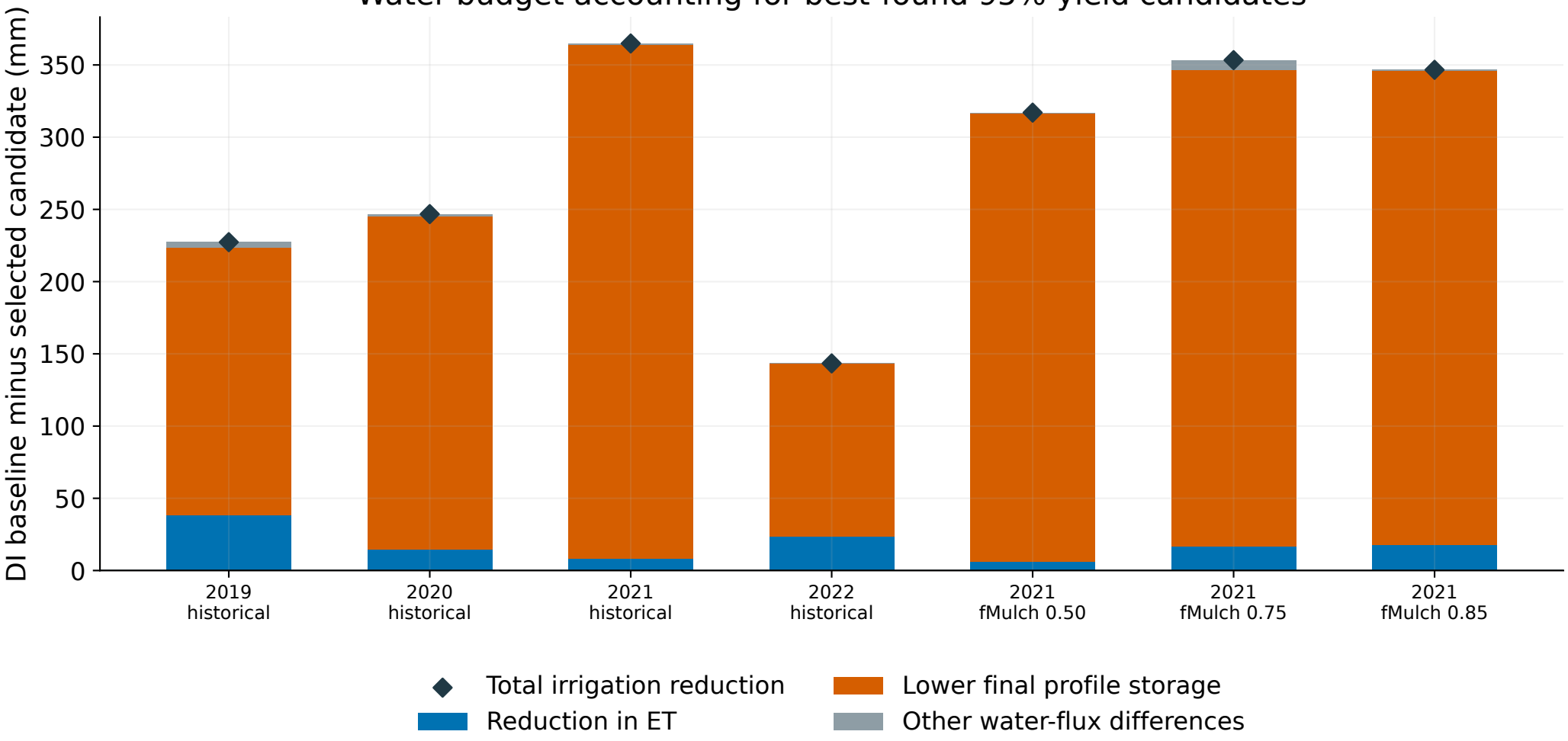


Separate conditional parameter sets; original GEE forcing. No added storage floor or event hardware cap.


2021, fMulch = 0.85 | actual evaluated solutions
Achieved grain yield at 13% moisture (t ha⁻¹)
Actual seasonal irrigation (mm)
0
2
4
6
8
10
12
14
16
0
100
200
300
400
500
600
GA
NSGA-II (one objective)
GSA
DE
Proportional DI response (prior run)
Historical NSGA-III front*
95% of published DI yield


Three-seed variability at the shared yield requirement
2019 | historical
2020 | historical
2021 | historical
2022 | historical
2021 | fMulch 0.50
2021 | fMulch 0.75
2021 | fMulch 0.85
220
240
260
280
300
320
Minimum irrigation found at 95% published yield (mm)
GA
NSGA-II (one objective)
GSA
DE
Historical NSGA-III*


Convergence at 95% published yield | independent phase
2021 | historical
2021 | fMulch 0.85
Best feasible irrigation so far (mm)
Objective queries in independent target run
GA
NSGA-II (one objective)
GSA
DE


Water-budget accounting for best-found 95%-yield candidates
DI baseline minus selected candidate (mm)
0
50
100
150
200
250
300
350
2019 historical
2020 historical
2021 historical
2022 historical
2021 fMulch 0.50
2021 fMulch 0.75
2021 fMulch 0.85
Total irrigation reduction
Reduction in ET
Lower final profile storage
Other water-flux differences